

FROM CREATURE TO CREATURE

A NEW TRANSFORMABLE MECHANICAL BILLBOARD SYSTEM



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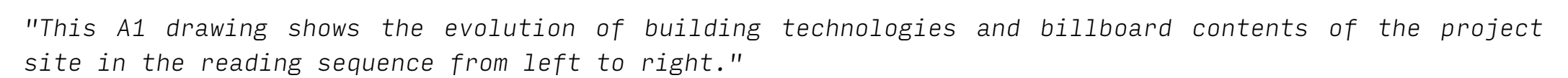
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How to build up a transformable billboard system that allows transforming from a gig stage to a telescoping ball?

Part I: Context

In this chapter, I will introduce my initial research, the development history of Times Square's architecture and billboard technology, the site analysis of the project and an overview of the entire project.

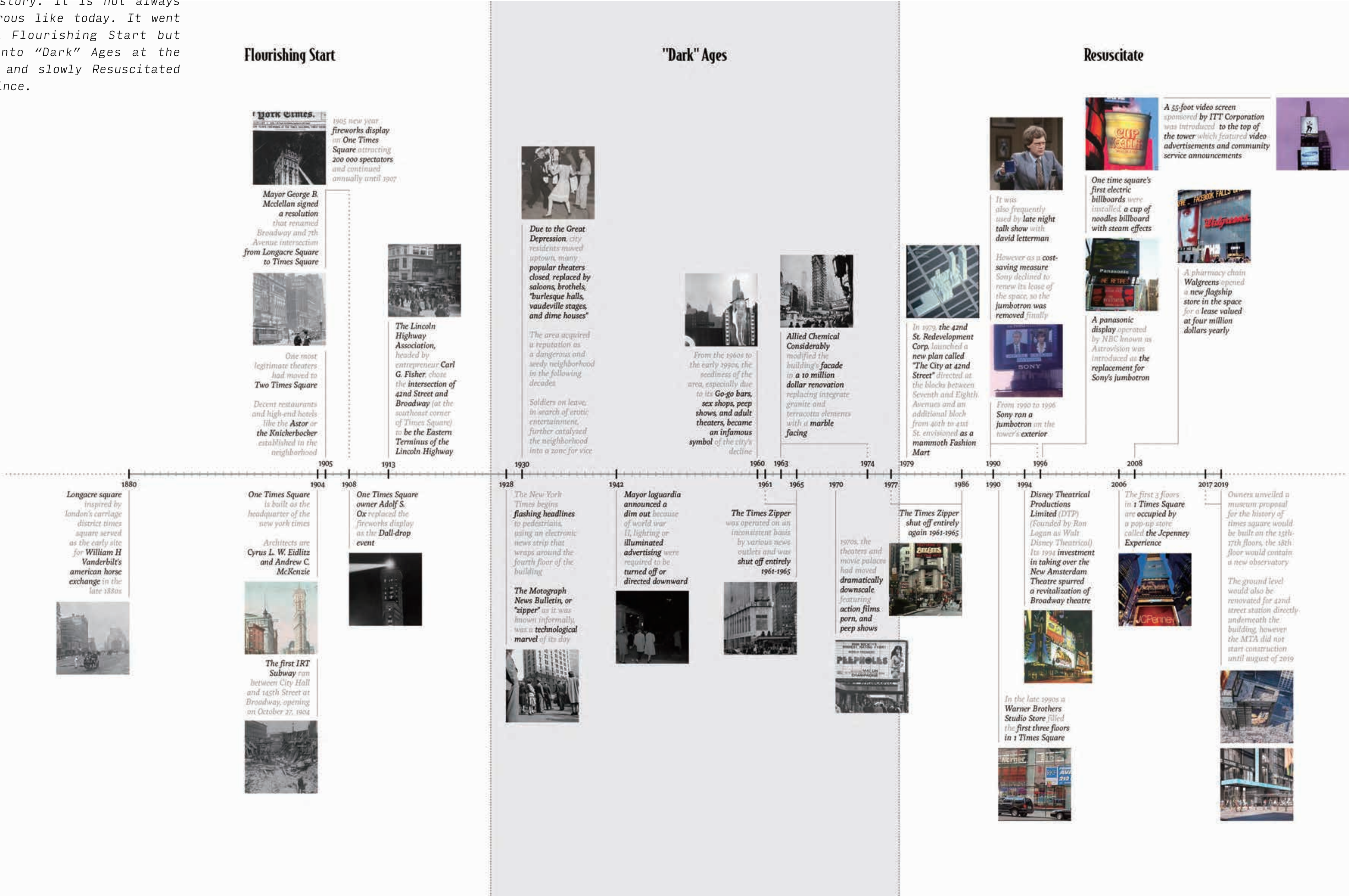
History of Times Square - Evolution



Part I: Context

History of Times Square – General Timeline

"Times Square as a public space, has held so many events in the history. It is not always prosperous like today. It went from a Flourishing Start but drop into "Dark" Ages at the middle and slowly Resuscitated ever since.



Part I: Context

History of Times Square – Ball-drop

"As a special example of billboard technology in Times Square, the Ball-drop has aroused my great research interest. After researching, I found that it is not just a special billboard because of its unique shape. It is also a dynamic monument which gives it the aggressivity to time and space."

Ball-Drop Evolution Time Line

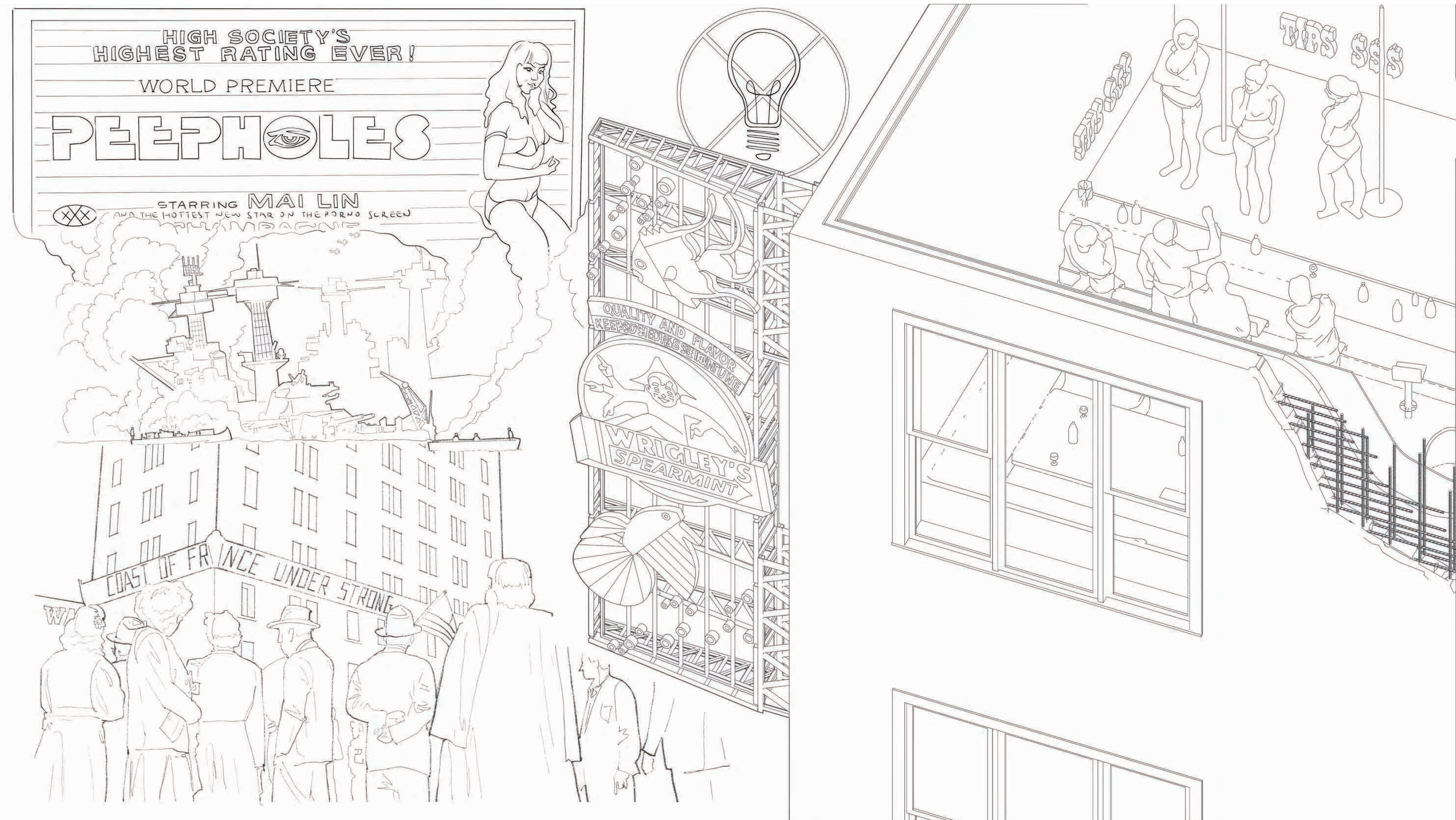




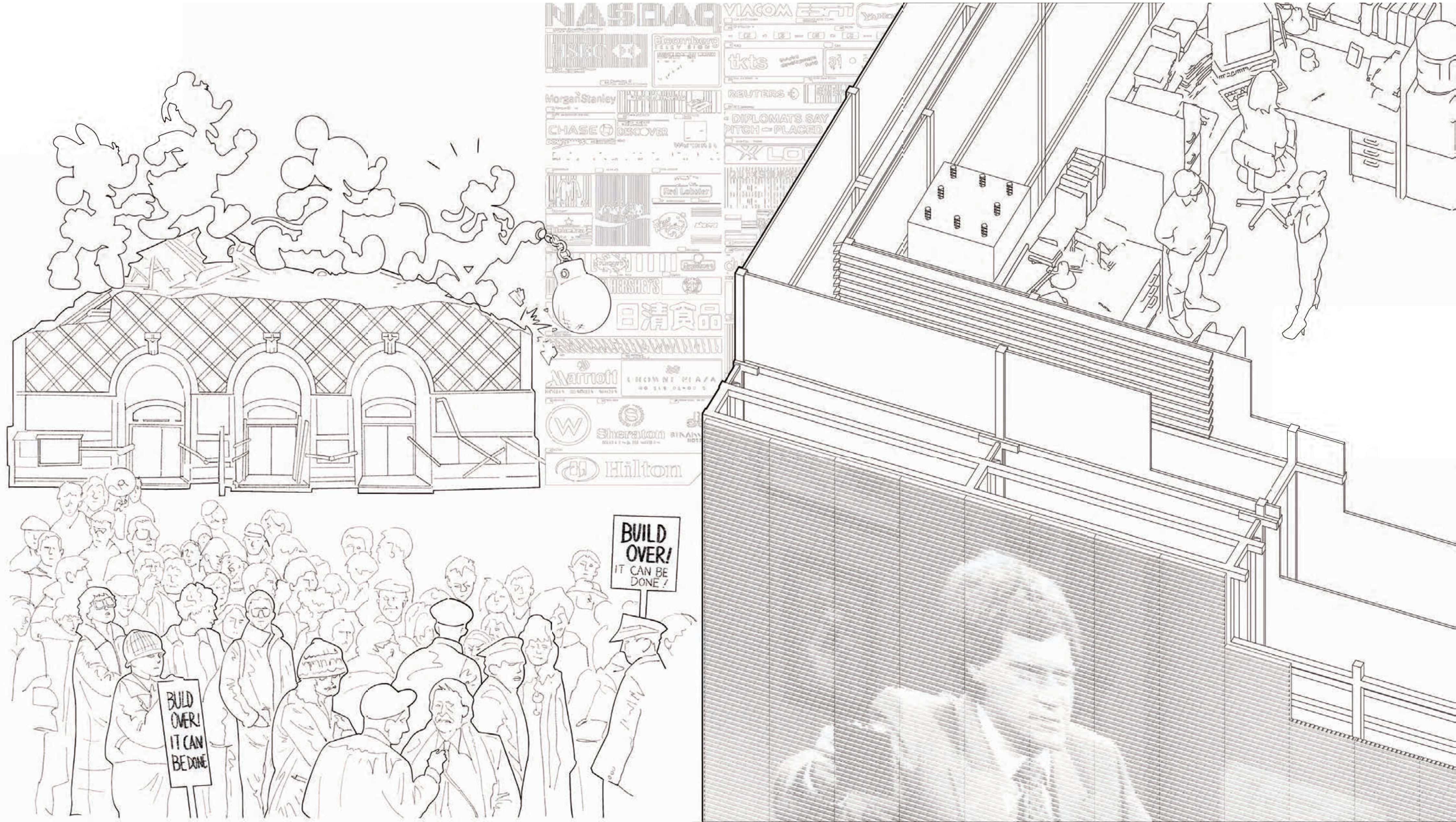
"During this period, the horse-drawn carriage industry brought here economic growth, buildings were generally brick structures, and billboards were mostly made of textile fabrics."

Part I: Context

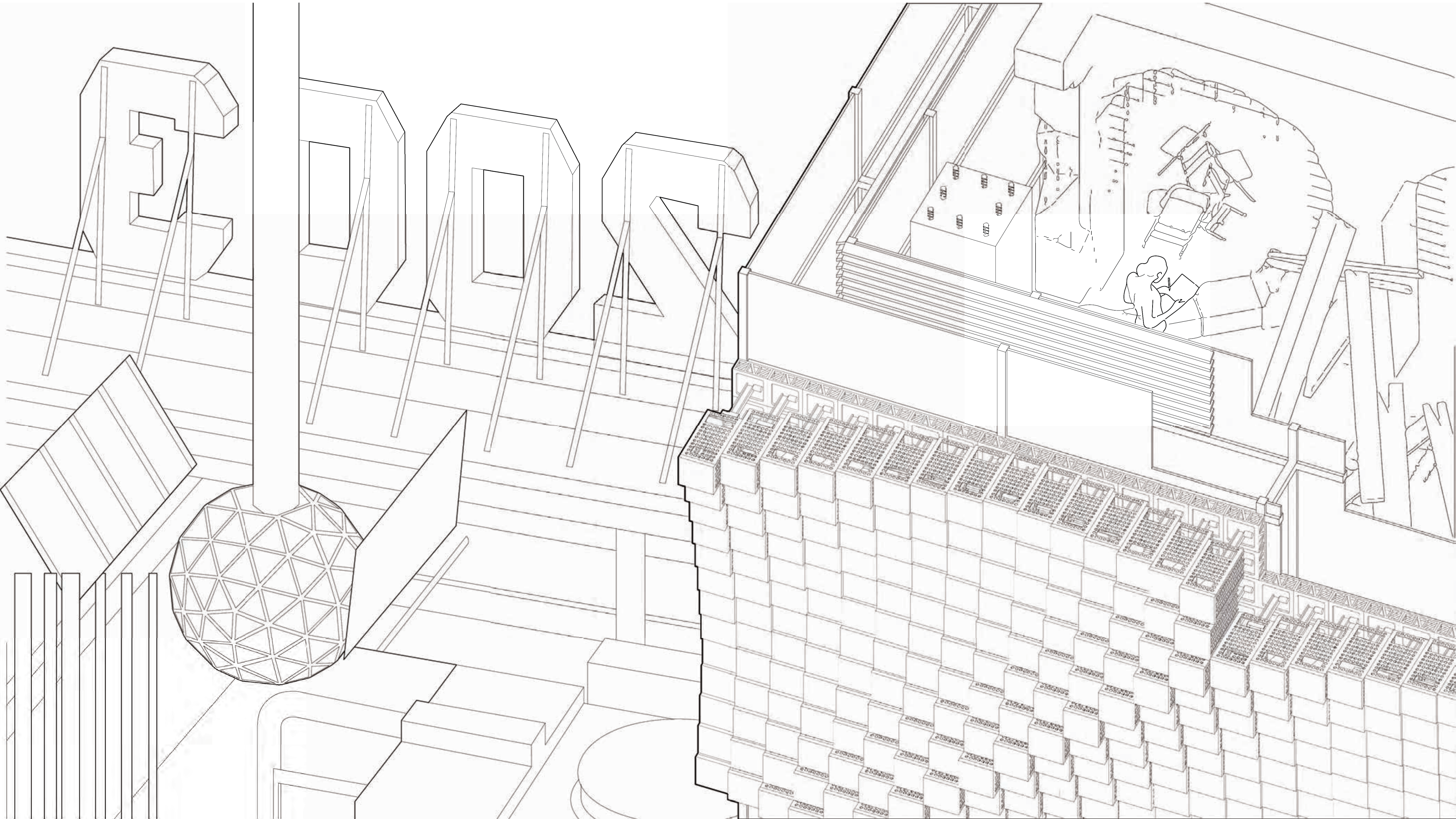
Technical Evolution - 1928-1979



"The economic turmoil before and after World War II, the domestic demand for film and stage plays shrank sharply, and the alternative vulgar entertainment industry directly affected the content of billboards. Technological development brought concrete buildings and neon lights."



"With the large-scale use of LED billboards, the vulgar industry and its affiliated billboards have also been replaced by any "healthy" content. After several updates, the building has also become an earthquake-resistant core tube + glass curtain wall structure."



"Emerging technologies were put into use and quickly spread around the world, such as Coca-Cola 3D. This originates from the technology gene including the ball-drop that was planted in the early days of the plaza's name change."

Part I: Context

Nowadays – Interior of One Times Square

"At the same time, abuse of architecture behind the billboards exists. This was the interior of one times square in 2019."



Inside the building in 2019. Joey Hadden, 2019.



Exposed steel inside One Times Square. Joey Hadden, 2019.



A mostly-abandoned floor in One Times Square. Joey Hadden, 2019.



Exposed steel on a mostly-abandoned floor in One Times Square. Joey Hadden, 2019.

Part I: Context

Nowadays – Scene of Times Square



"Now the entire interior of the square has become a pure media space, which is highly informationized and electronic. However, the surrounding buildings have been de-named, they became pure structures to support the billboards in the dark."



"In my conception, all the covering media on Times Square will be removed, which reveals the look of the beneath buildings, the spatiality of Times Square is back. The advertisement's media converged into a sphere, for the sponsorships as usual."

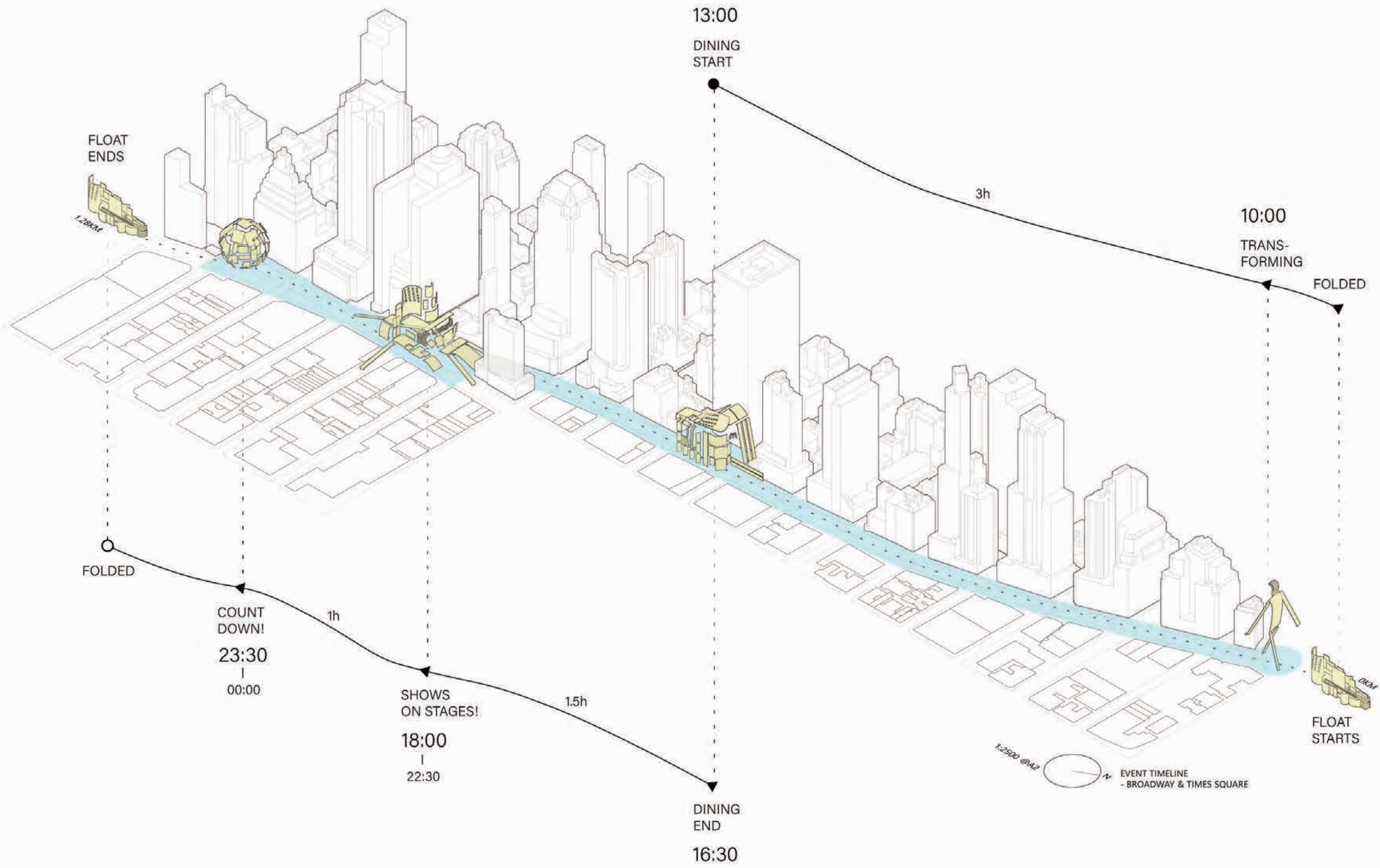


"A billboard suddenly having an interior, being a roof, becoming a monument. These give it the capabilities of other architectural agencies, more than just a billboard. To compare with how the billboards used to be, they are now becoming one single object, it represents the whole plaza."

Part I: Context

Project Overview - Float Event Timeline

"Upon this, I propose to expand both time and space of the Ball-drop moment in Times Square's new year eve, to create a brand-new, new year celebration on Times Square. In the form of a float parade."



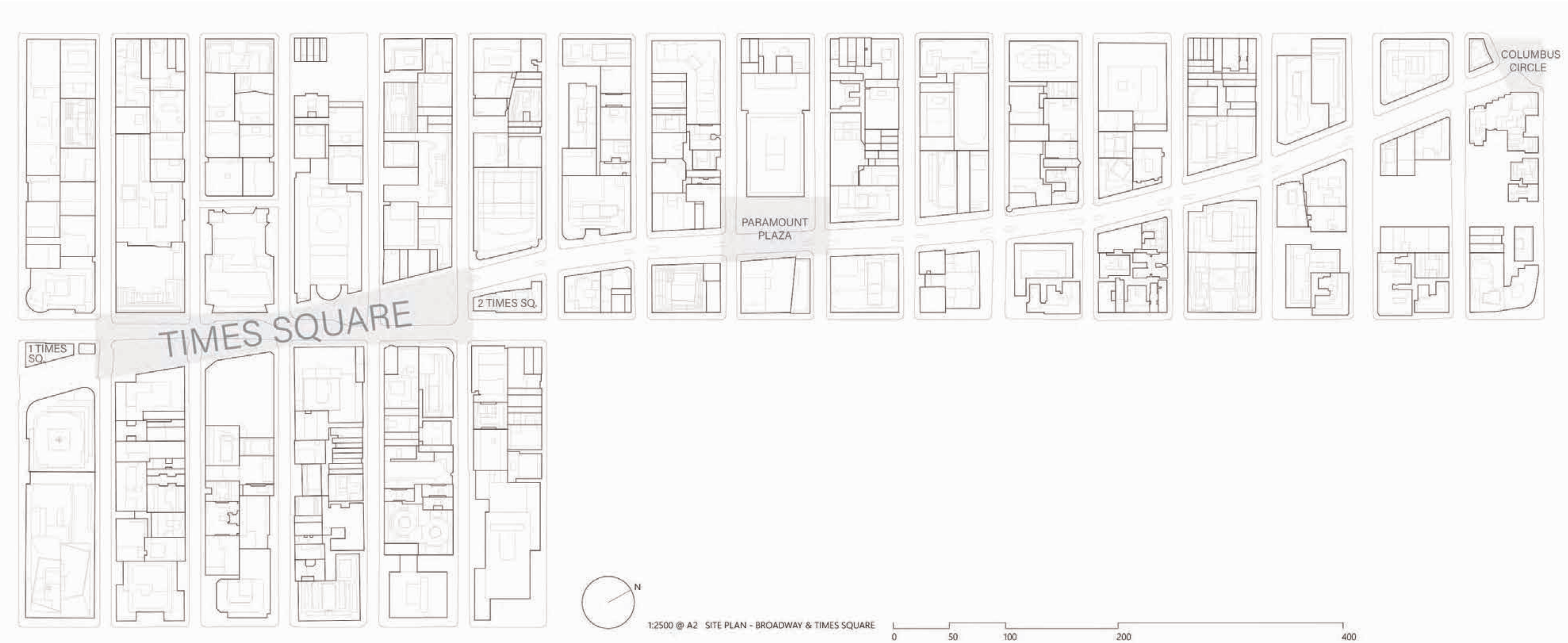
Part II: Site Analysis

The float parade will take place in Times Square, starting from Columbus Circle and ending at Times Square One. To introduce this huge-mass plan, it needs investigation and analysis of the local town space, billboard sponsors, etc.

Part II: Site Analysis

Site Analysis – Site Plan

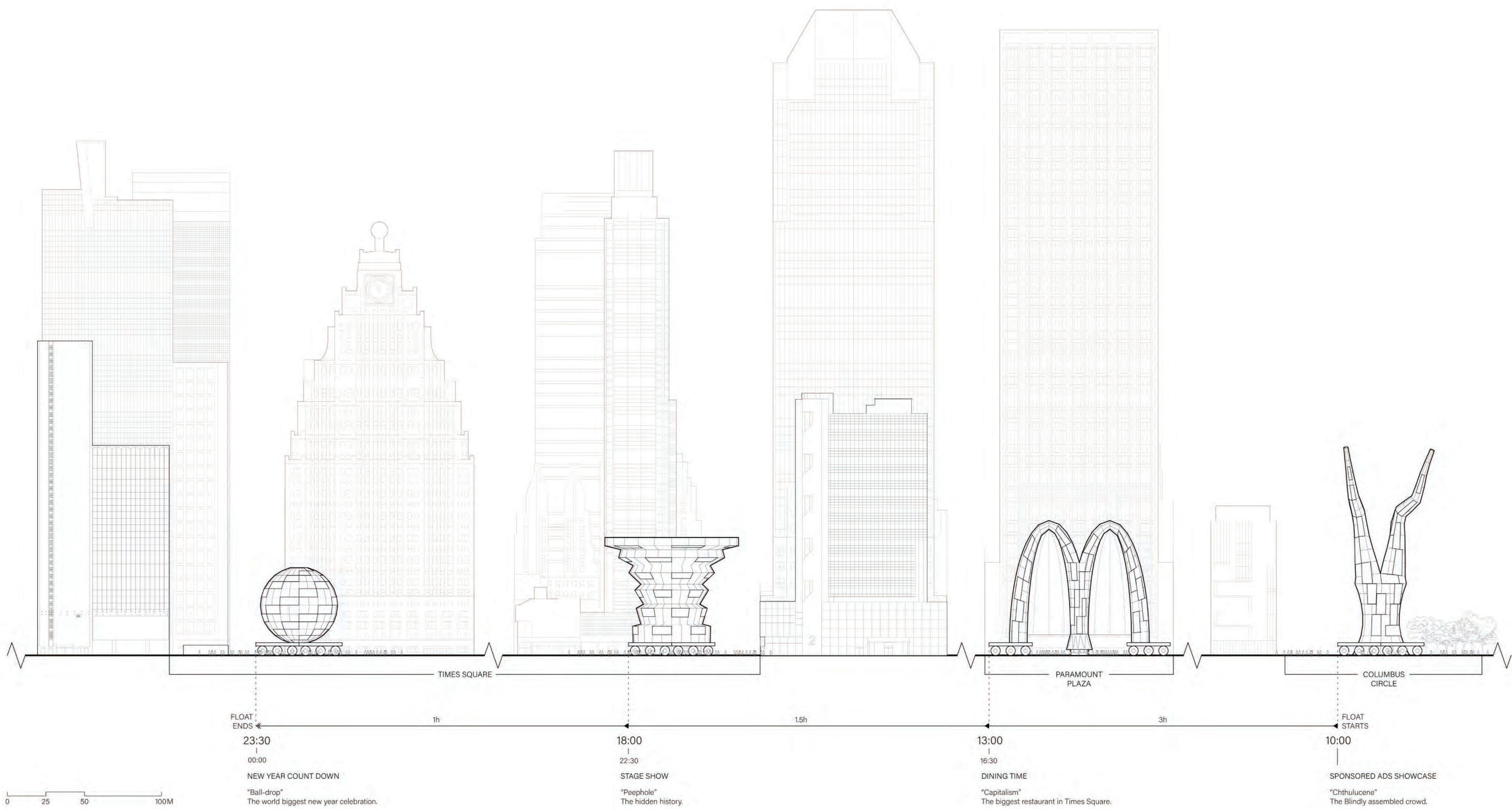
"The site plan shows the open areas from Broadway all the way down to Times Square, that capable for the float's major activities."



Part II: Site Analysis

Site Analysis – General Section with Timeline

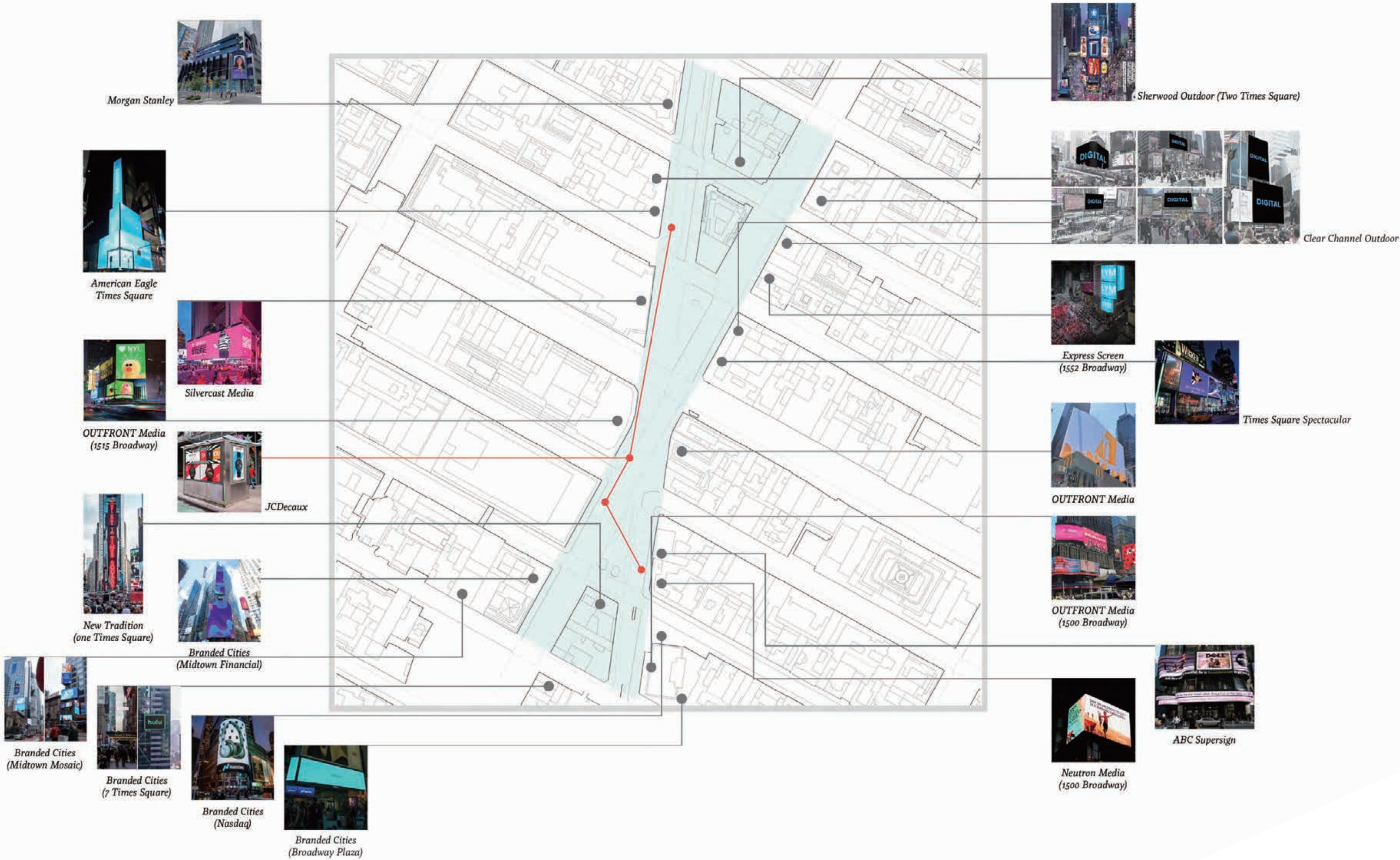
"The float will move slowly along Broadway. Because of movable modules, it can transform into four themed structures at different stages of the parade."



Part II: Site Analysis

Site Analysis – Stakeholders/Sponsorships

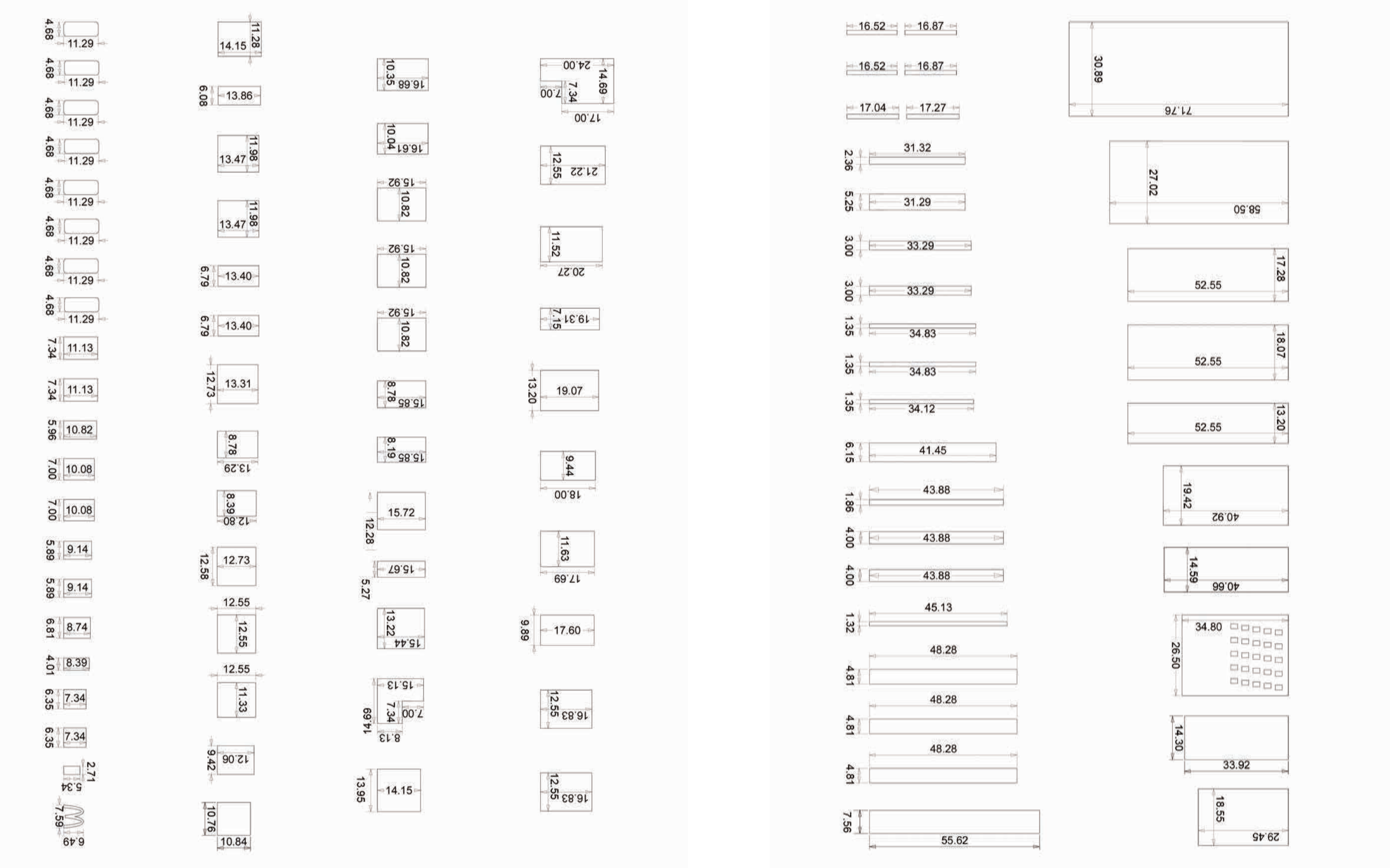
"The map shows nowadays the stakeholders of each billboard on times square. They are occupying all building surfaces in times square, and also the potential sponsors of the project."



Part II: Site Analysis

Site Analysis – Major Billboards

"In the analysis, there are totally around 90 major existing billboards on Times Square. It will be a very eco-friendly design consideration to reuse them."



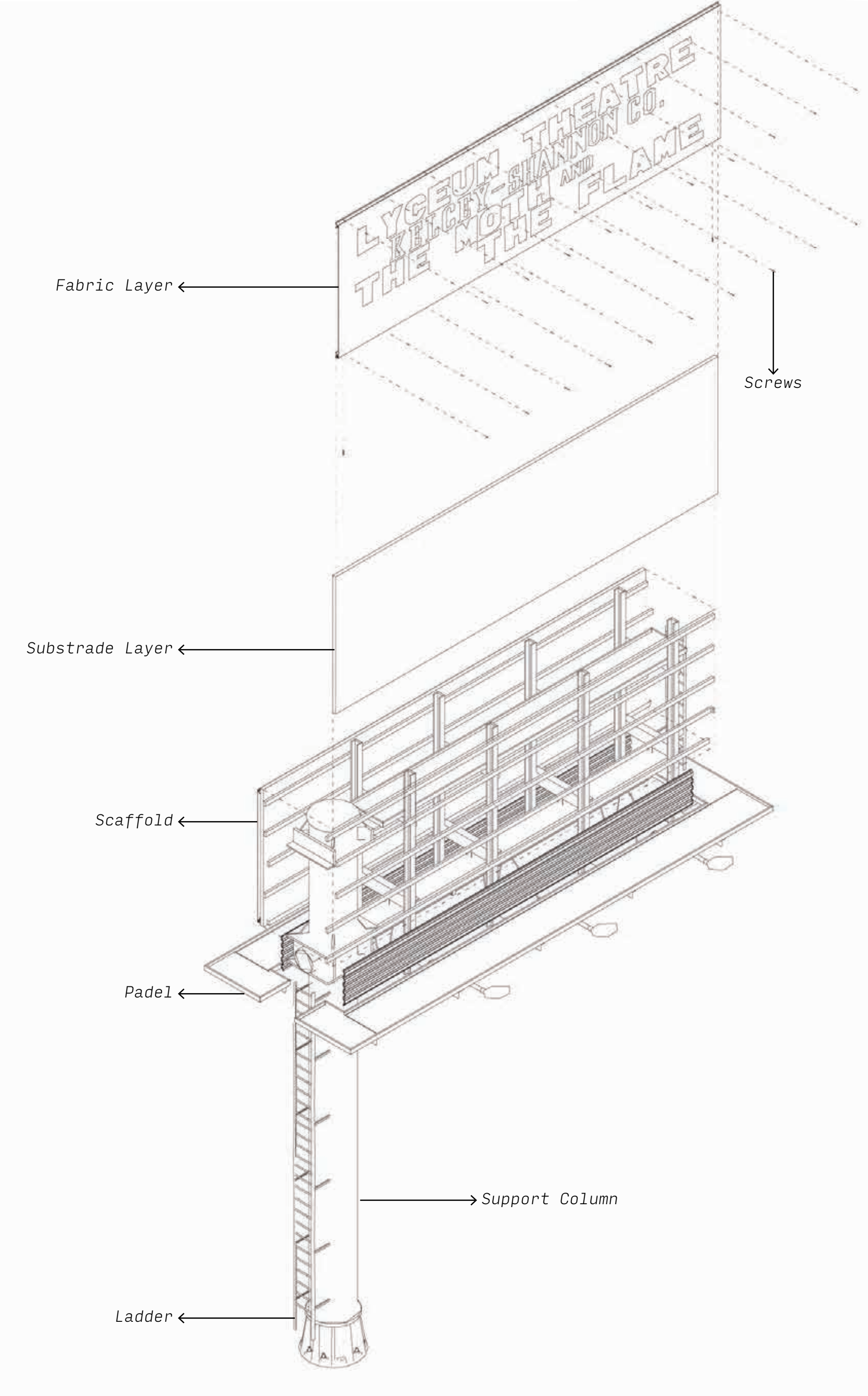
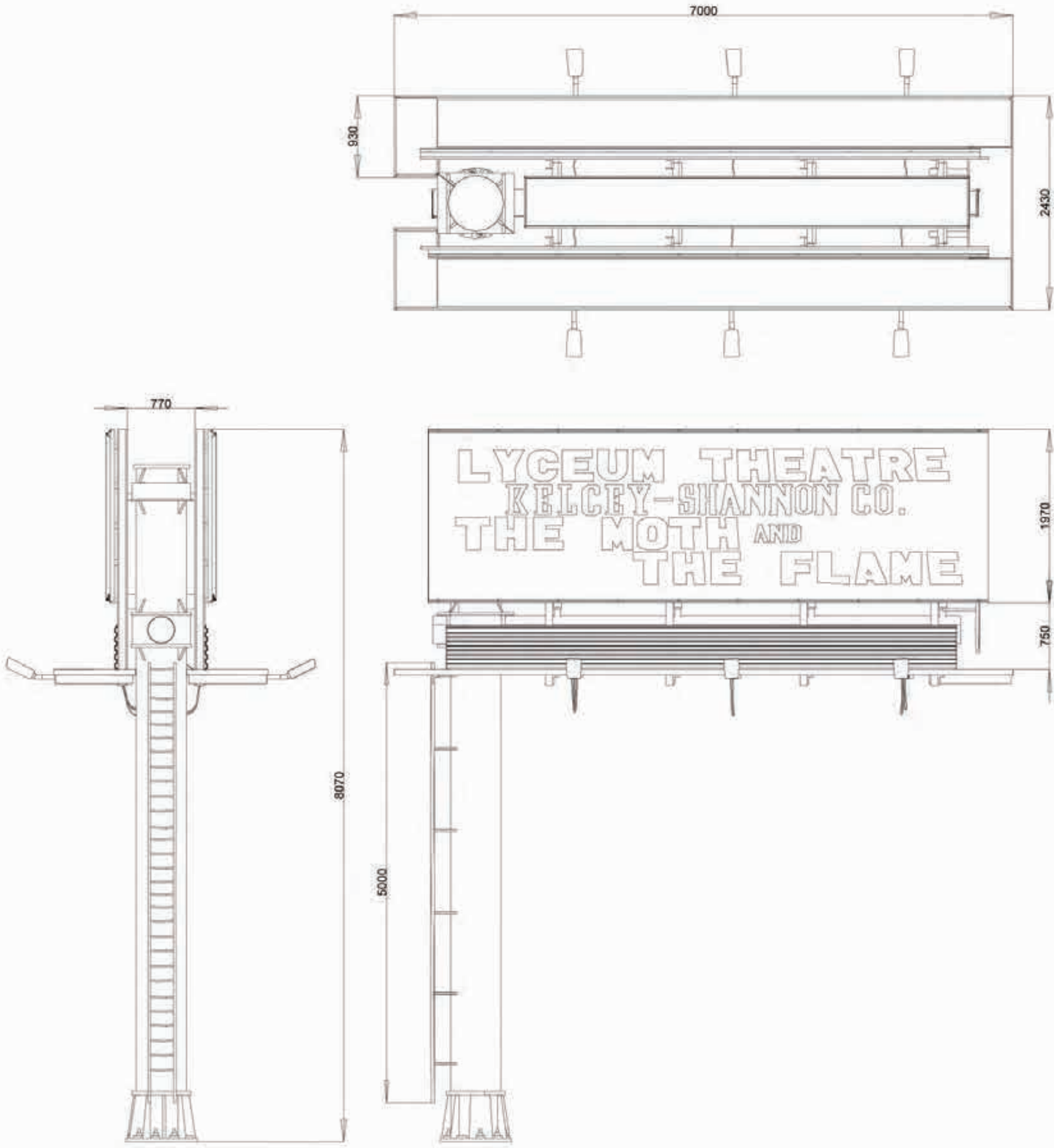
Part III: Related Case Studies

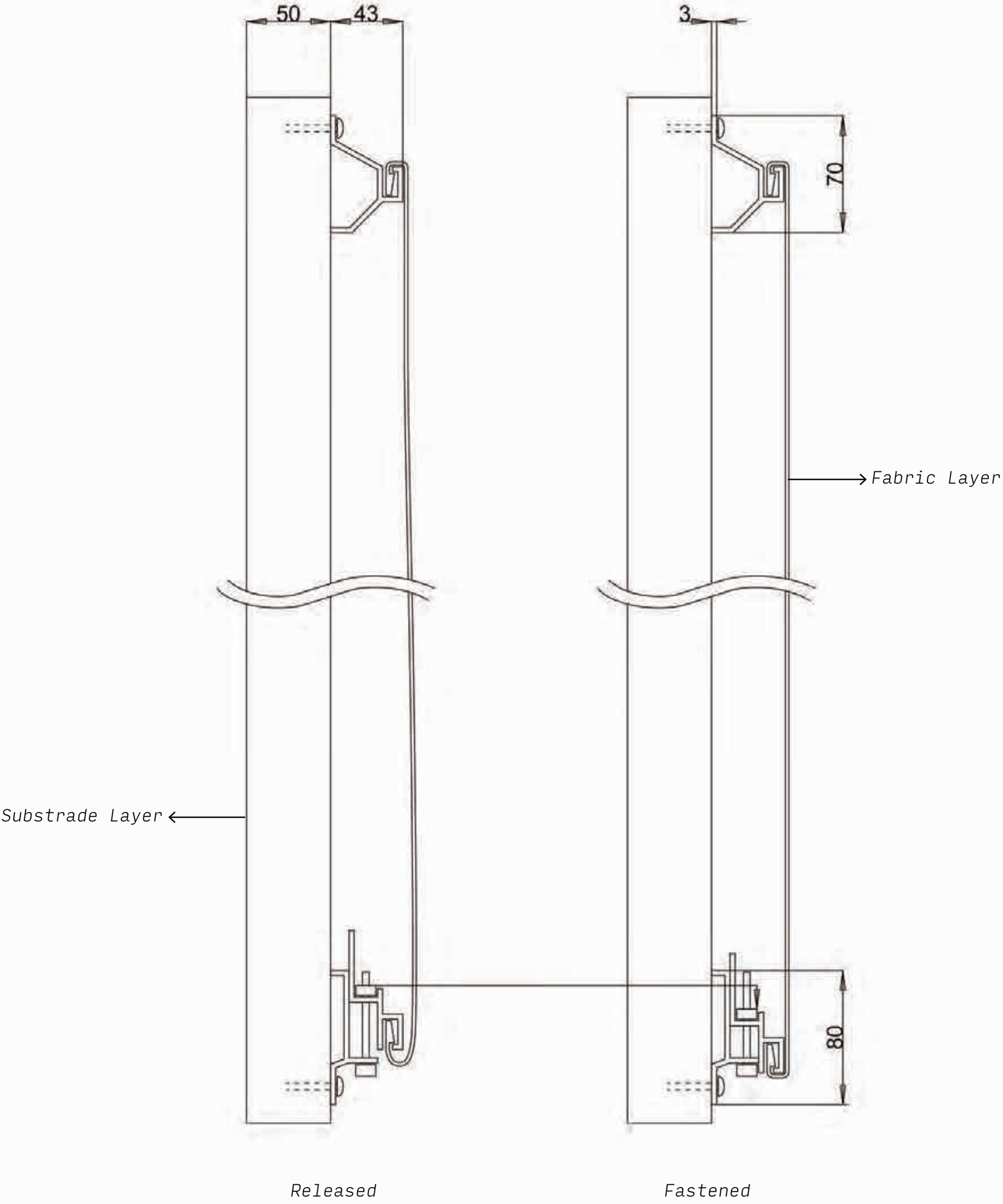
This chapter presents an extensive research and analysis from different billboards, field research, and actual cases related to display devices.

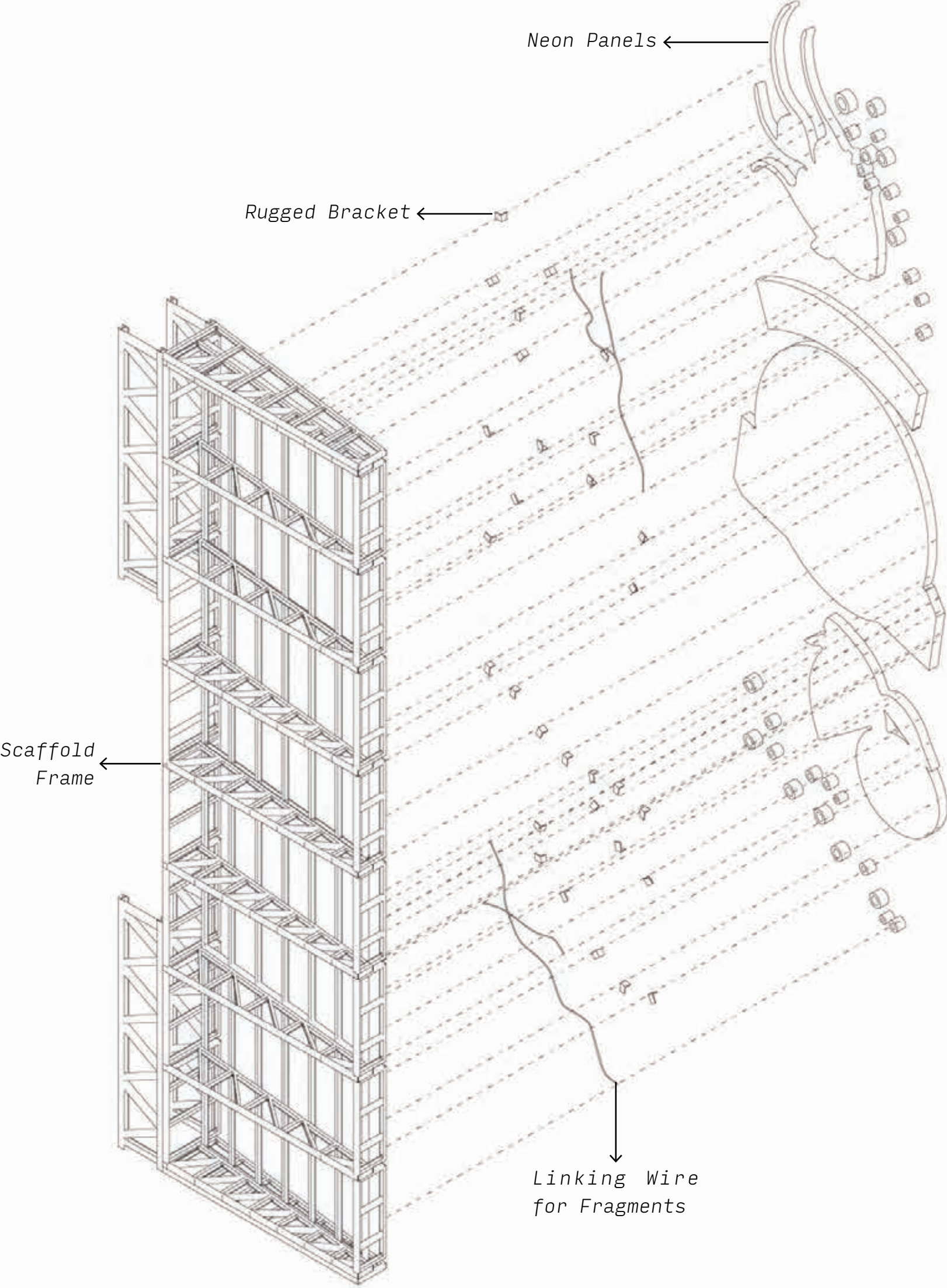
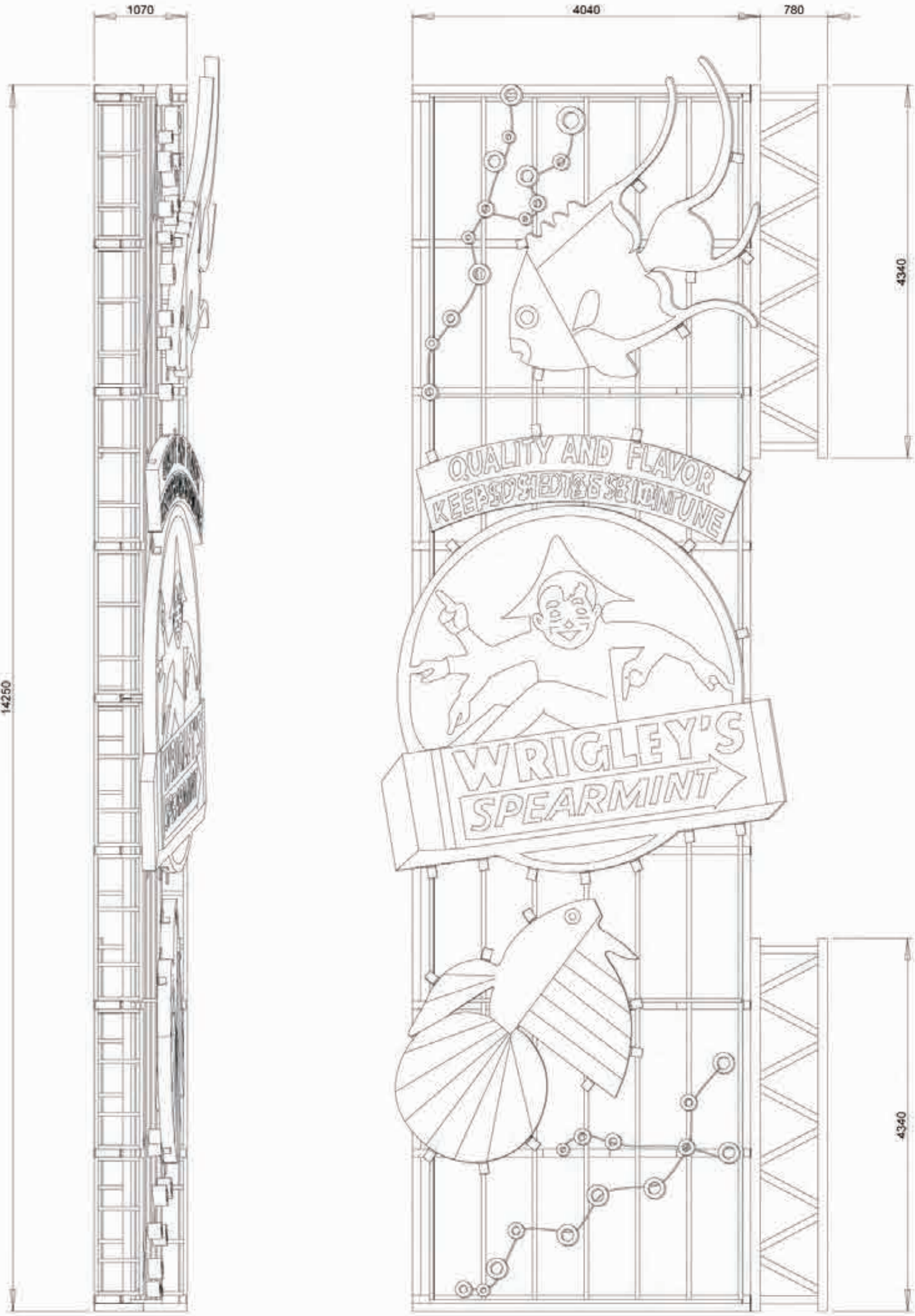
Part III: Related Case Studies

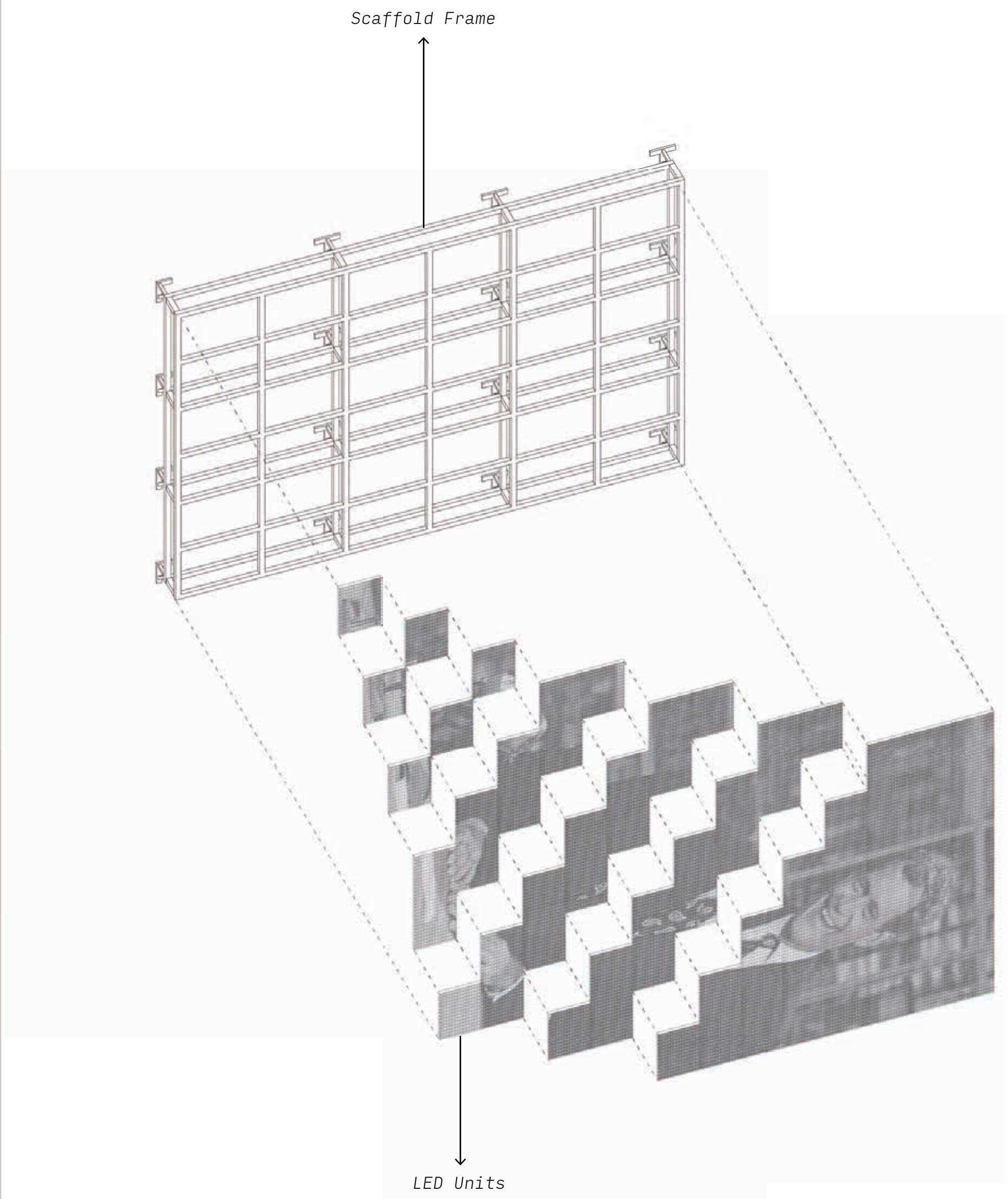
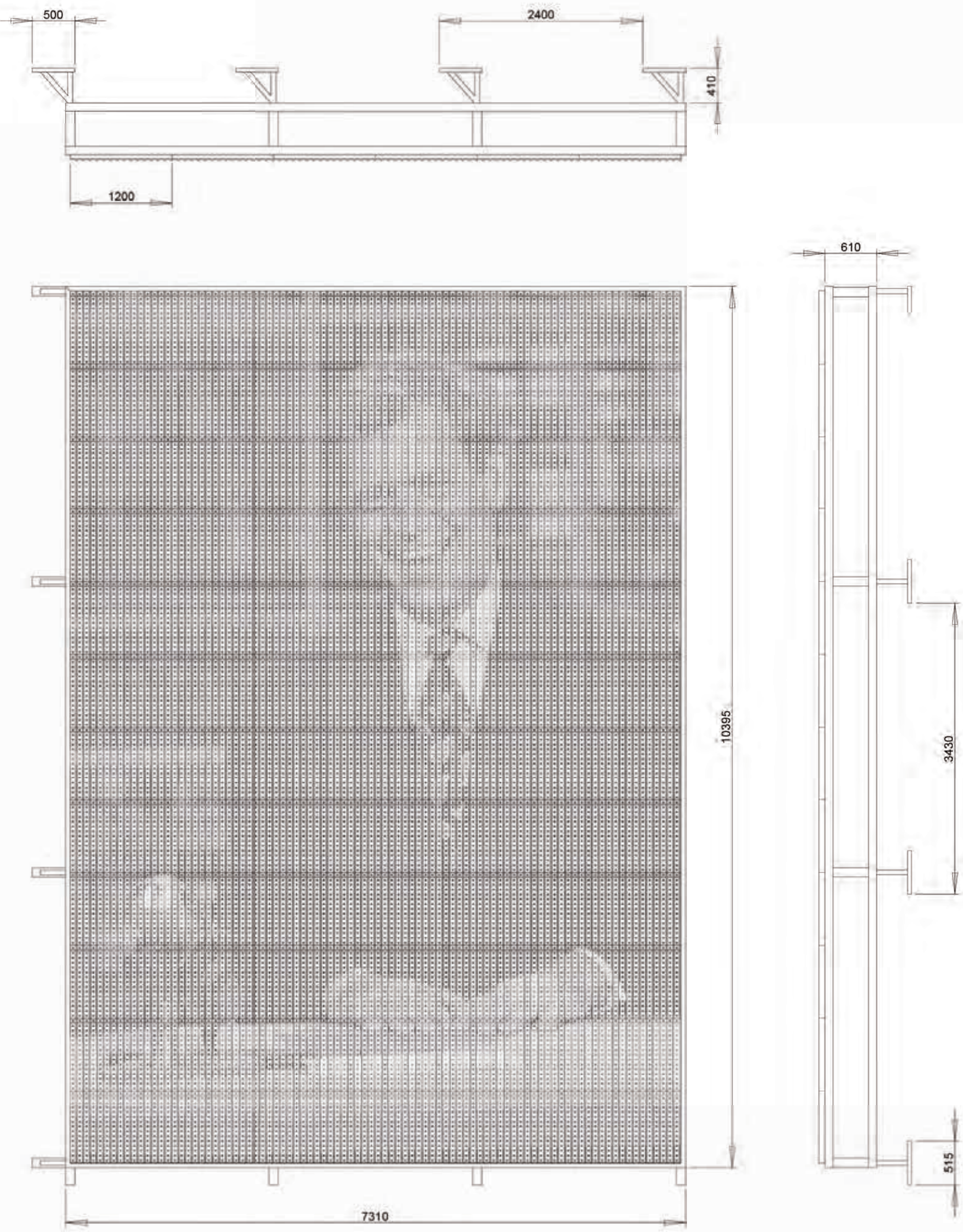
Billboards Typology – Fabric Billboard

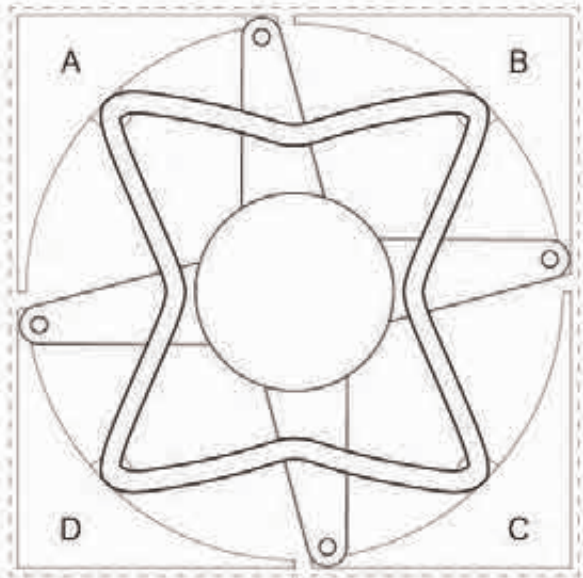
"To understand deeper about the mechanisms of billboards, the set of analysis concludes most of the current types."



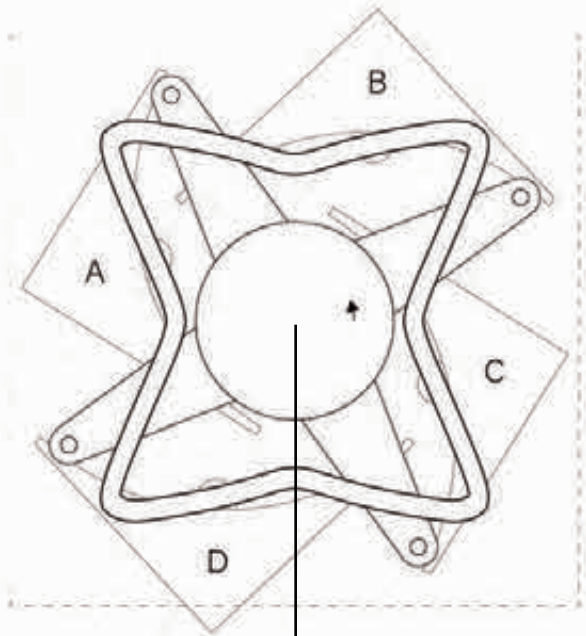




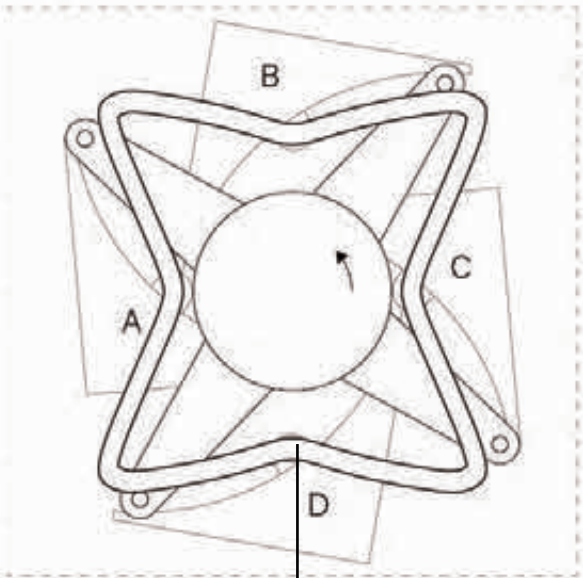




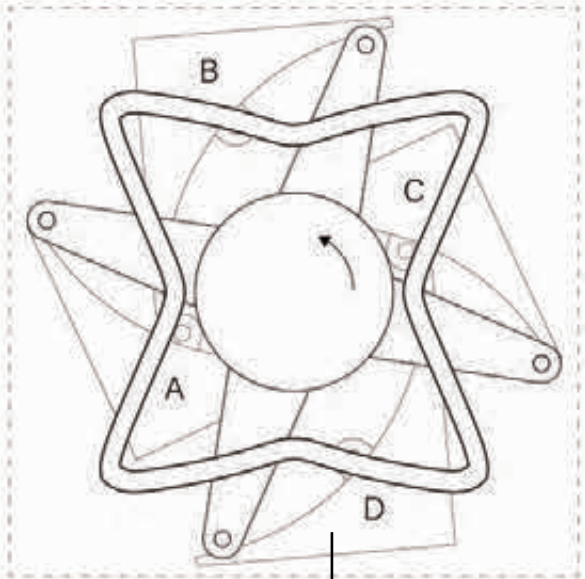
Showing Face AB



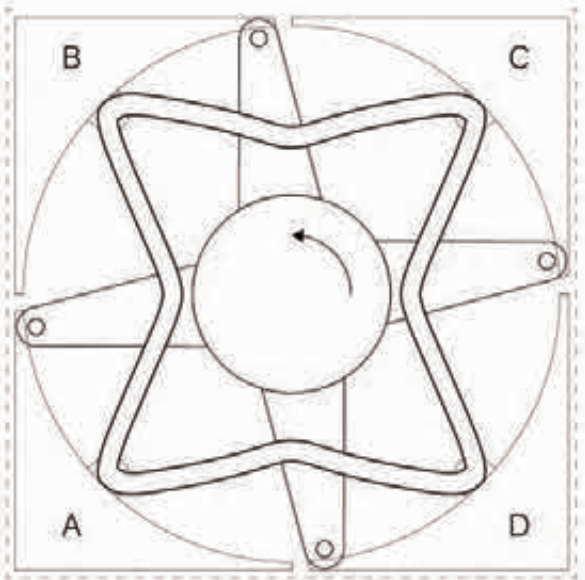
Shaft



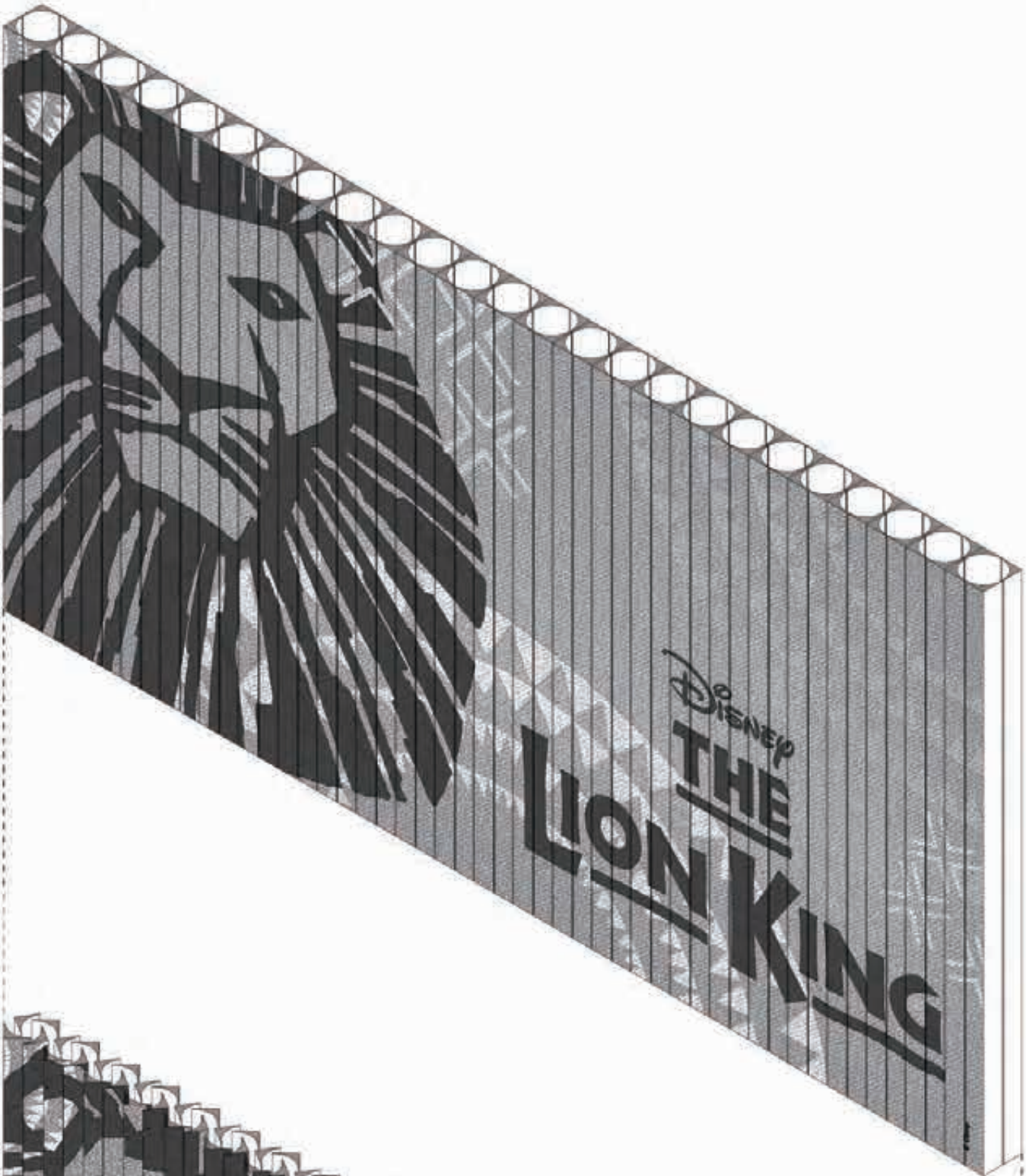
Steel Trace Rail



Fence Blade



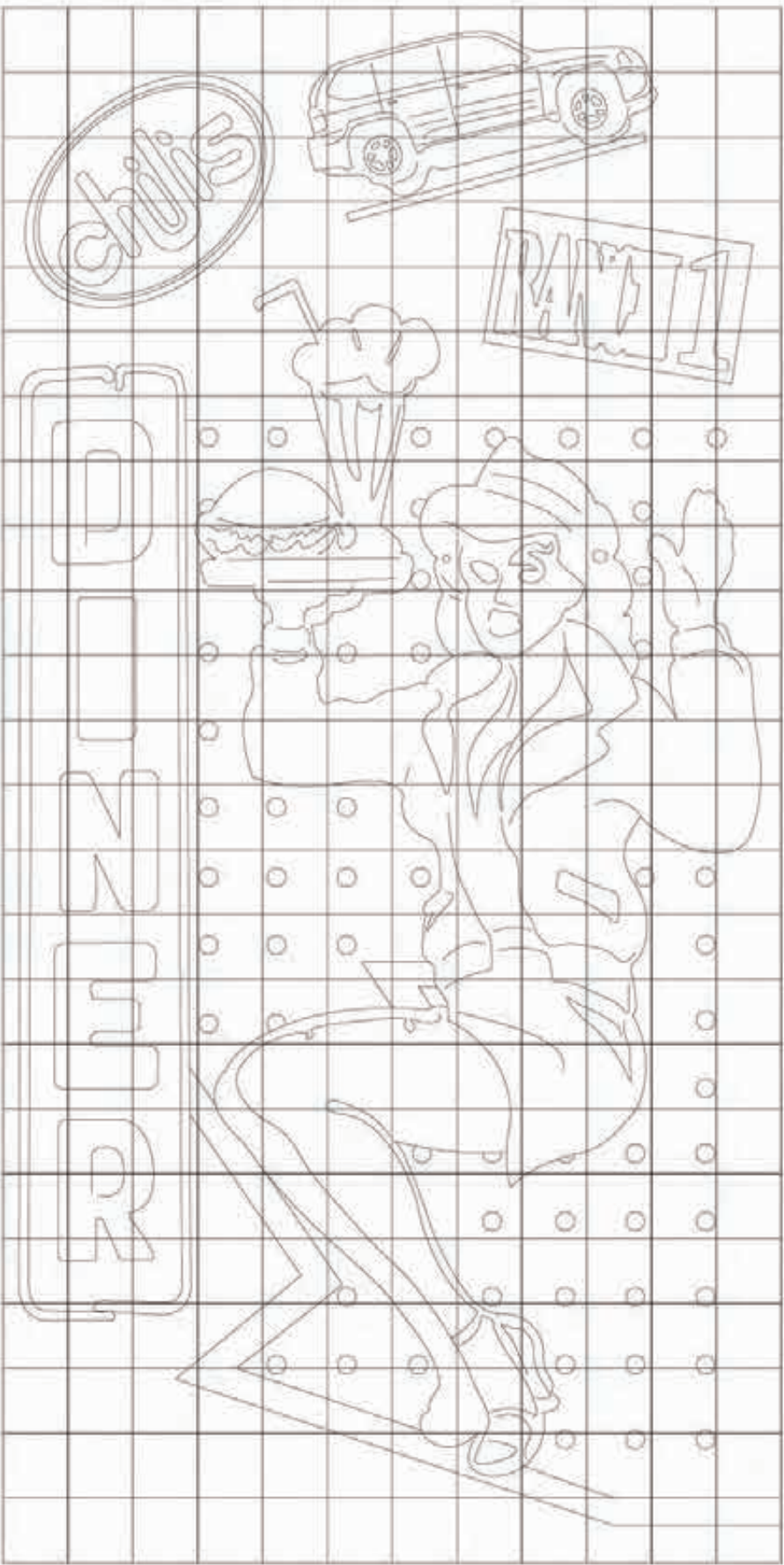
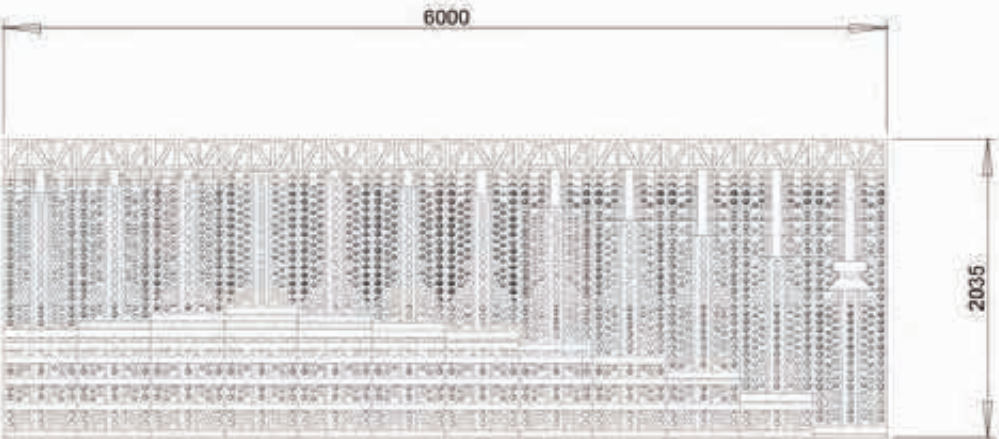
Showing Face BC



Static



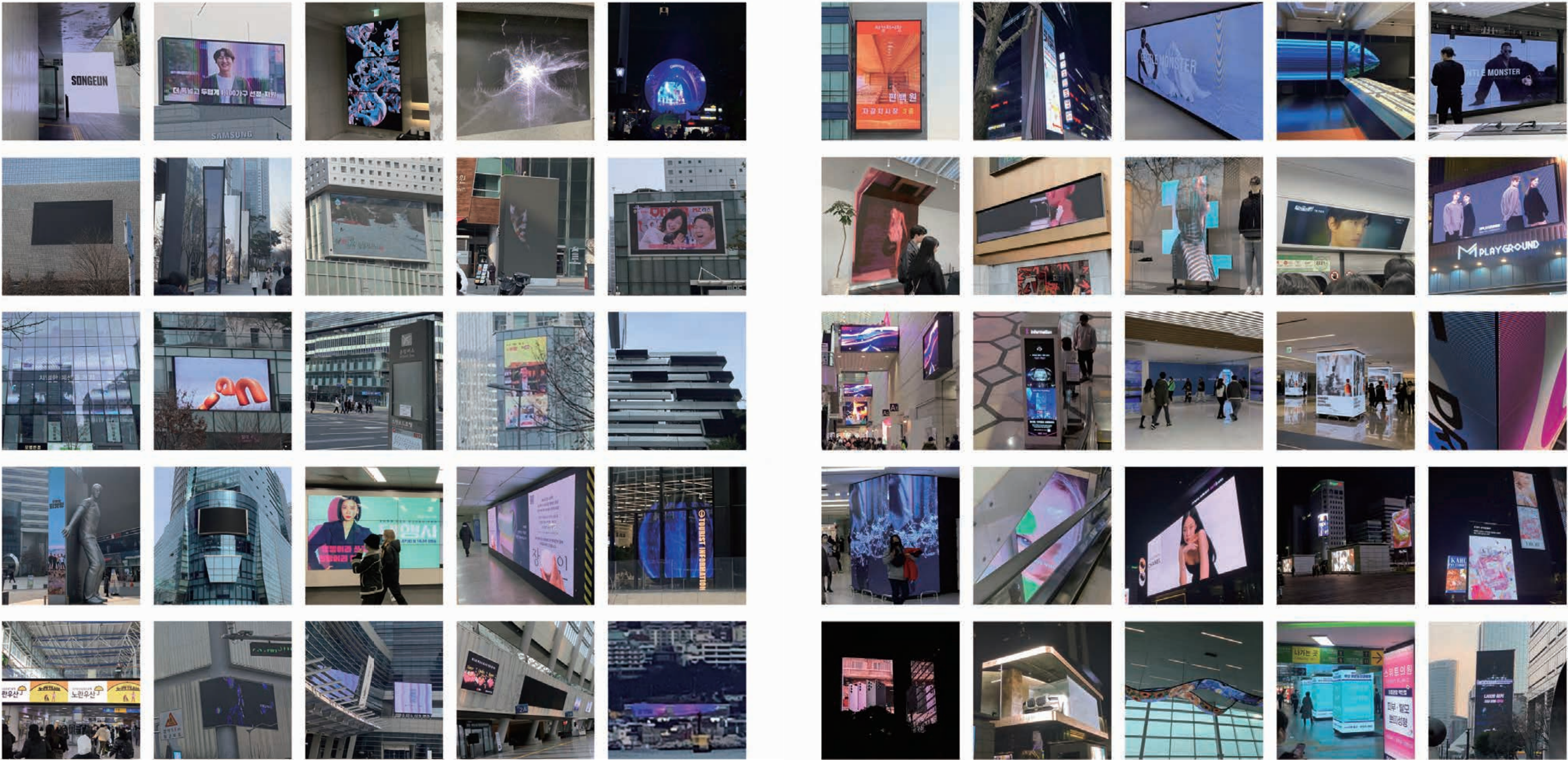
Dynamic



Telescopic Rods
& Servomotors

LED Circuits

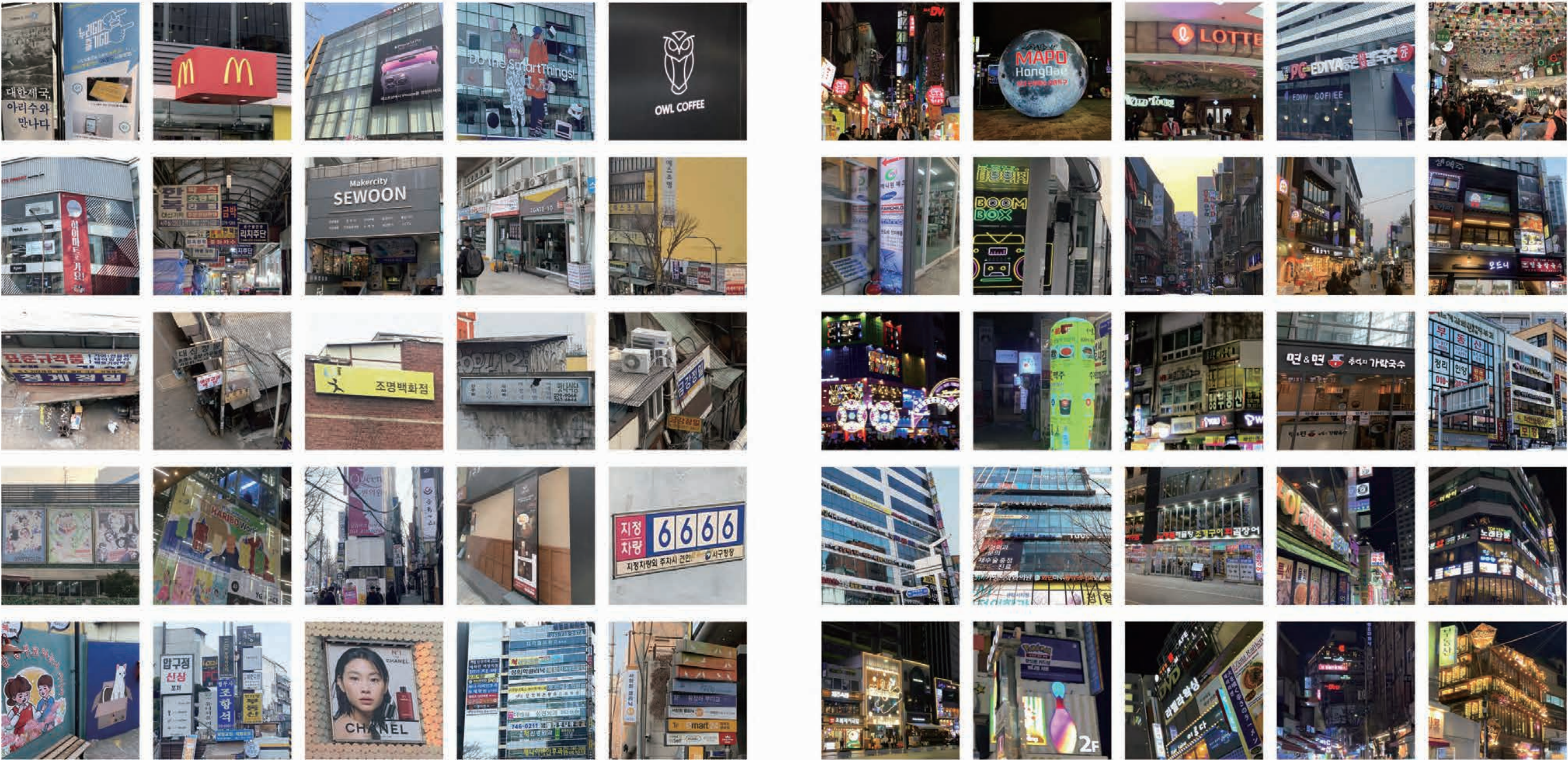
LED Boxes



"During my trip to Seoul+Busan, I analyzed all the billboard examples I saw, and found that the emerging types of LED and display media should account for the largest proportion."

Part III: Related Case Studies

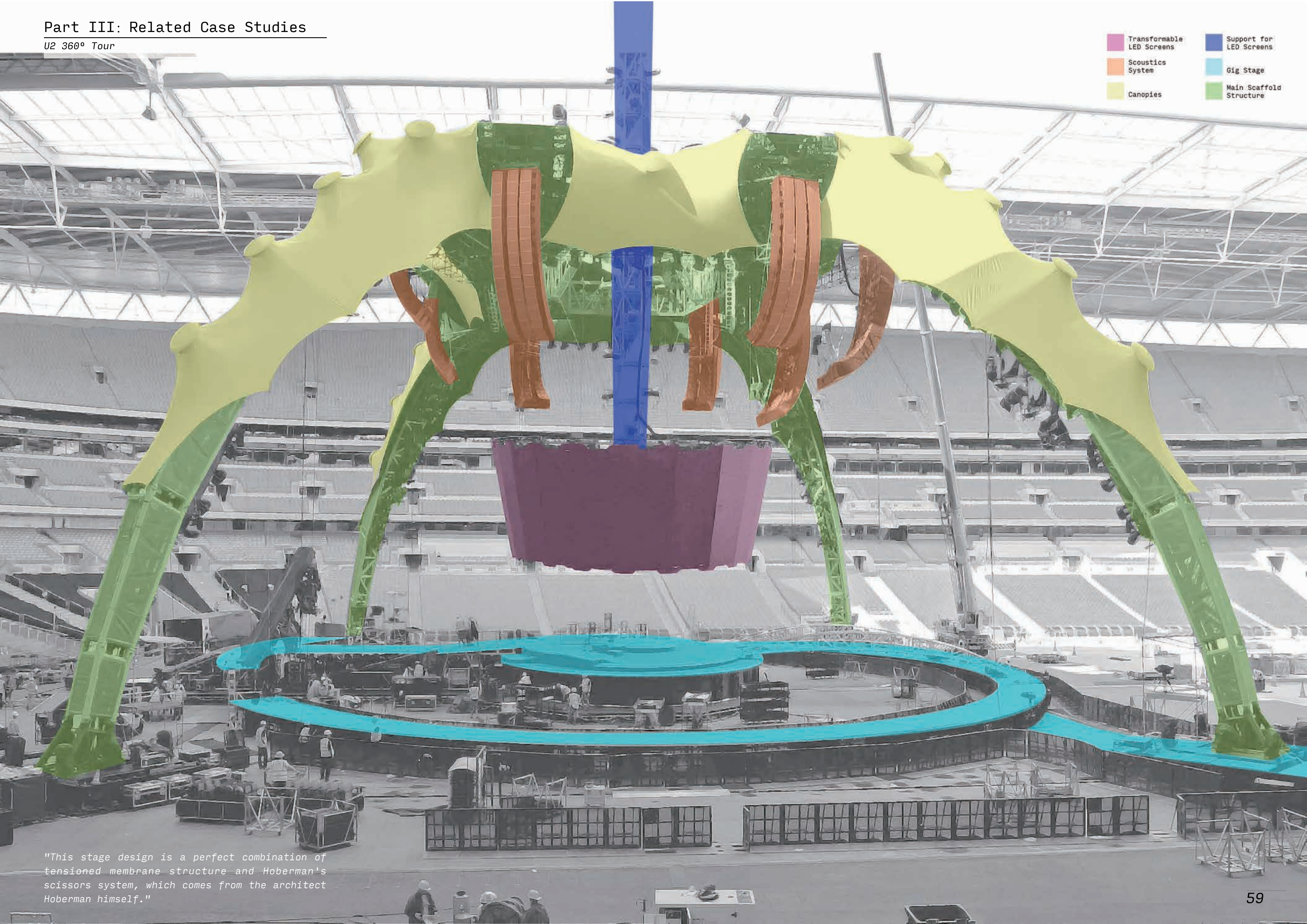
Billboard Case Study – Seoul & Busan Billboard Research



"However, the traditional fabric billboards also occupy a considerable proportion here, presumably because of their low cost and easy installation and disassembly. But they're mostly old and faded, left unattended."

"Surprisingly, in South Korea (especially Busan), neon billboards are used in large numbers, which is similar to Japan. Most of these colorful advertising installations are installed by local restaurants, gathered in the neighborhood where dining and entertainment are popular."

- | | |
|------------------------------|----------------------------|
| Transformable
LED Screens | Support for
LED Screens |
| Scoustics
System | Gig Stage |
| Canopies | Main Scaffold
Structure |



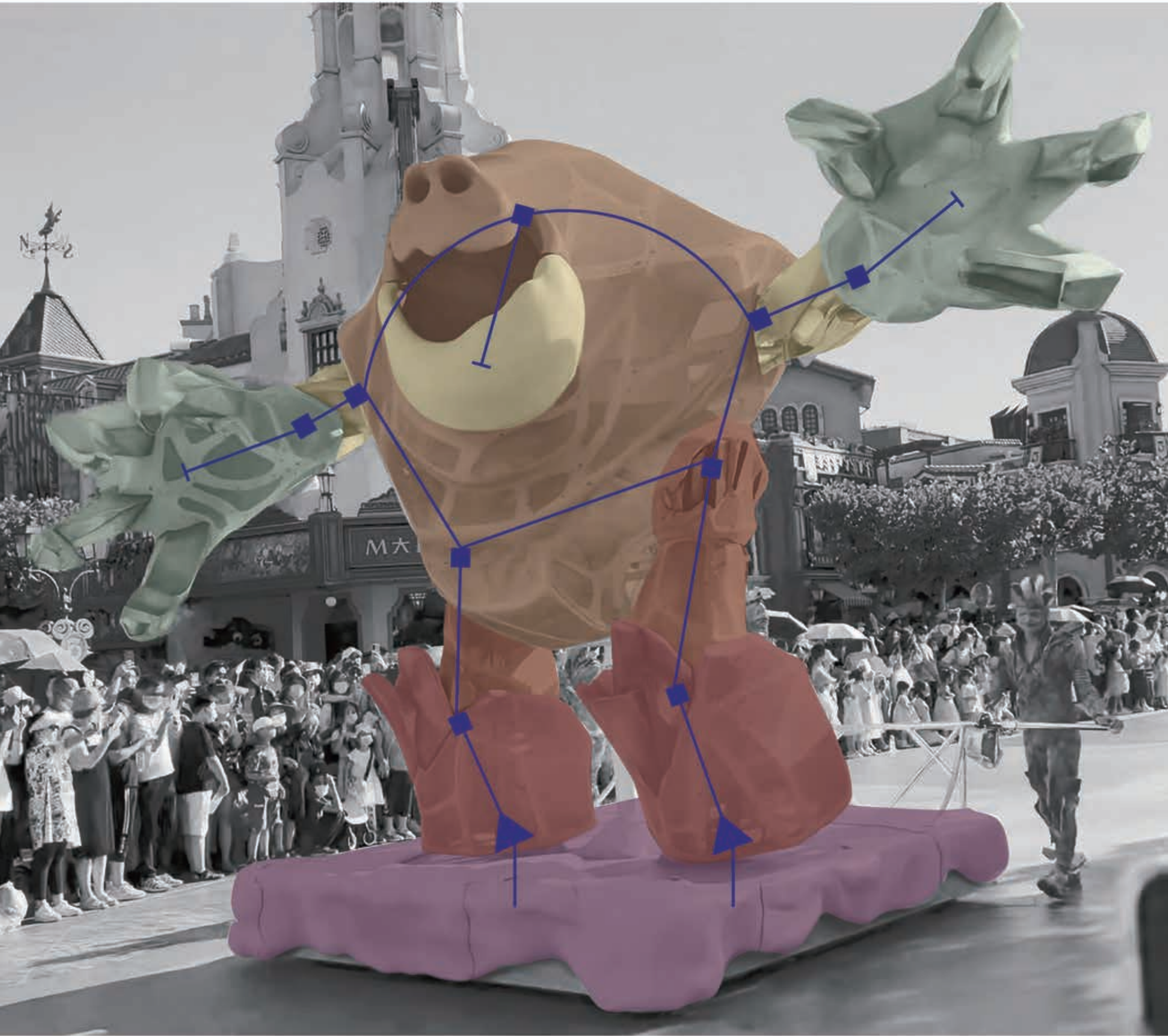
"This stage design is a perfect combination of tensioned membrane structure and Hoberman's scissors system, which comes from the architect Hoberman himself."

Part III: Related Case Studies

Disney New Dynamic Float (2021)

"As an important design reference, Disney's dynamic floats provide an excellent design strategy: the combination of the main support in the center and the separate dynamic joints."

Dynamic Analysis



Load Path



Root



Joint



End



Base
Plate

First
Layer

Second
Layer

Third
Layer

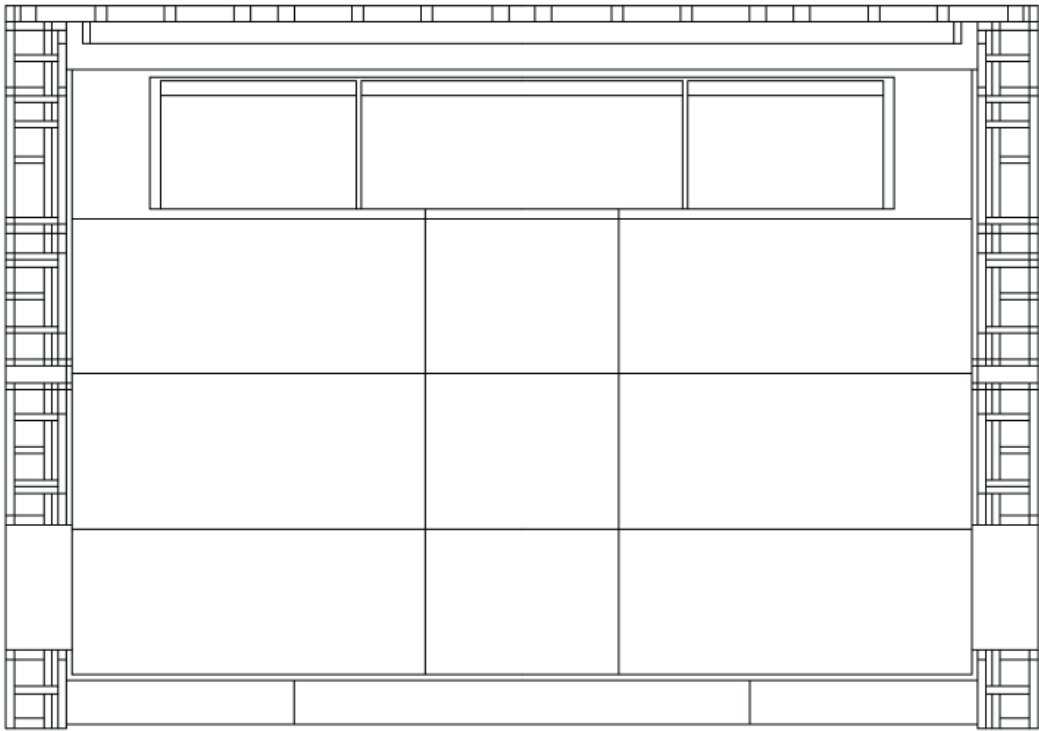
Fourth
Layer

Fifth
Layer

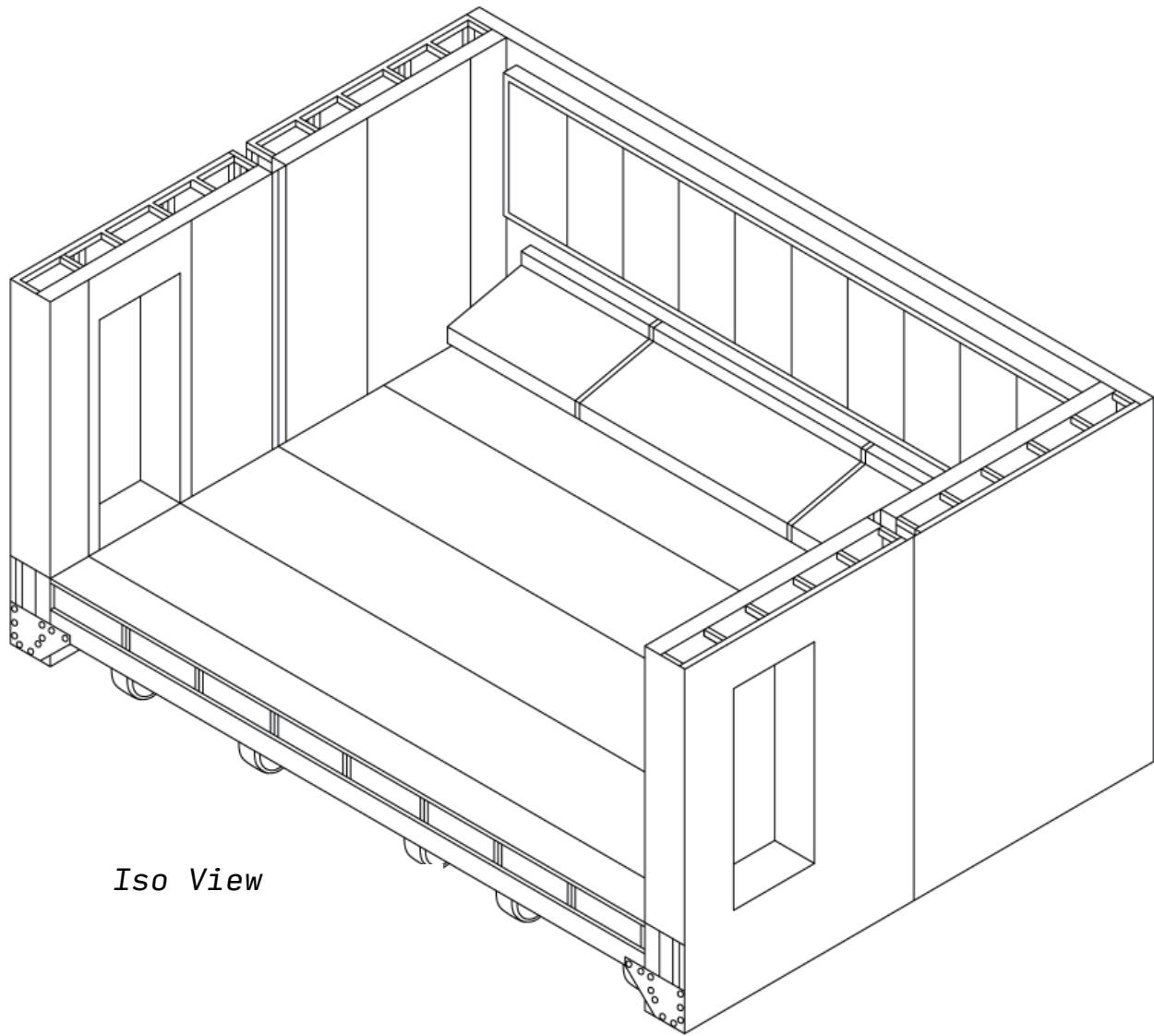
Part III: Related Case Studies

Movement Strategy – Super Bowl Stages (Es Devlin)

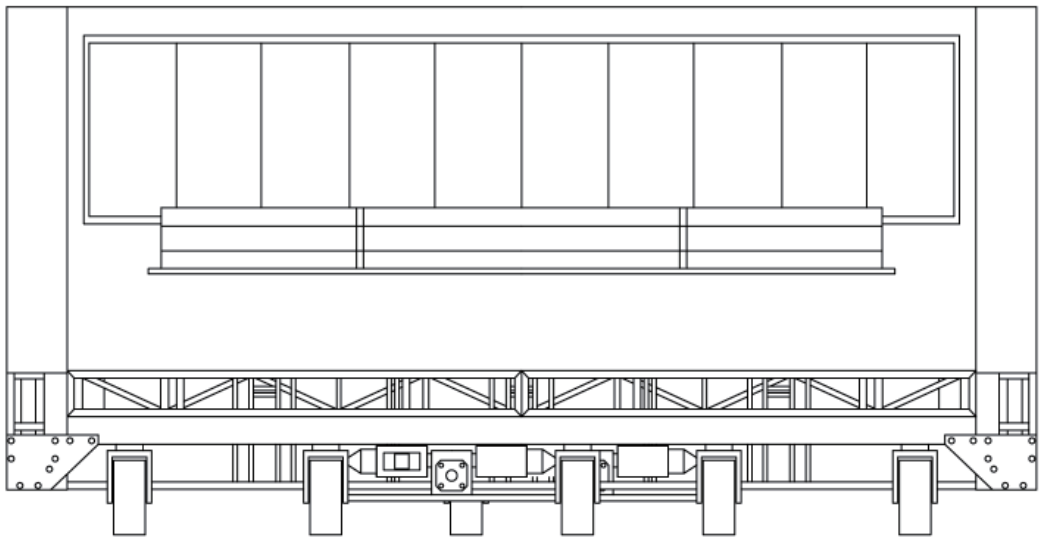
"Super Bowl mobile gig stage design provides a reasonable strategy for my giant float's transportation."



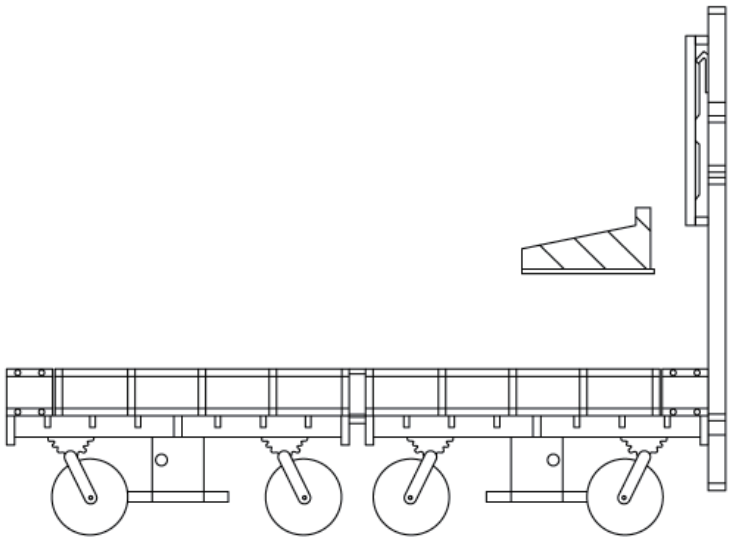
Top View



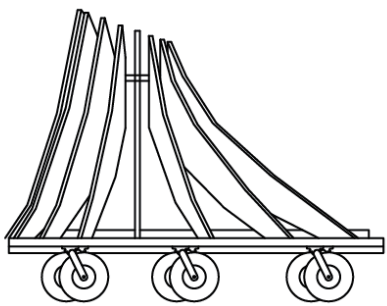
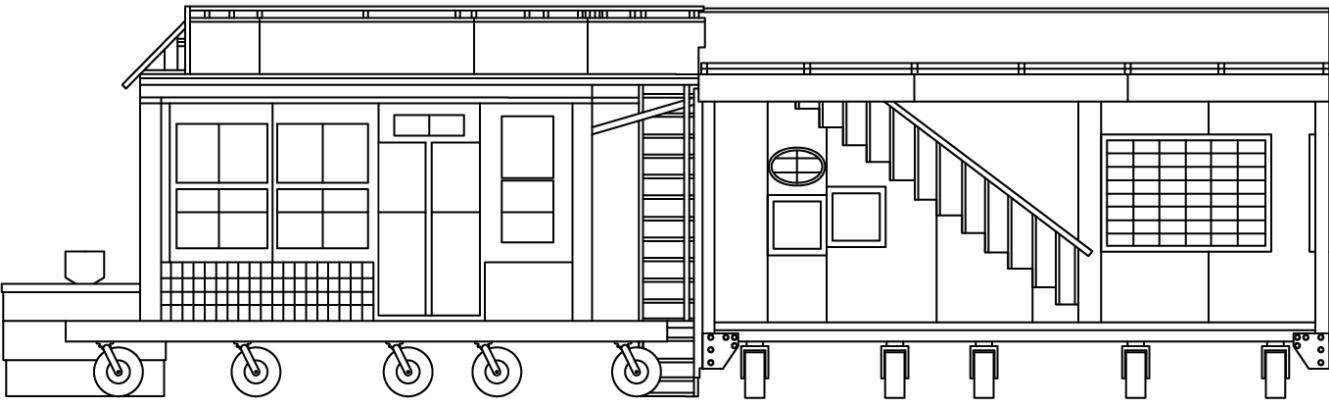
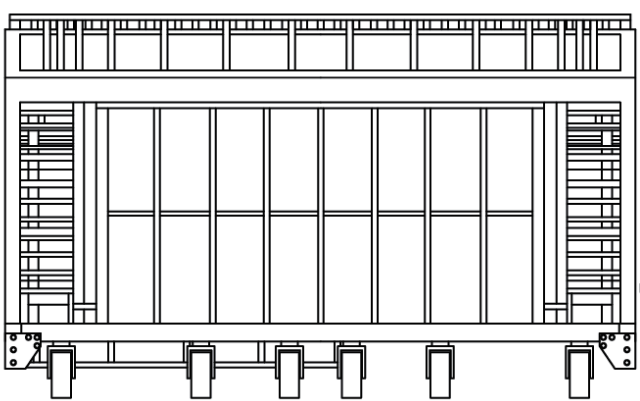
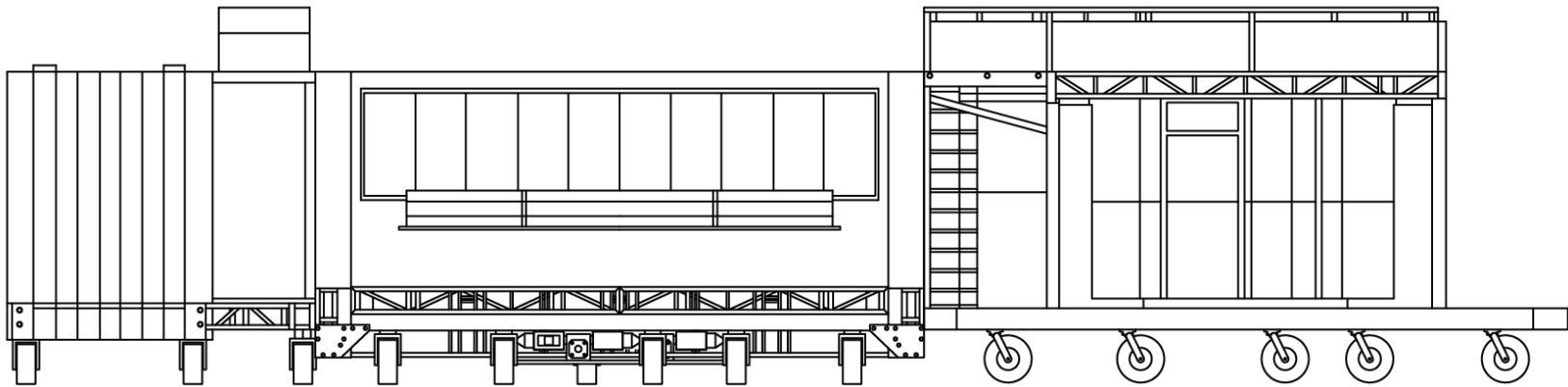
Iso View



Front View



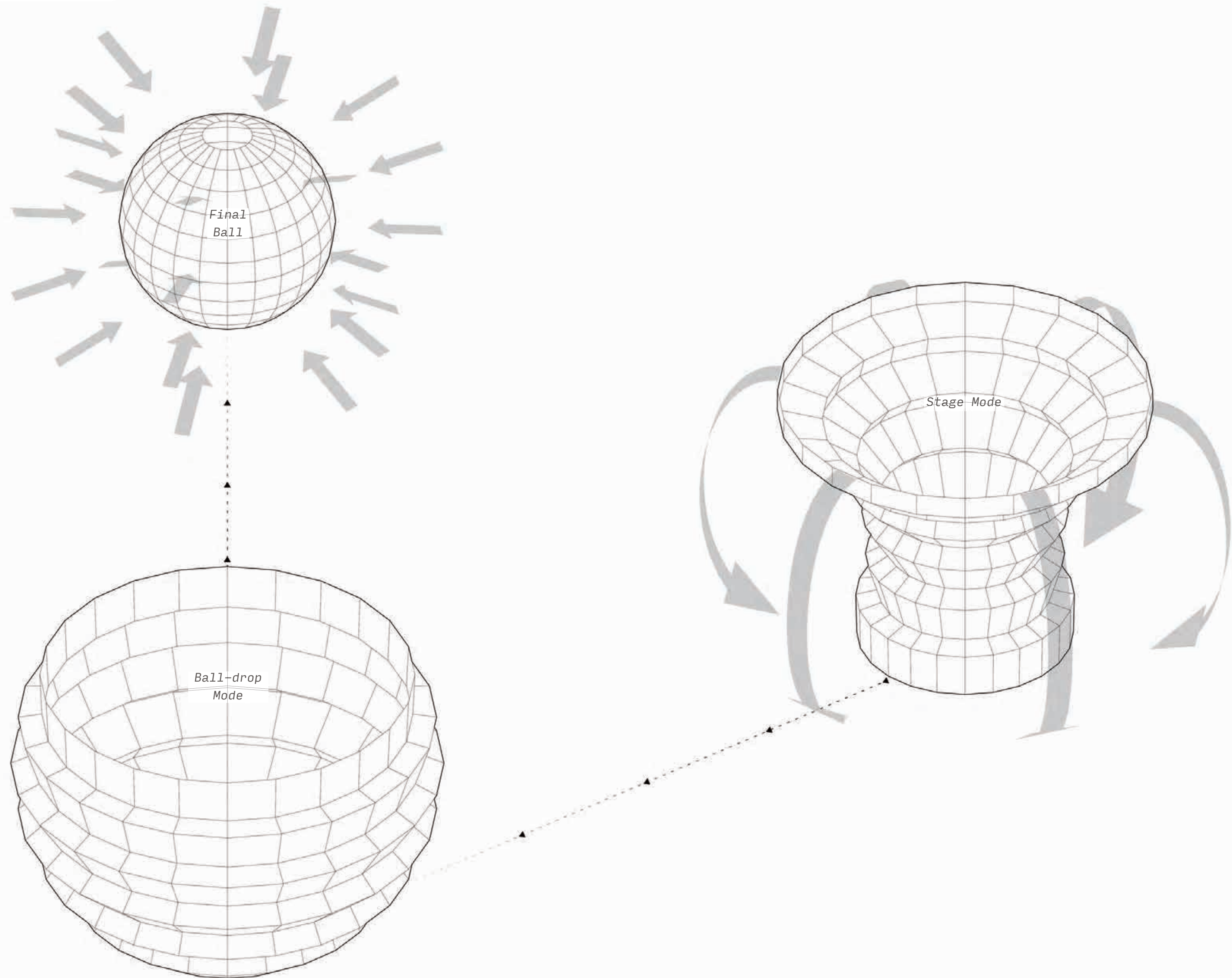
Right View



Long Elevation

Part IV: Design of Dynamic

By a series of model explorations, the spatiality of the design is expanded. This chapter determines the research interval of Technical Study in the whole unit project, and also determines three structural states and two dynamic conversion structural movement strategies.

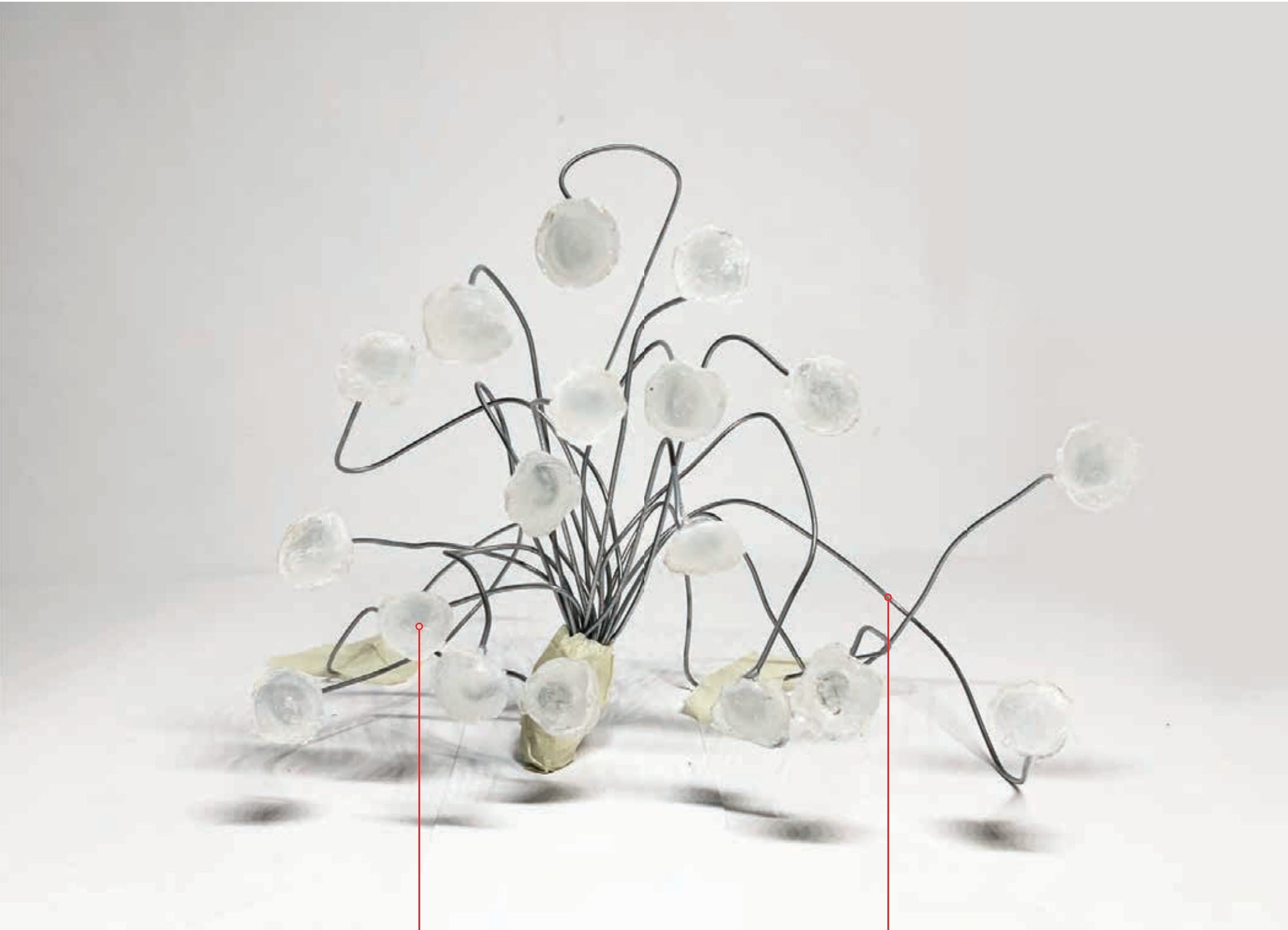


"The TS design is focusing on the dynamic progress when the float is transforming from the stage mode to the ball-drop mode, and then shrinking in to the final ball."

Part IV: Design of Dynamic

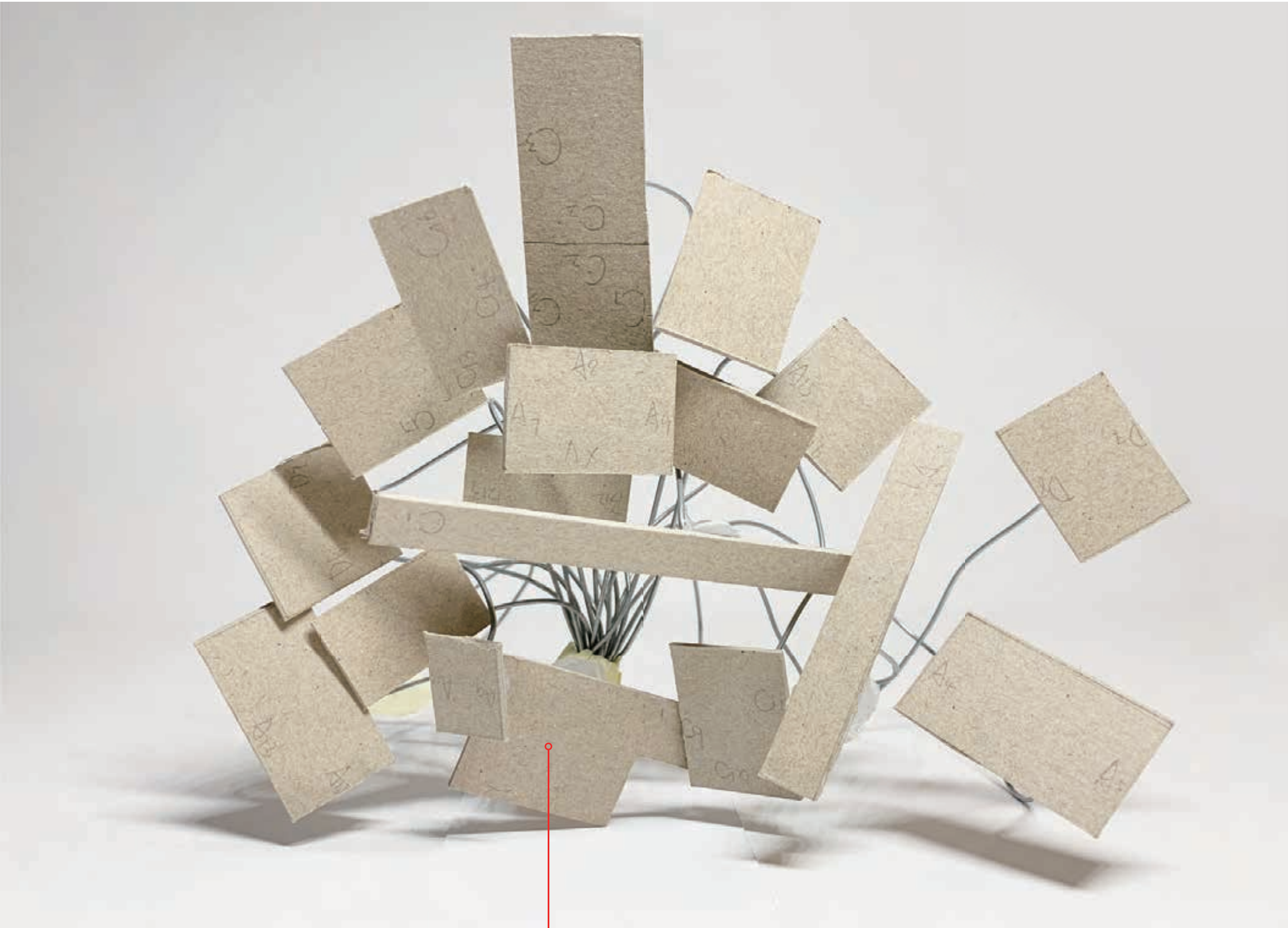
Model Exploration - Spatial Composition Model

"The first model exploration on the strategy of billboard spatial composition. Reusable adhesive joints enable the possibility of repeating experiments." 1:200 Model



Sticky Joint

Adjustable Frame

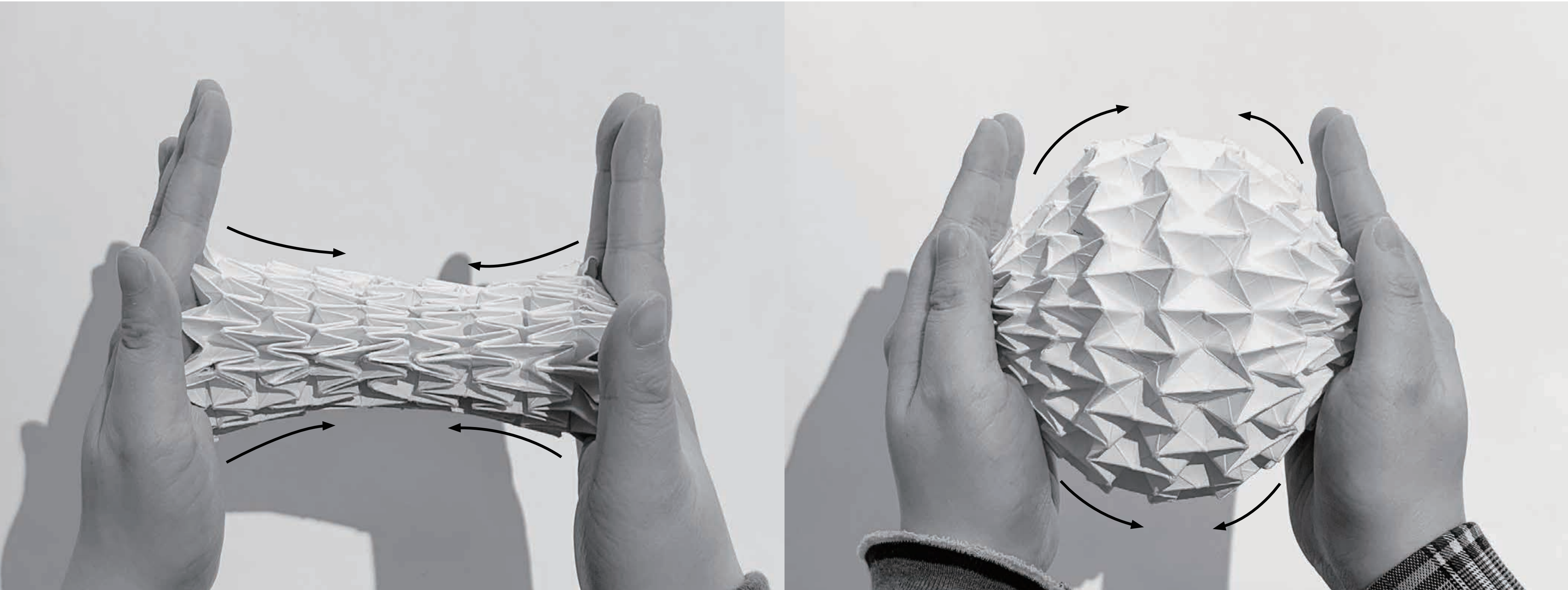


Replaceable Panel

Part IV: Design of Dynamic

Origami Explorations – Telescopic Sphere

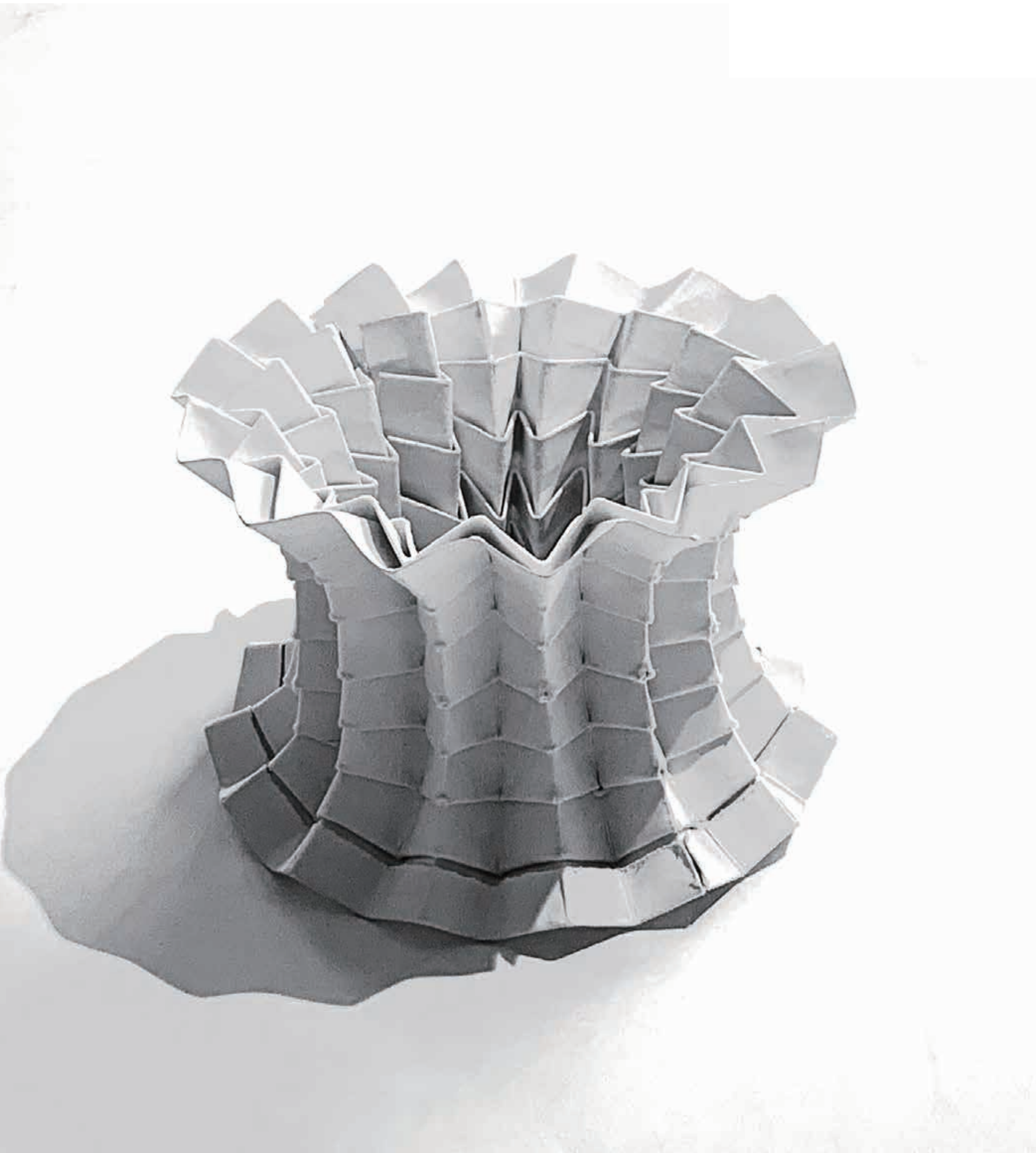
"Then I found that Origami can be very helpful for learning and understanding the structure. The first iteration had rich surface extensibility, but the structure was too complex." 1:200 Model



Part IV: Design of Dynamic

Origami Explorations – Revolving Sphere

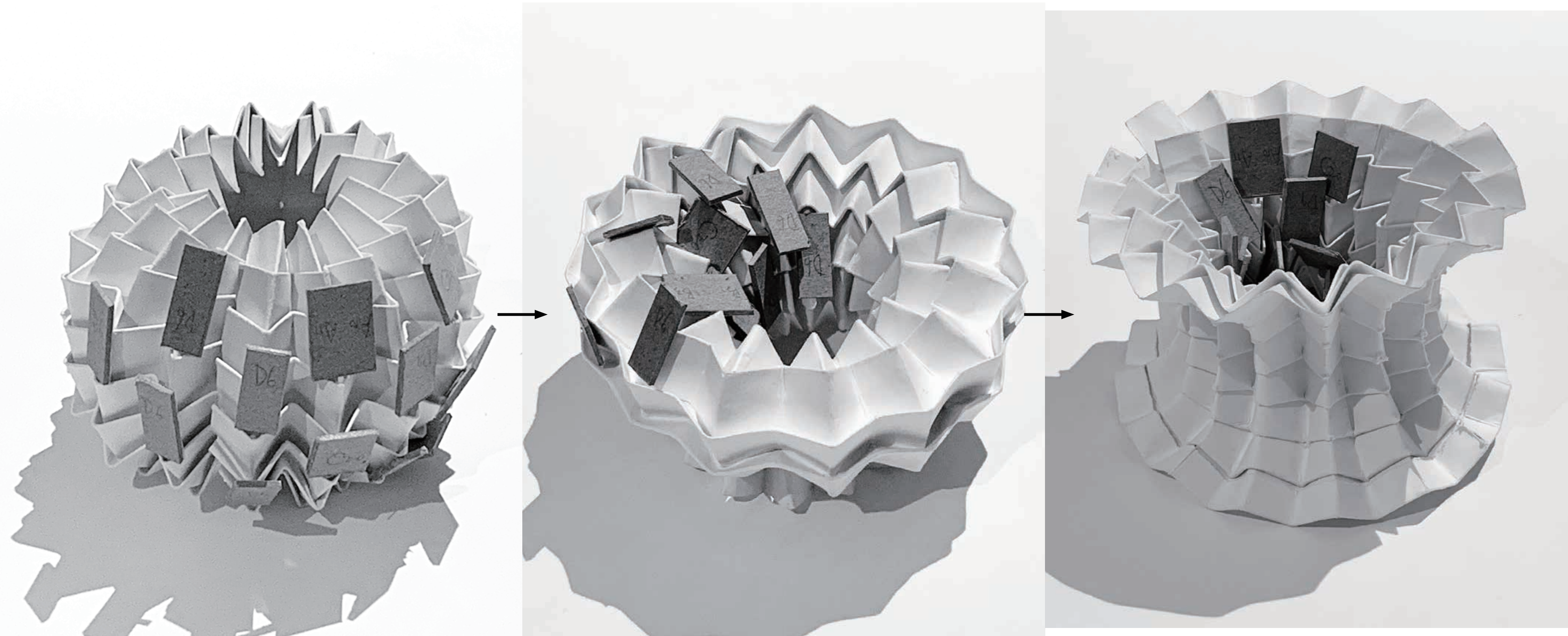
"The structure of the second iteration is relatively fixed, easier to generalize, and has infinite spatial revolving topology."



Part IV: Design of Dynamic

Origami Explorations – Revolving Sphere With Billboards

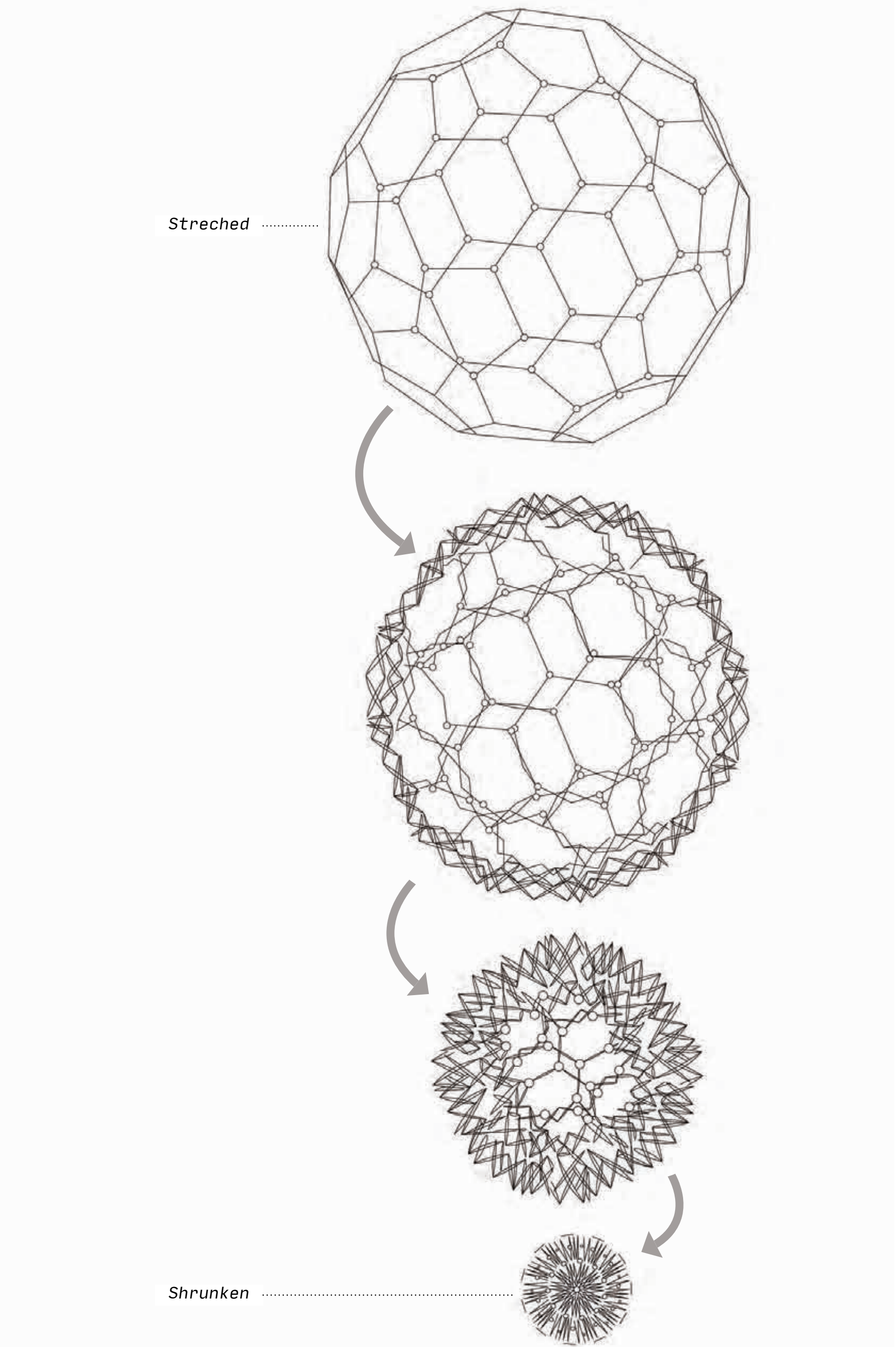
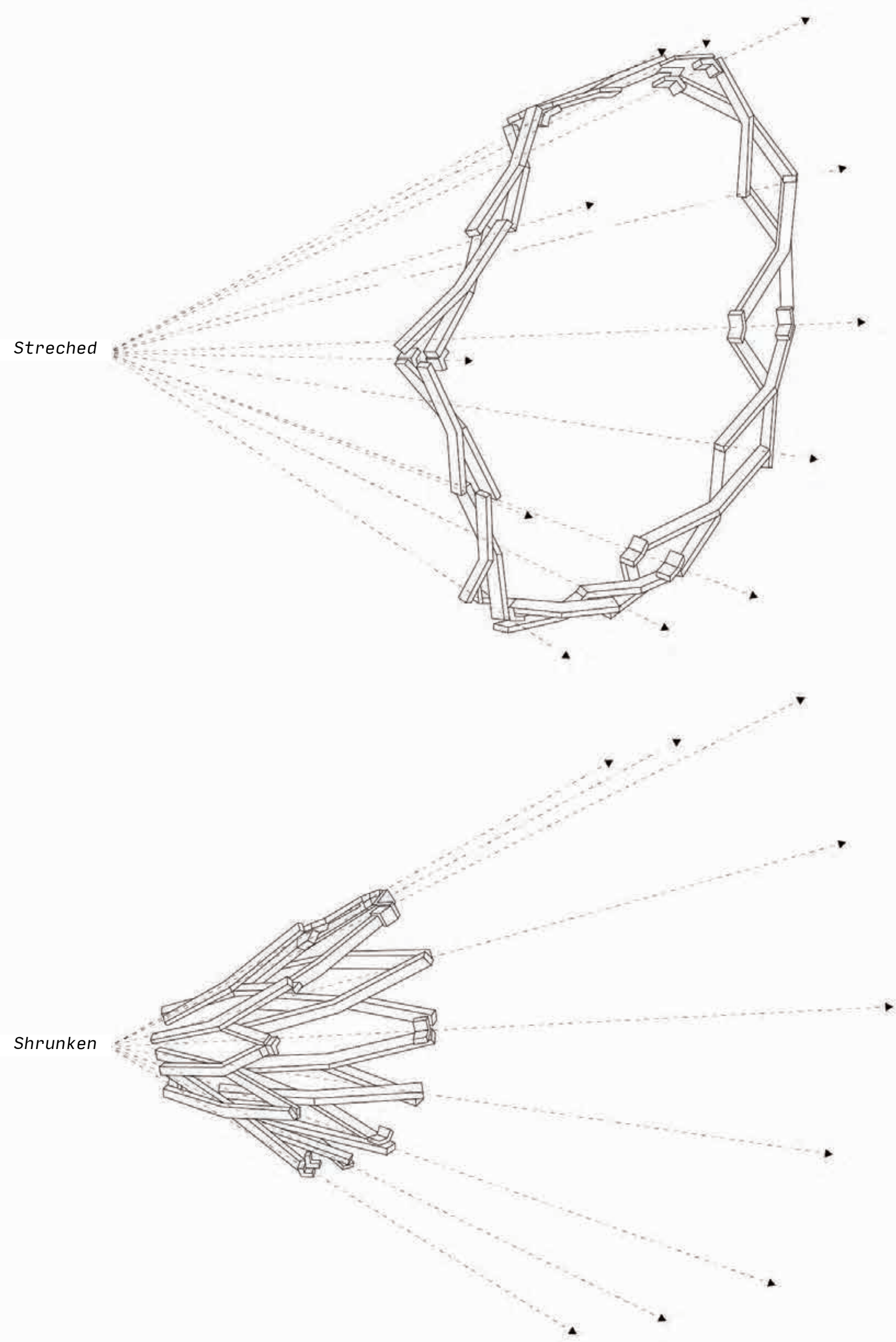
"This idea can support a dynamic structure with infinite flips." 1:200 Model



Part IV: Design of Dynamic

Movement Strategy - Hoberman sphere

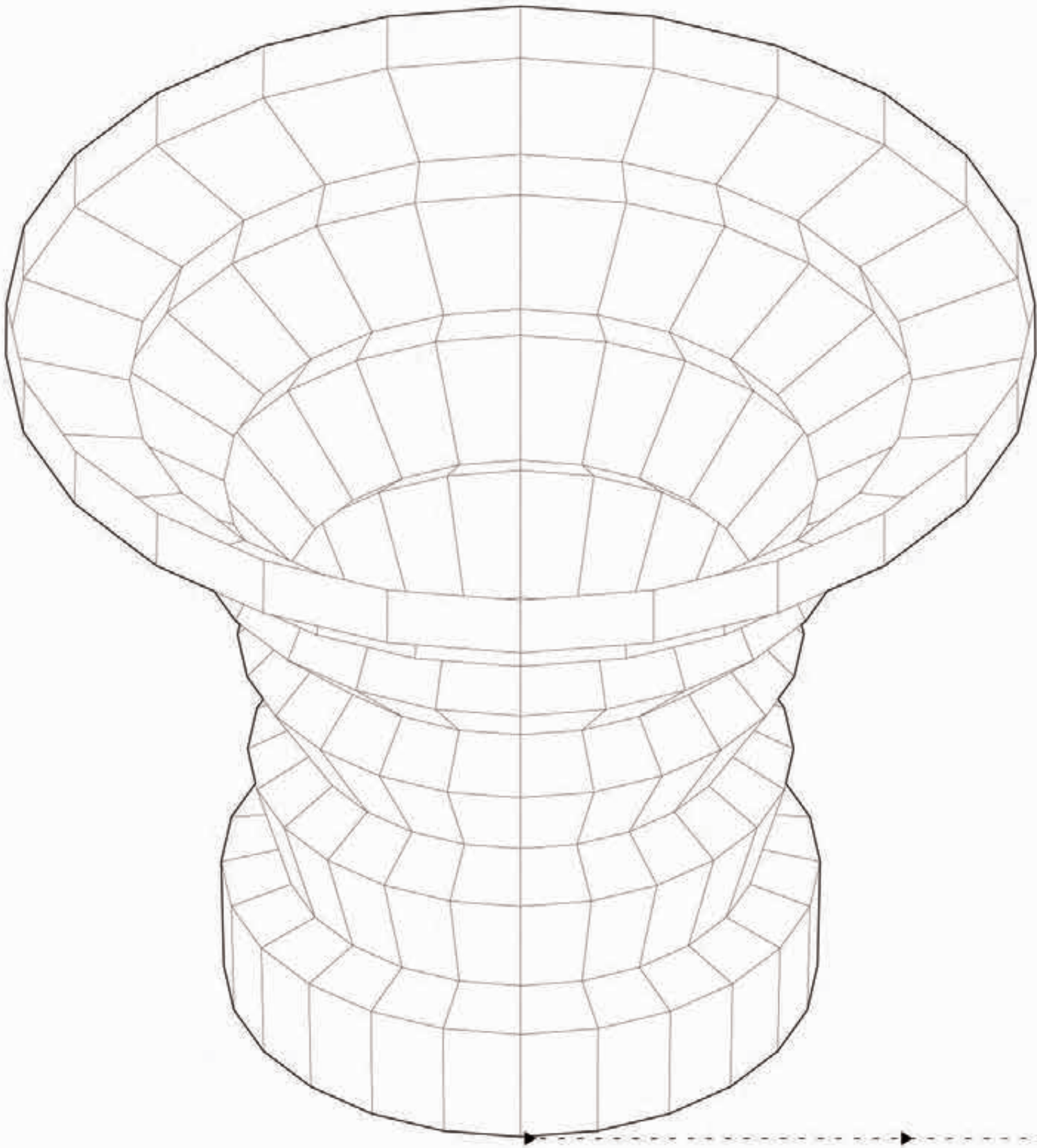
"By learning Hoberman Sphere from two different scales: Local and global. I had a better understanding of how this complex motion system functions."



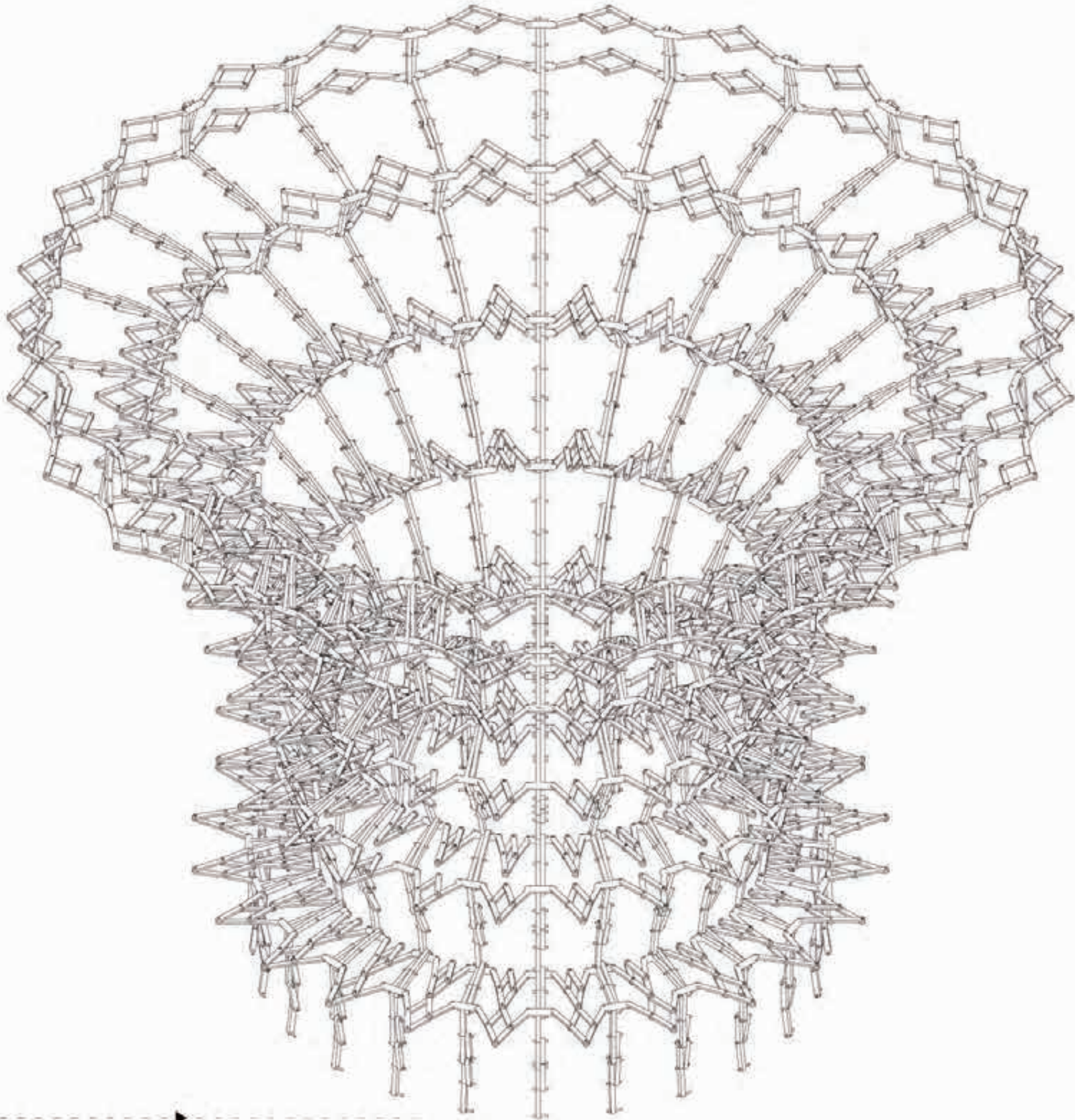
Part IV: Design of Dynamic

Compositional Strategy - Stage Mode

"To combine the revolving origami with Hoberman sphere, I figured out a brand new dynamic structure that supports spatially, not only to revolve, but also to shrink."



Simplified Structure Illustration

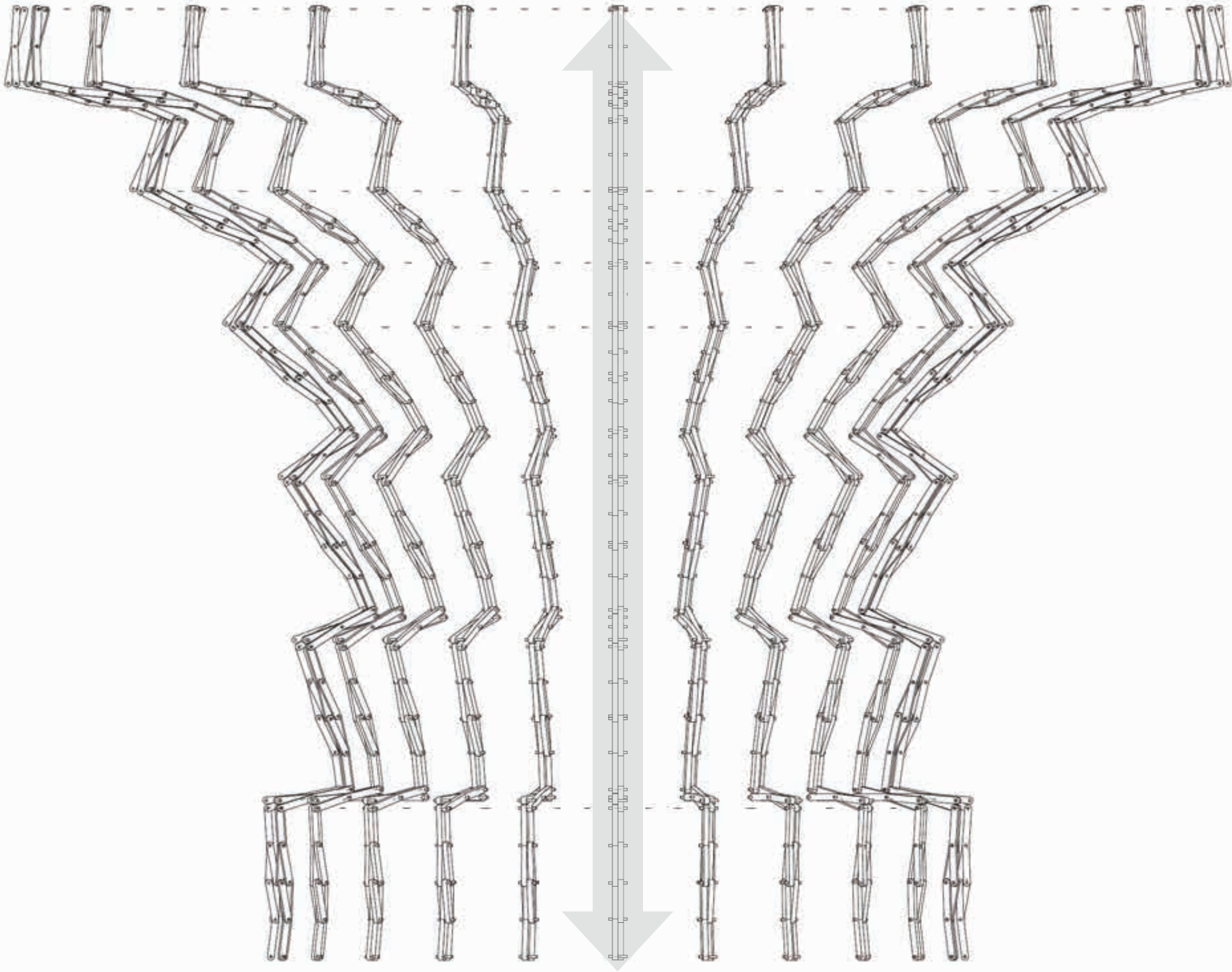


Actual Structure

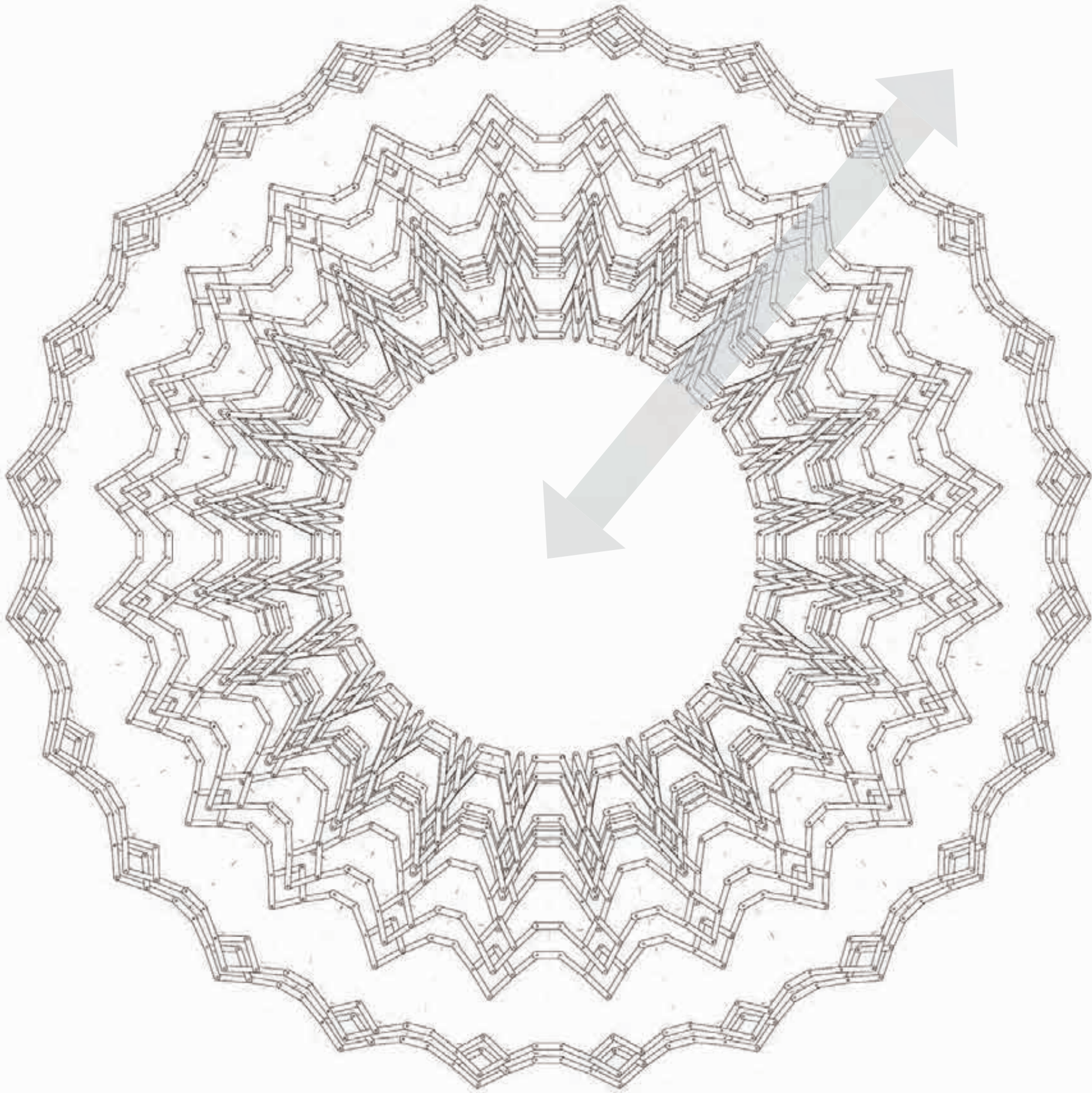
Part IV: Design of Dynamic

Compositional Strategy - Stage Mode

"The figure below decomposes the entire structure into two parts, the longitudinal structure and the transverse structure, which are respectively responsible for the movement of the overall structure in their respective directions."



Longitudinal Structure - Stage Mode

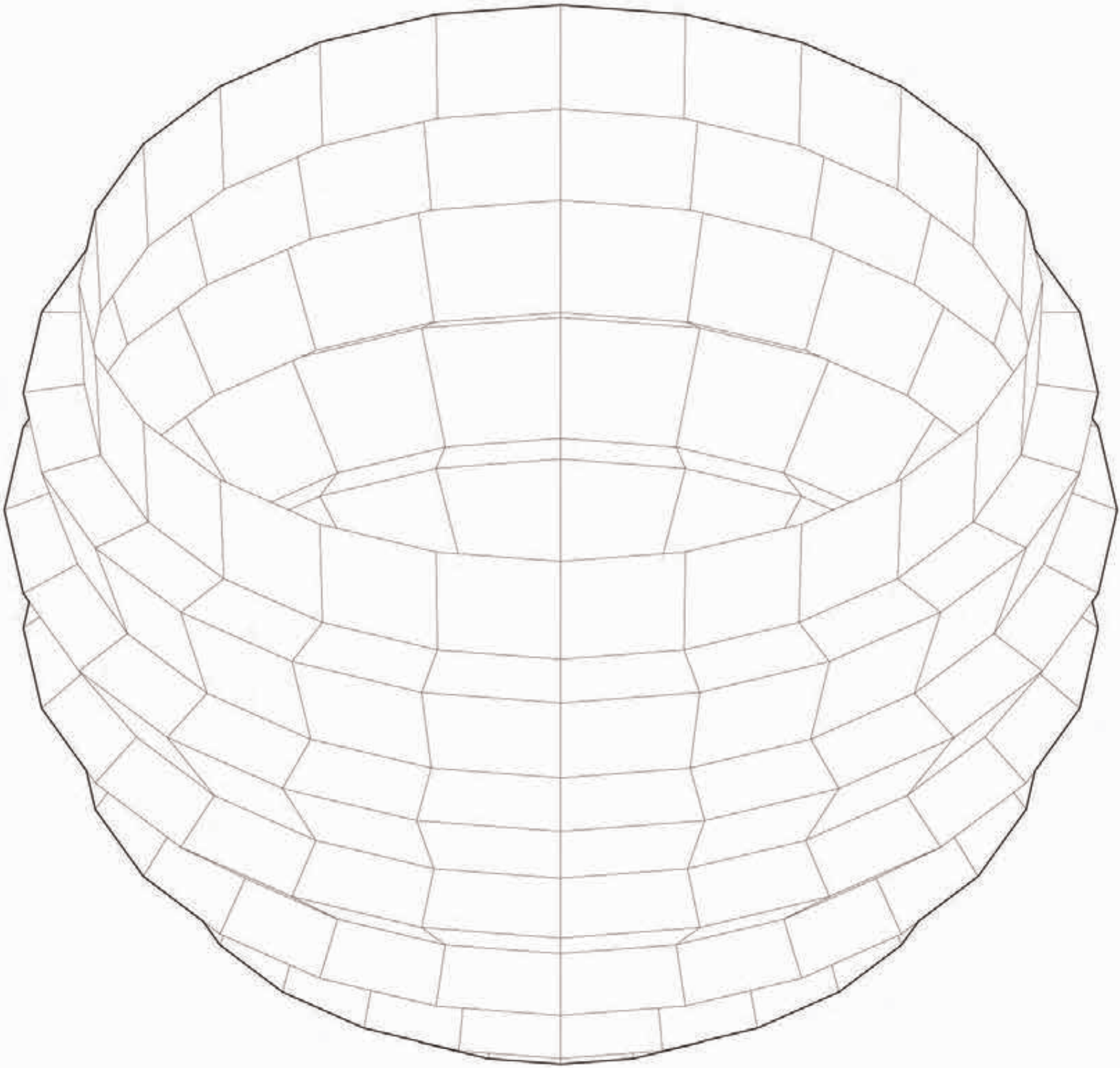


Transverse Structure - Stage Mode

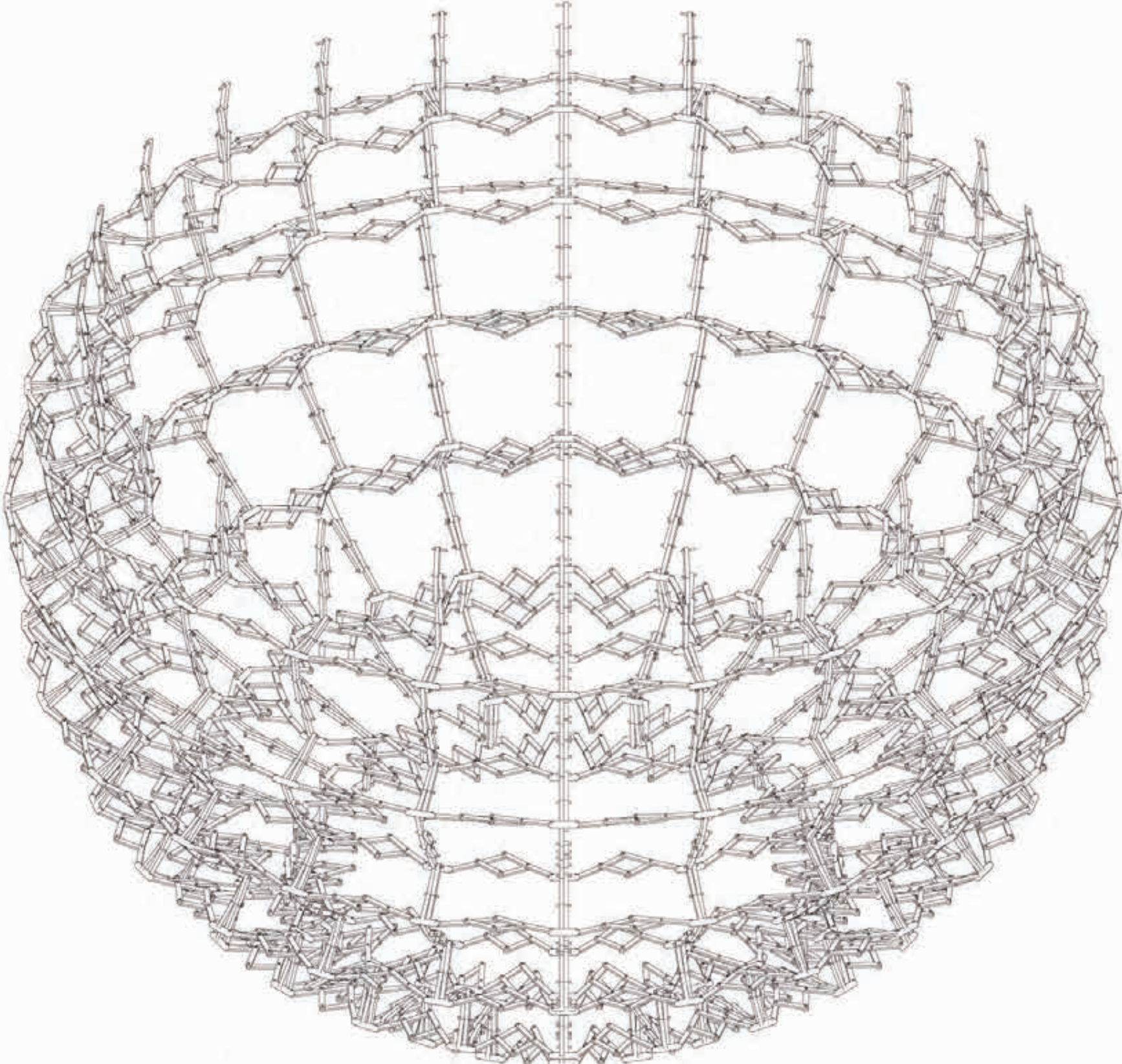
Part IV: Design of Dynamic

Compositional Strategy – Ball-drop Mode

"At this stage, the structure of Stage Mode is gradually restructured into its reverse topology due to the flipping of widgets. See the long diagram at the end of this book for the detailed flipping process."



Simplified Structure Illustration

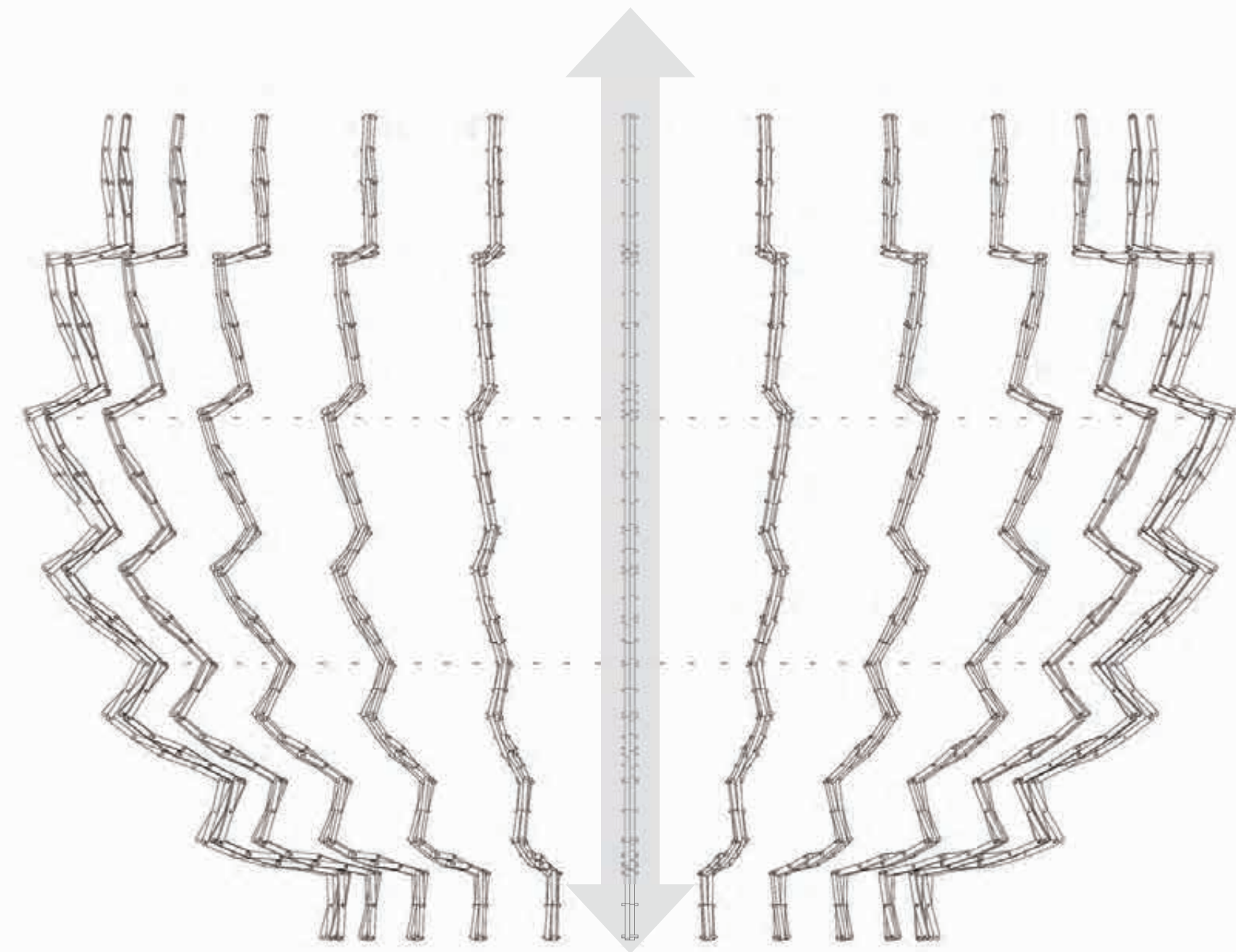


Actural Structure

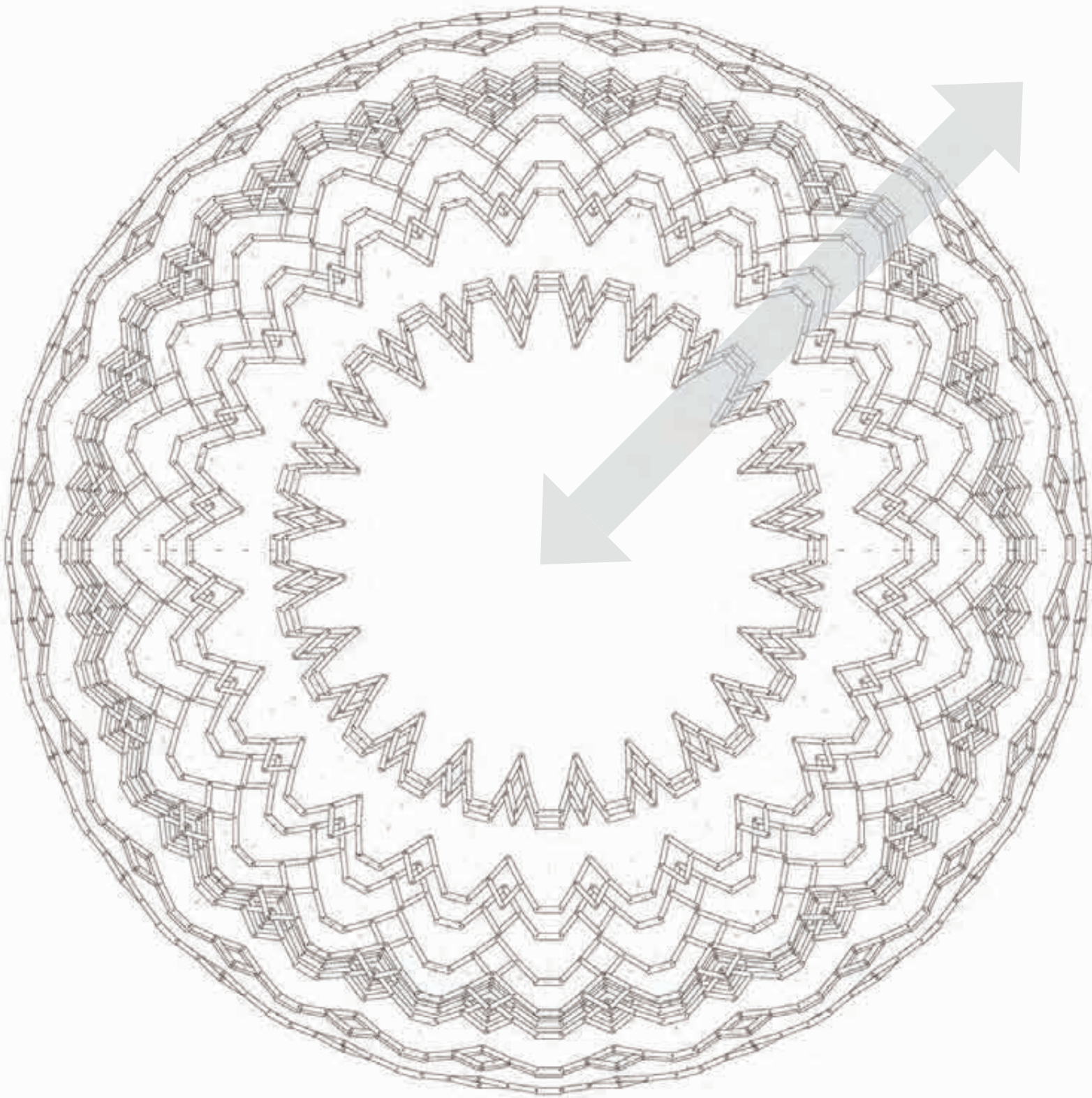
Part IV: Design of Dynamic

Compositional Strategy – Ball-drop Mode

"The figure below decomposes the entire structure into two parts, the longitudinal structure and the transverse structure, which are respectively responsible for the movement of the overall structure in their respective directions."



Longitudinal Structure – Ball-drop Mode



Transverse Structure – Ball-drop Mode

Part IV: Design of Dynamic

Dynamic Sequence - Final Ball

"At this stage, the overall structure will slowly shrink from the bowl-shaped structure of the Ball-drop Mode to the final spherical shape."



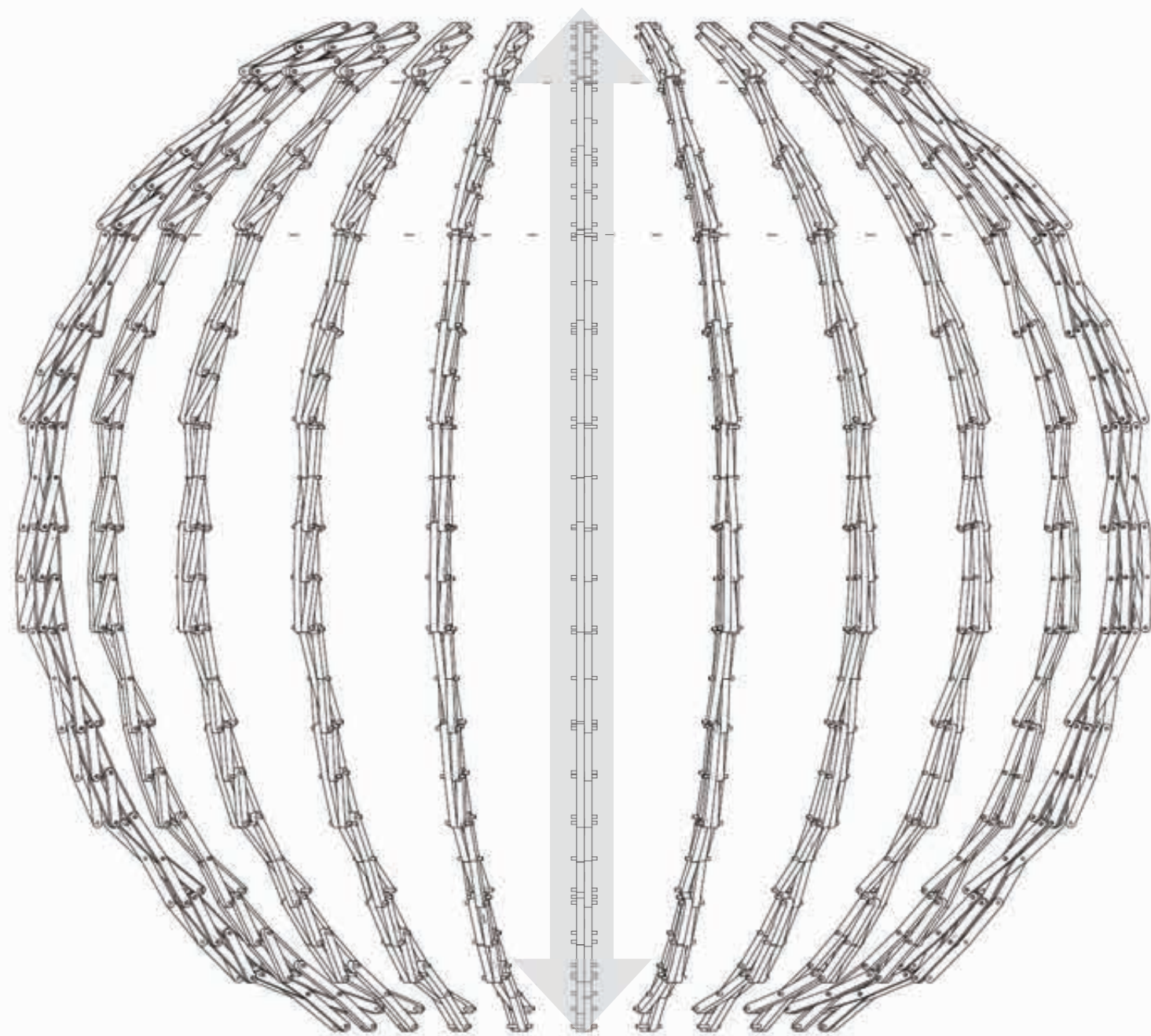
Simplified Structure Illustration

Actural Structure

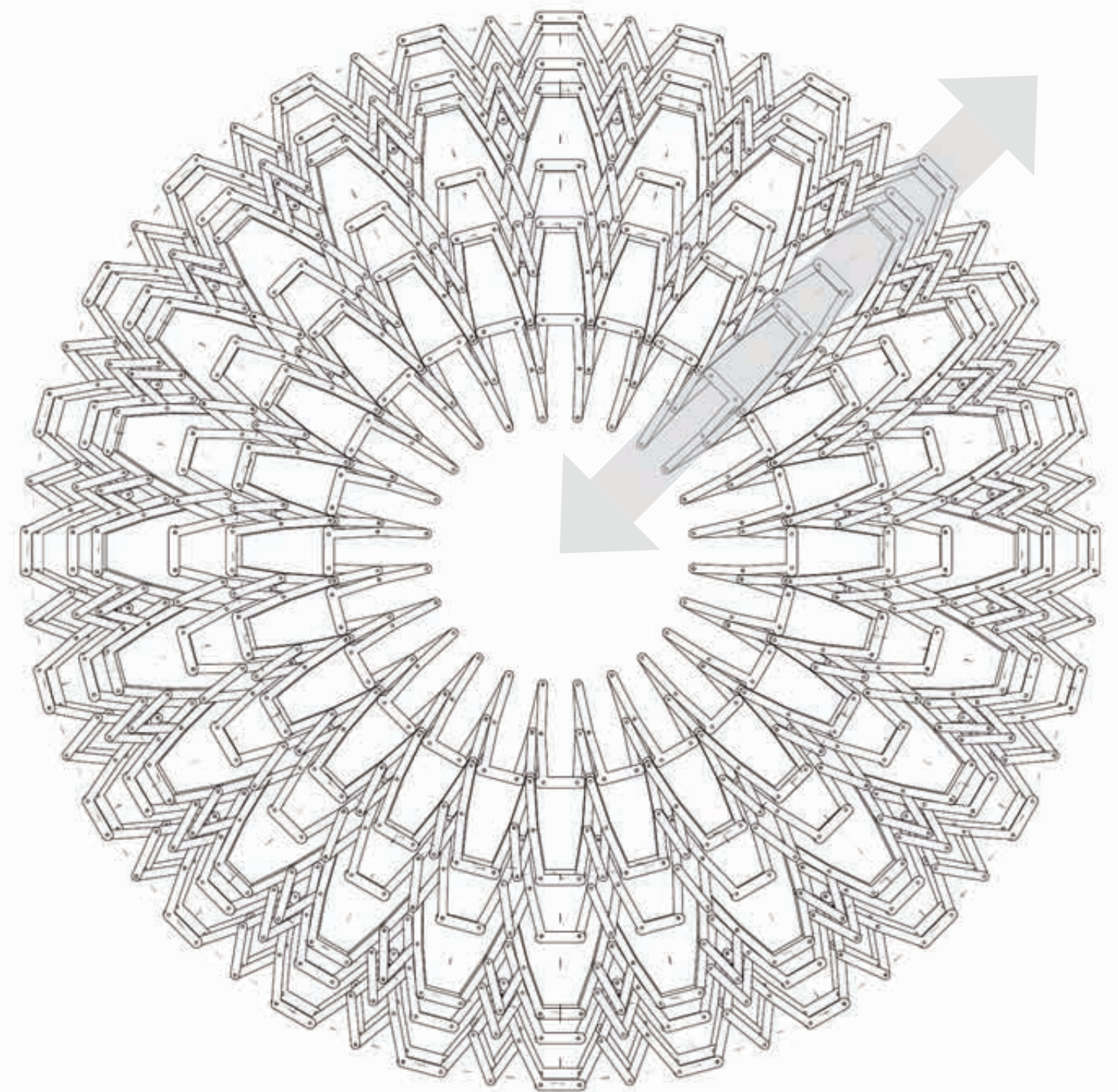
Part IV: Design of Dynamic

Compositional Strategy - Final Ball

"The figure below decomposes the entire structure into two parts, the longitudinal structure and the transverse structure, which are respectively responsible for the movement of the overall structure in their respective directions."



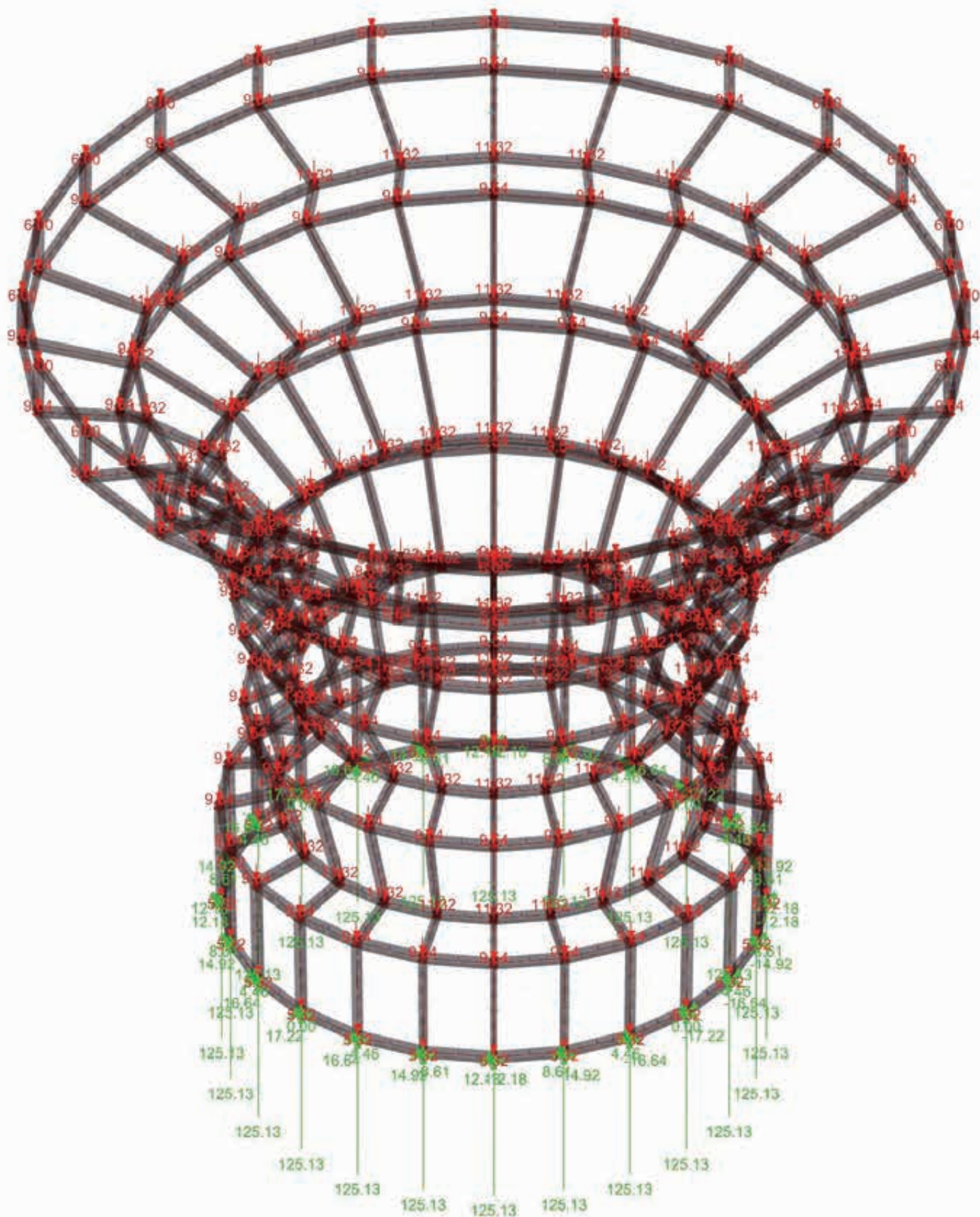
Longitudinal Structure - Final Ball



Transverse Structure - Final Ball

Part V: Main Structure

As the load-bearing core of the entire dynamic building, the main structure system requires a well-thought-out load baring strategy. This chapter attempts to solve design problems in scales from global to local.



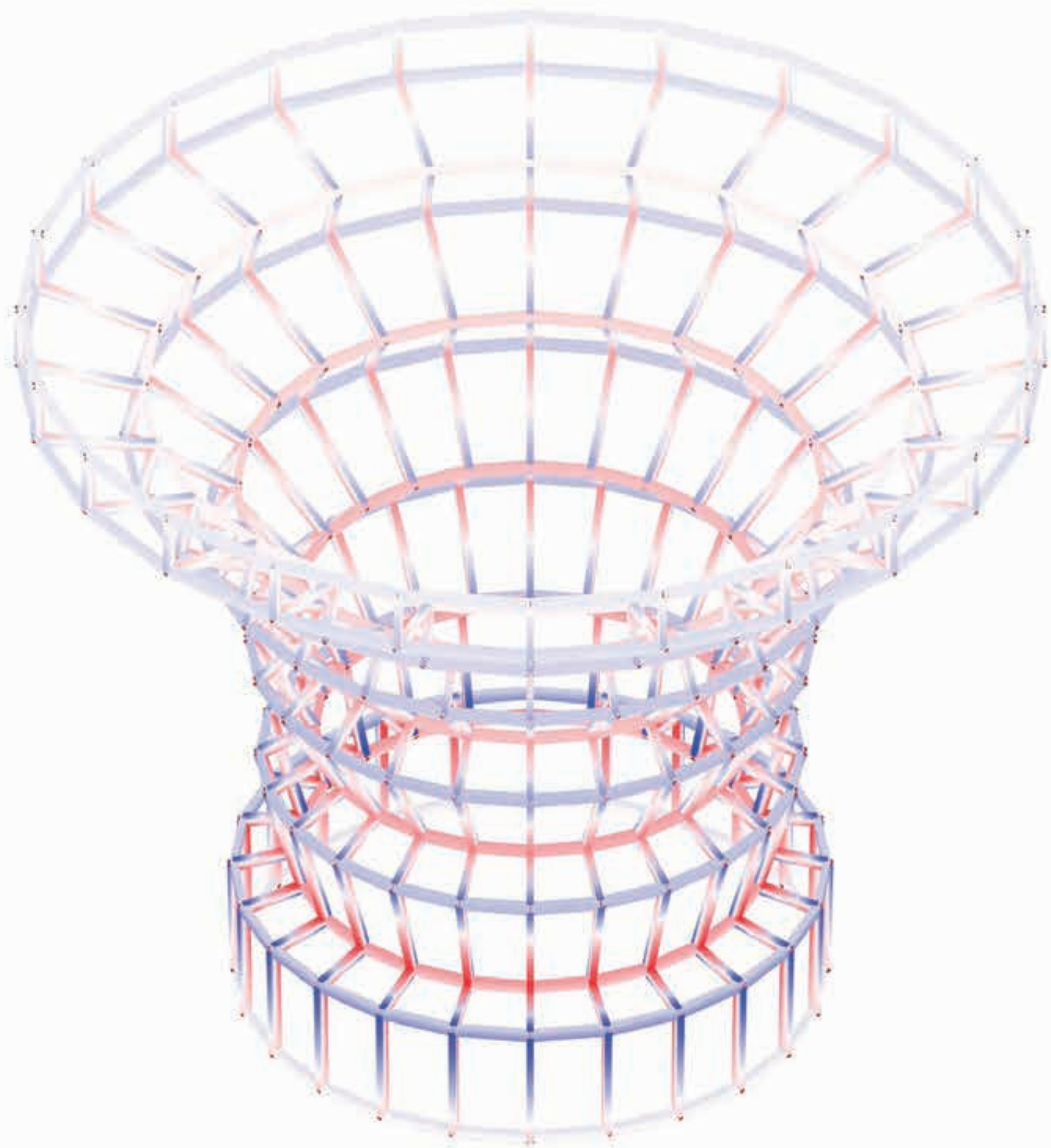
Force-KN

"Through the force analysis, it can be found intuitively that without the overall support of the structure, the load of each component increases layer by layer from top to bottom."



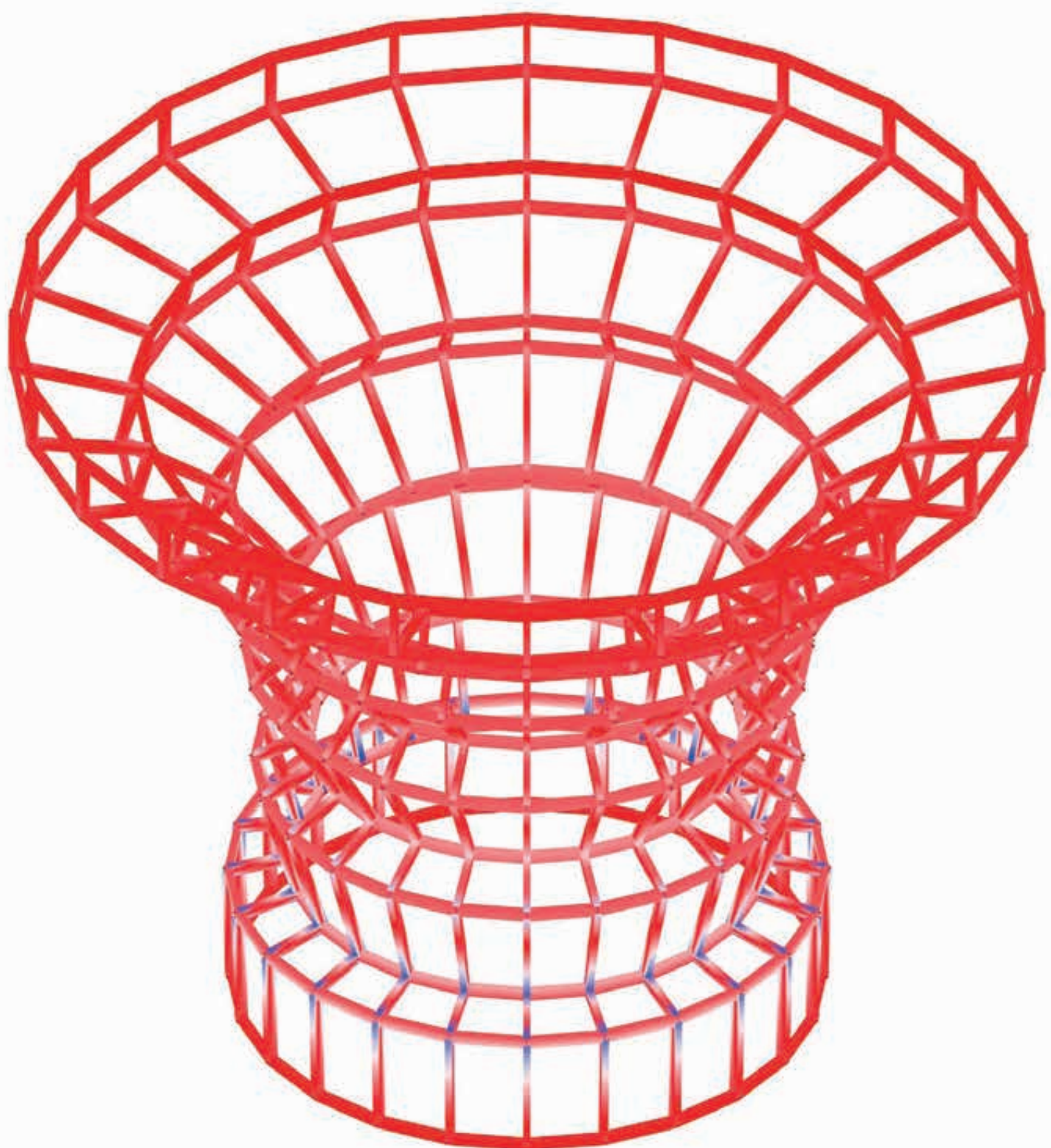
Displacement

"The most serious deformation of the overall structure will be at the upper part. This is due to the gradual accumulation of small deformations in the lower layers, which will cause a larger displacement of the upper part than the original position."



Axial Stress

"According to the axial stress analysis, the longitudinal structure bears a large shear force. In contrast, the transverse structure is relatively stable in force and does not require additional support."



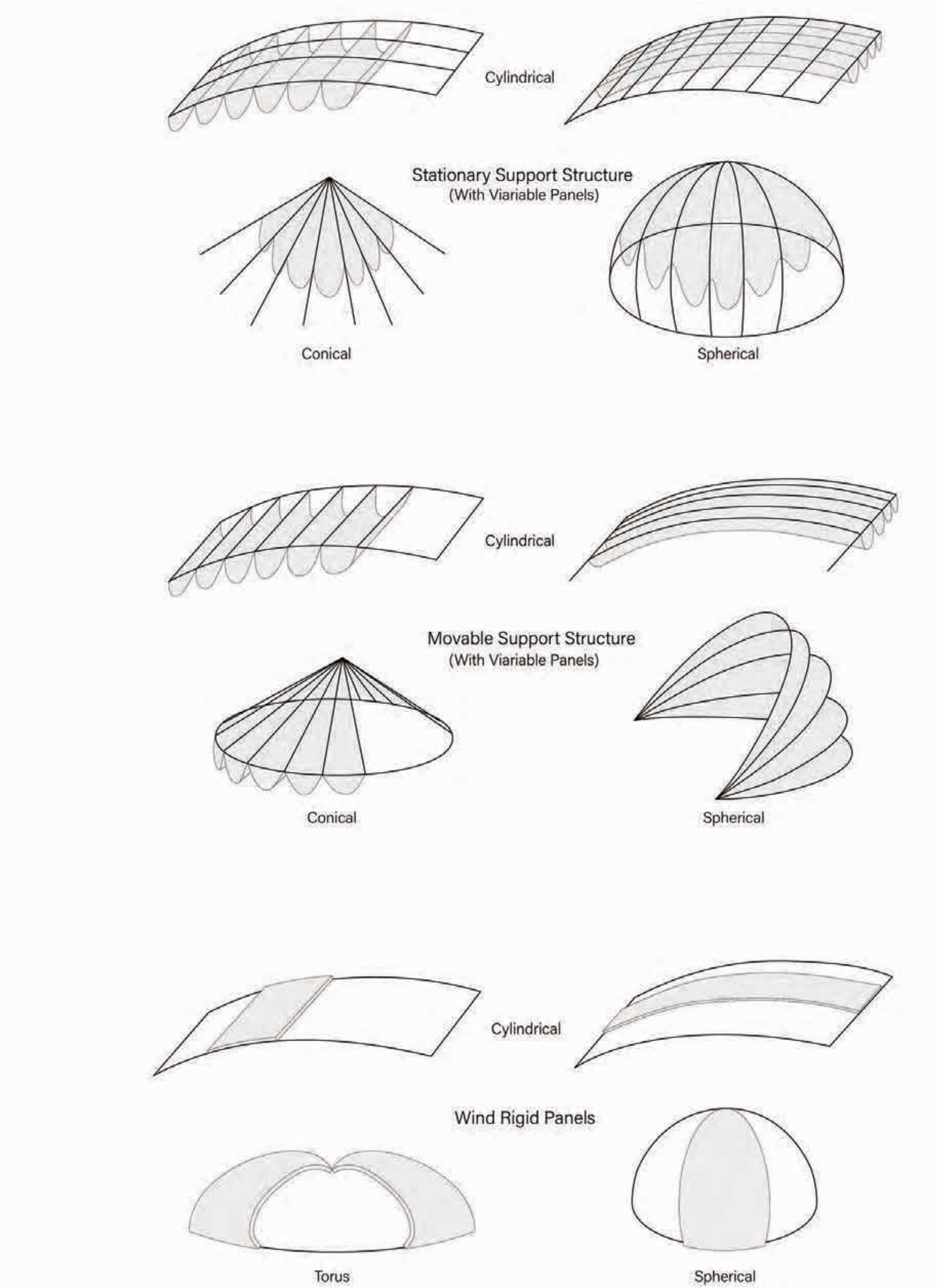
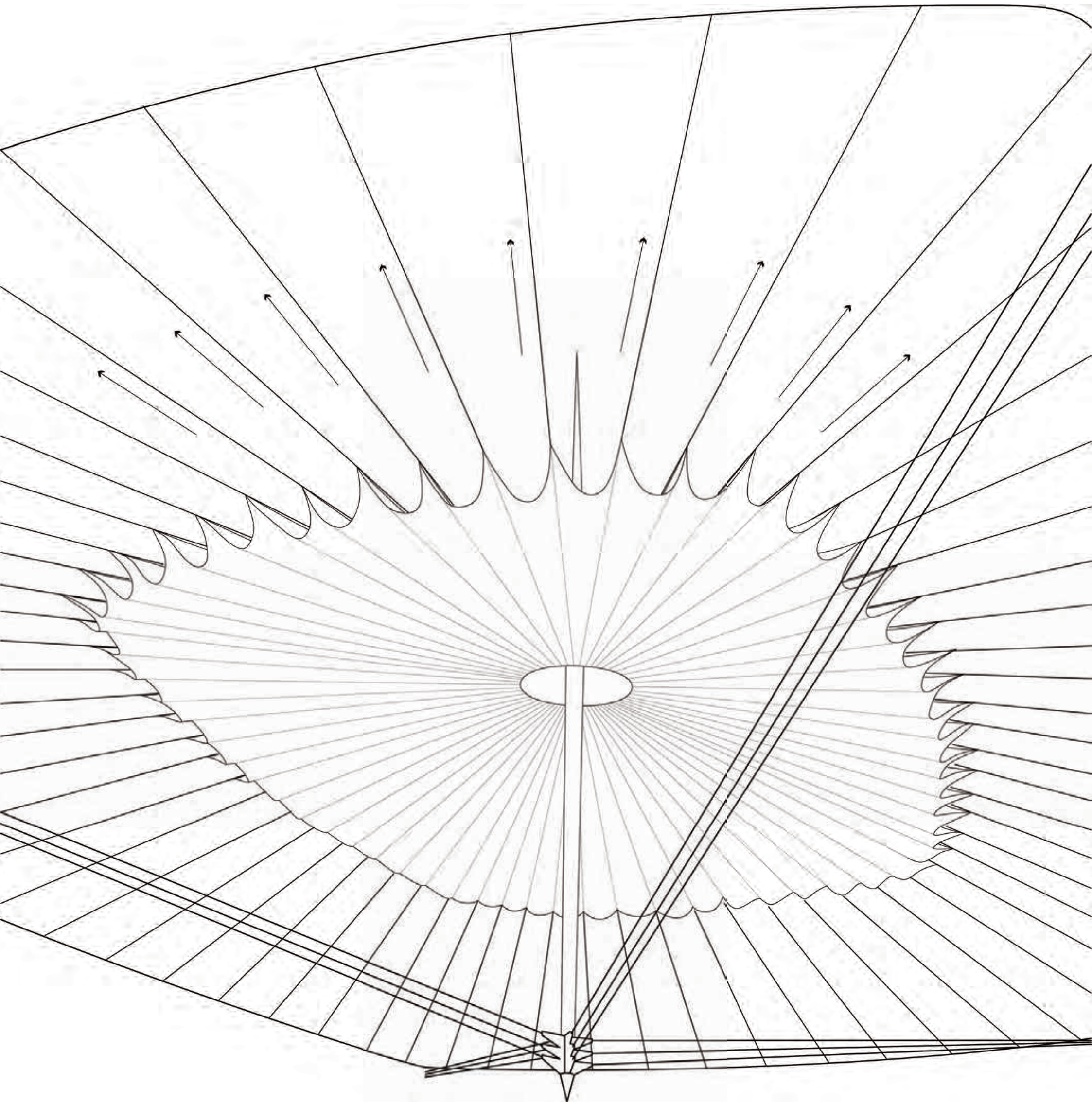
Utilization

"In the utilization diagram, we can see that some positions of the lower structure with the heaviest load-bearing tasks are still not fully utilized, and the load-bearing capacity of the overall structure is also in a saturated state."

Part V: Main Structure

Movement Strategy - Retractable Roof Typology (Frei Otto)

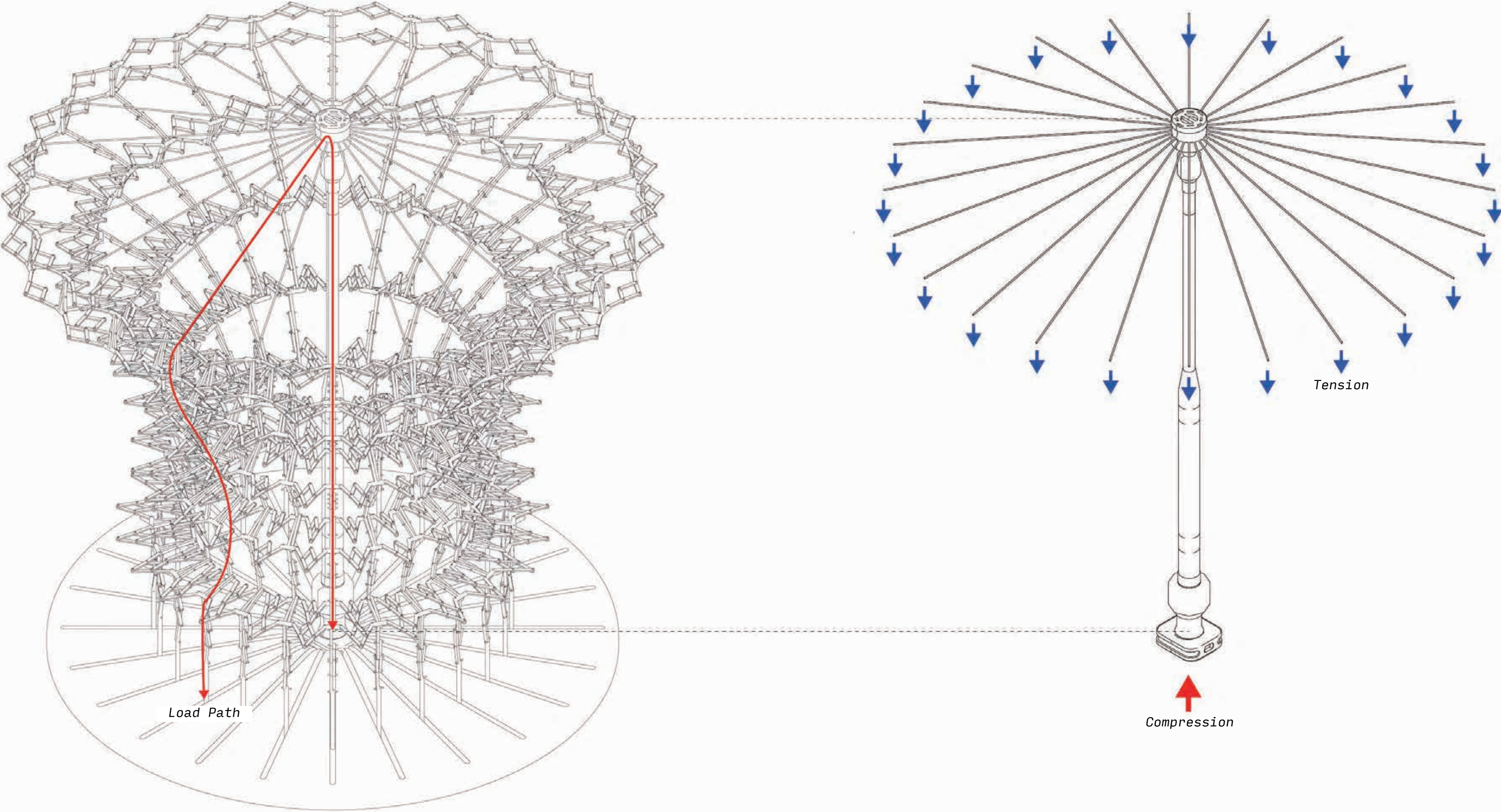
"In order to design a support scheme that can adapt to the dynamic structure, I studied a case of Frei Otto and related retractable roof typology."



Part V: Main Structure

Global Support – Hydraulic Central Support & Stretching System

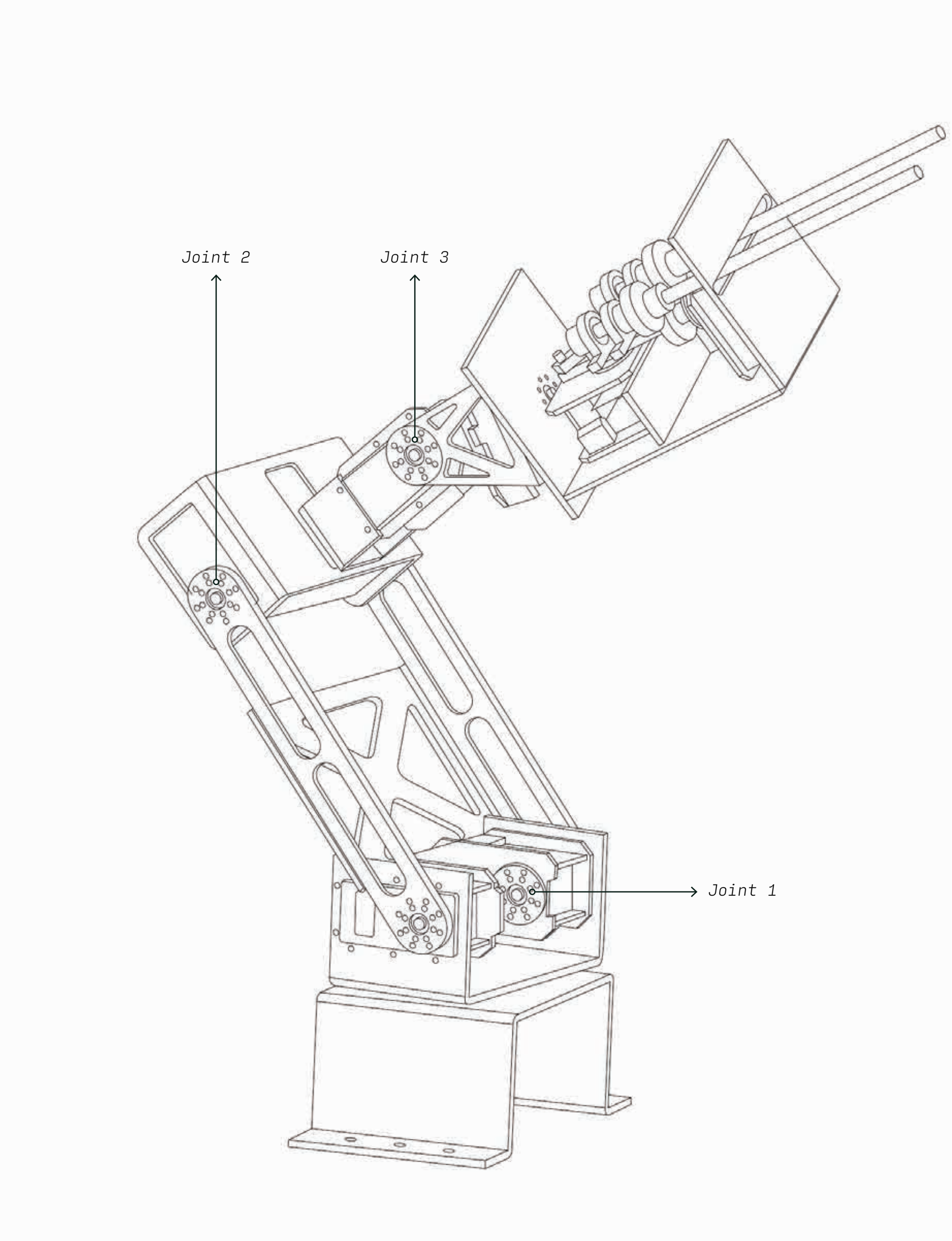
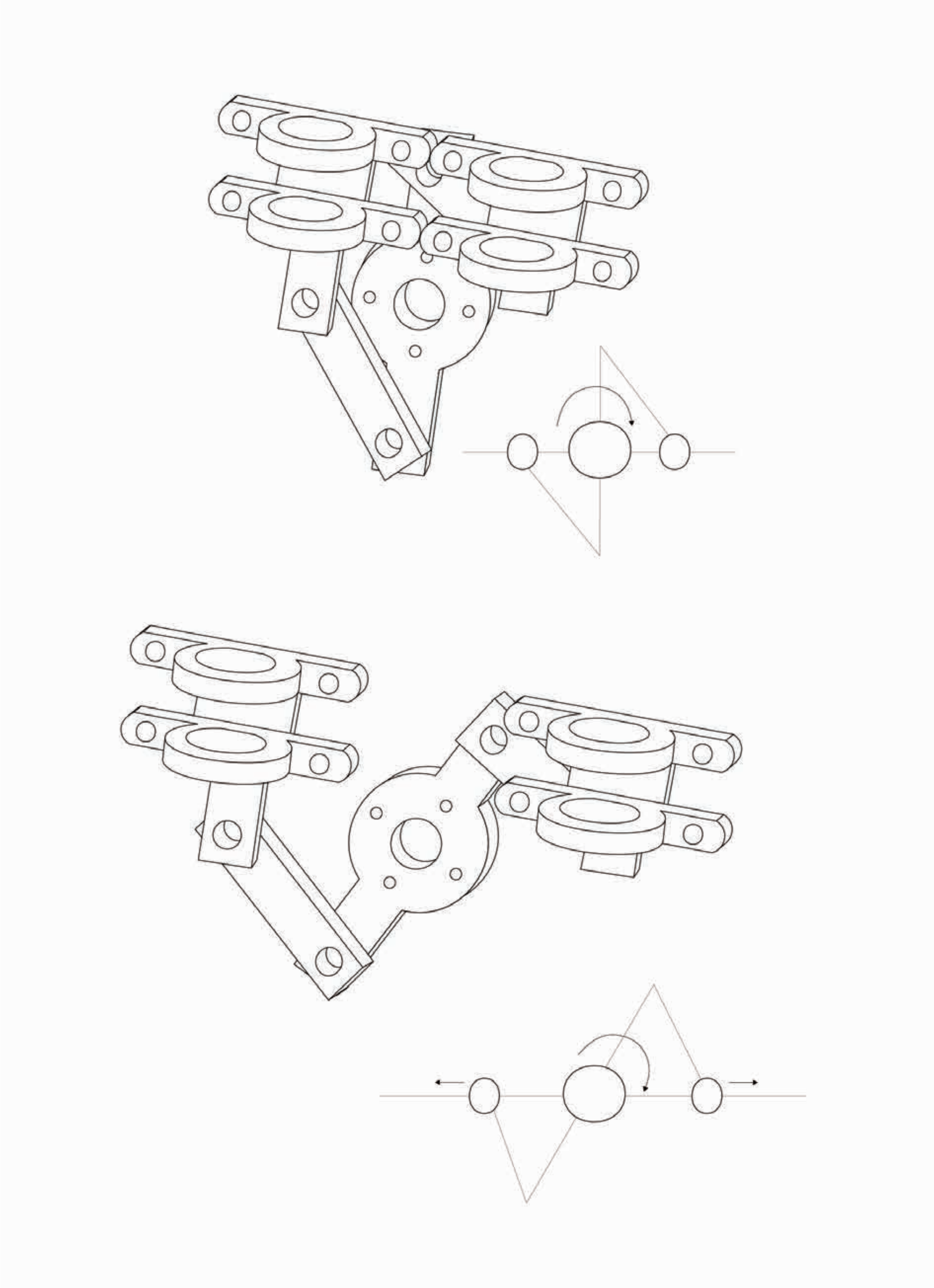
"As the main support of the float, the structure will pull and share the load of each longitudinal structure from the top. It is a telescopic rod structure that can adapt to different heights of the float."



Part V: Main Structure

Movement Strategy – Expansion Joint & Robotic Arm

"In order to increase the spatial diversity of the local units of the float, I studied the principles of expansion joint and robotic arm."



Part V: Main Structure

Local Support – Spring System & Ratchet Gear

"In order to refine the design of local unit rotation direction locking and structural stability, I designed a ratchet + spring system."

Single Rotatory Junction

30W4CR2VA Spring Steel

Surface Layer

Axis

Horizontal Support

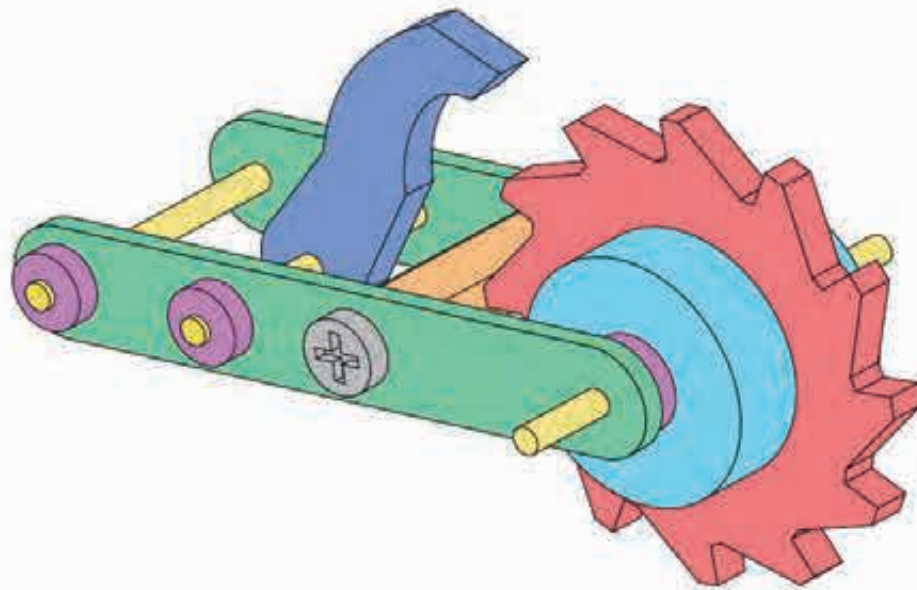
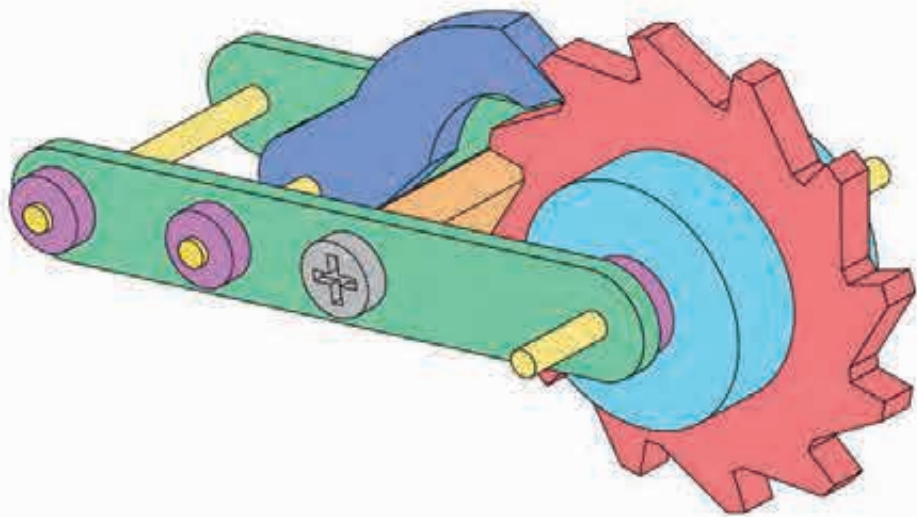
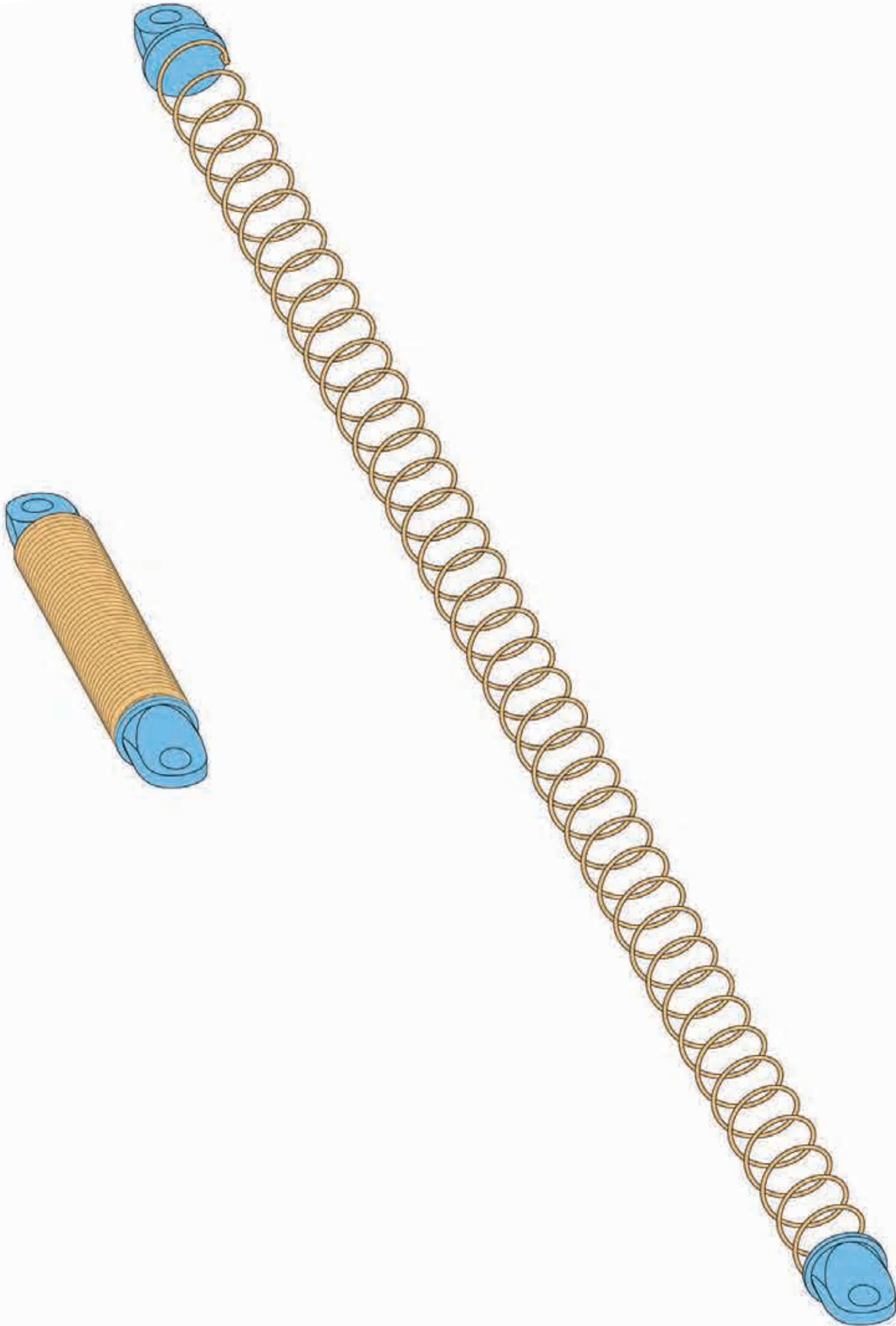
Screw

Ratchet Gear

Gasket

Adjustable Lock

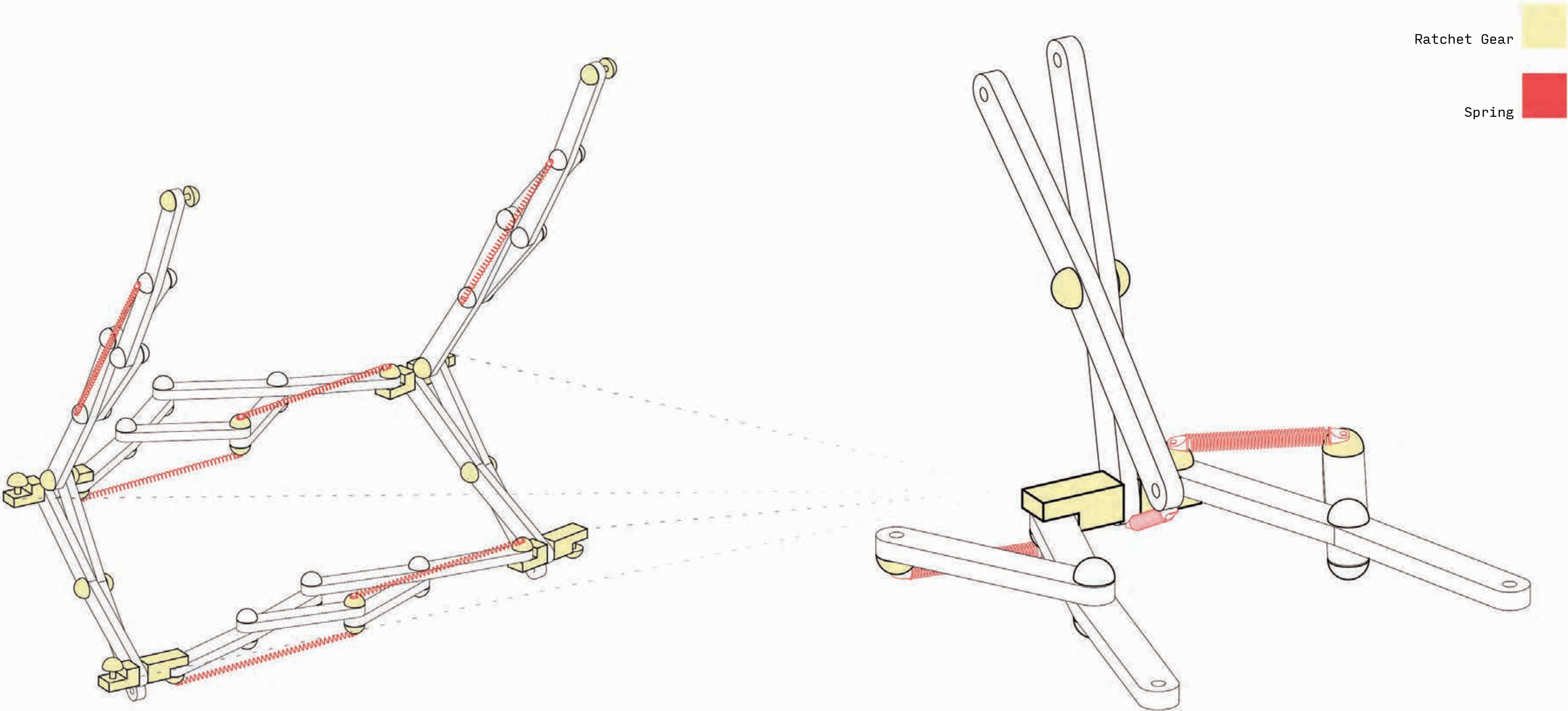
Servo Motor



Part V: Main Structure

Local Support – Spring System & Ratchet Gear

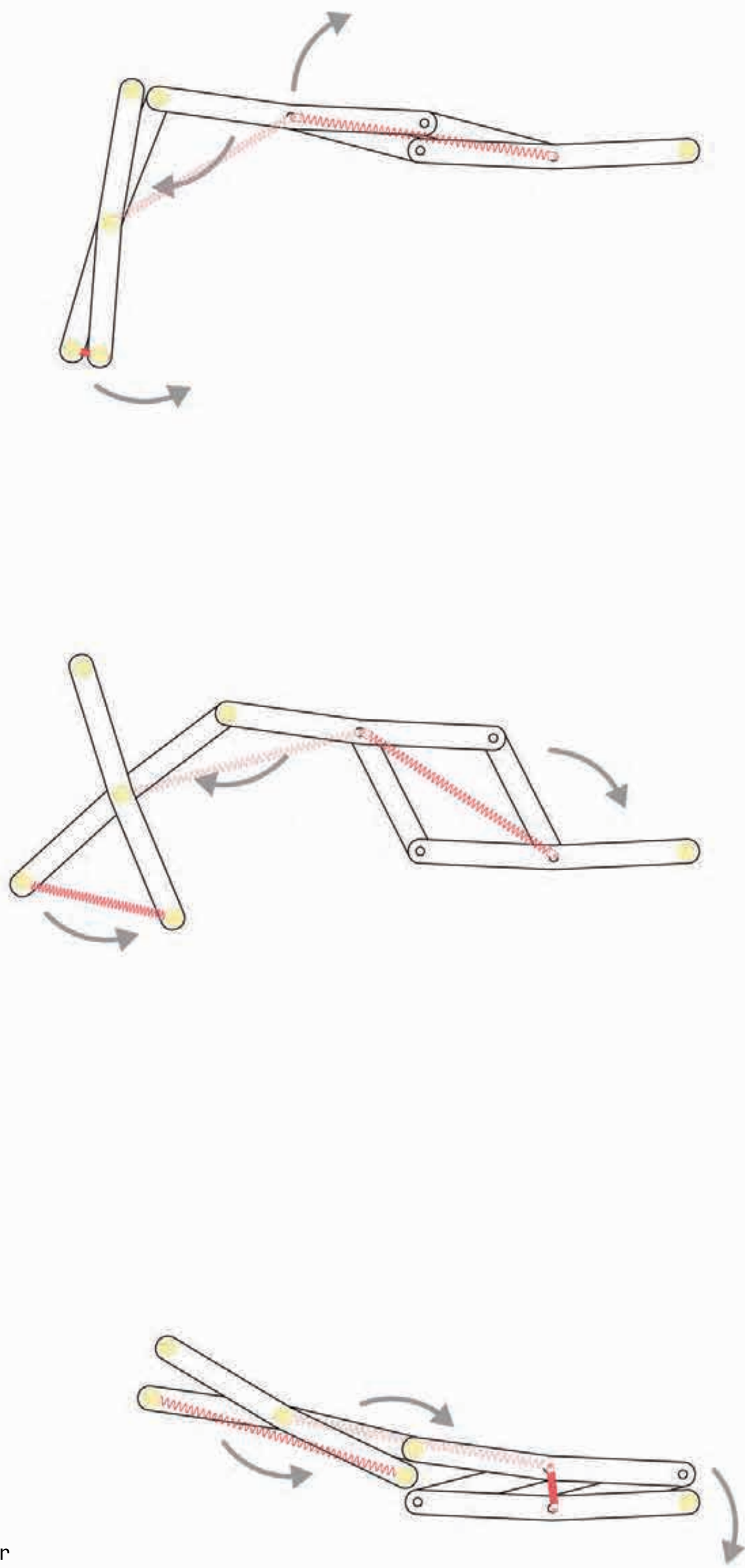
"This is where the springs and ratchets will appear in the units."



Part V: Main Structure

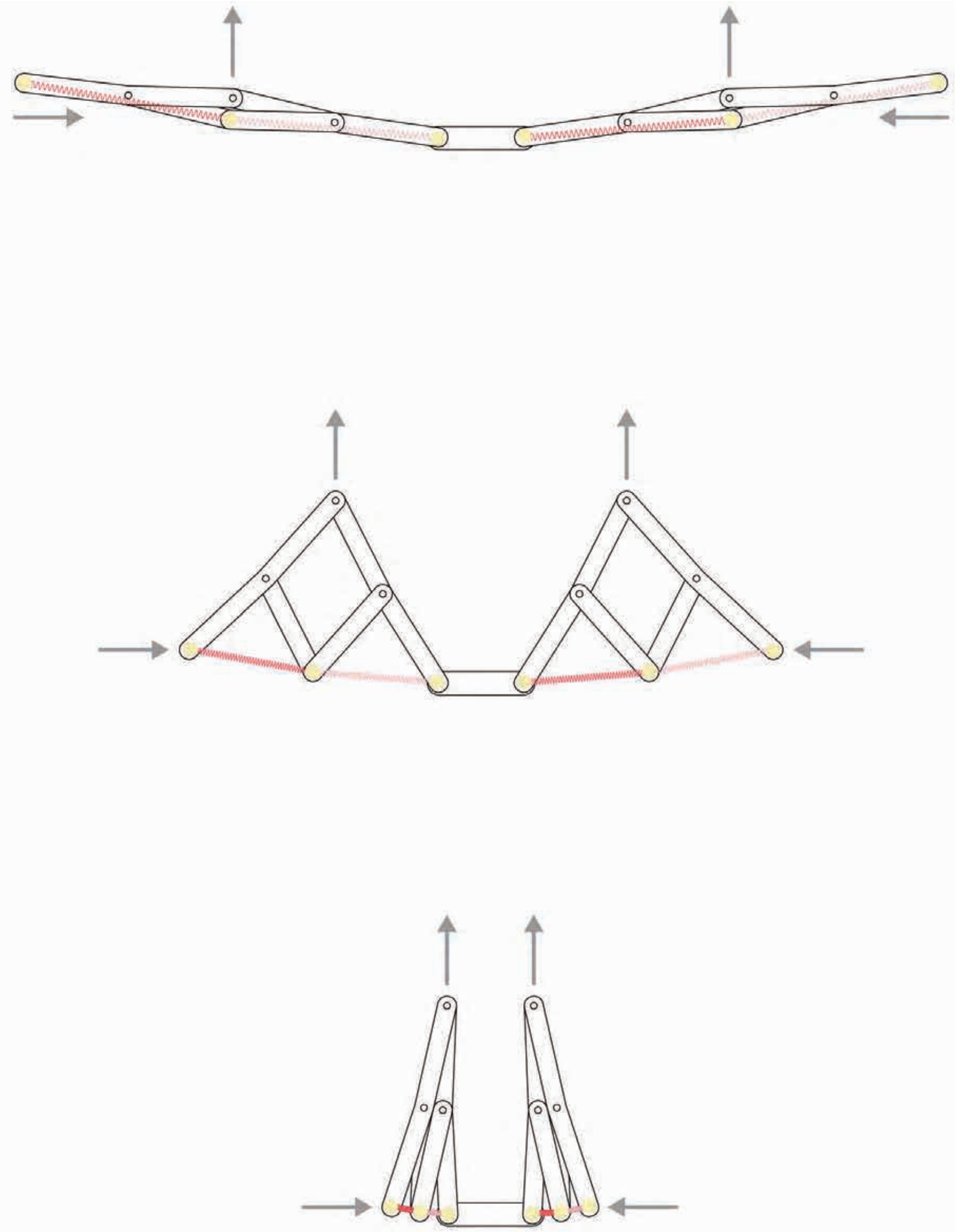
Local Support – Spring System & Ratchet Gear

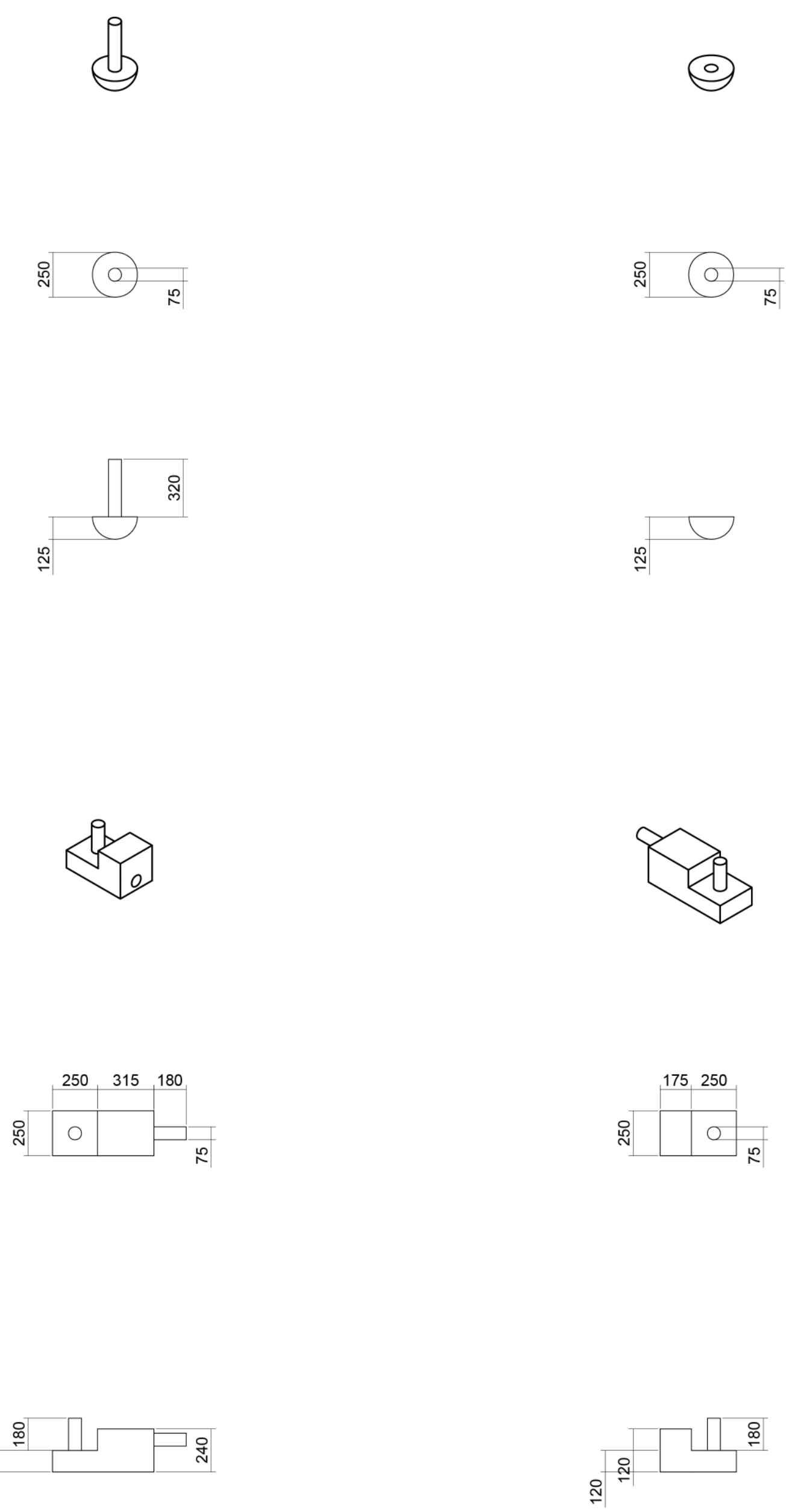
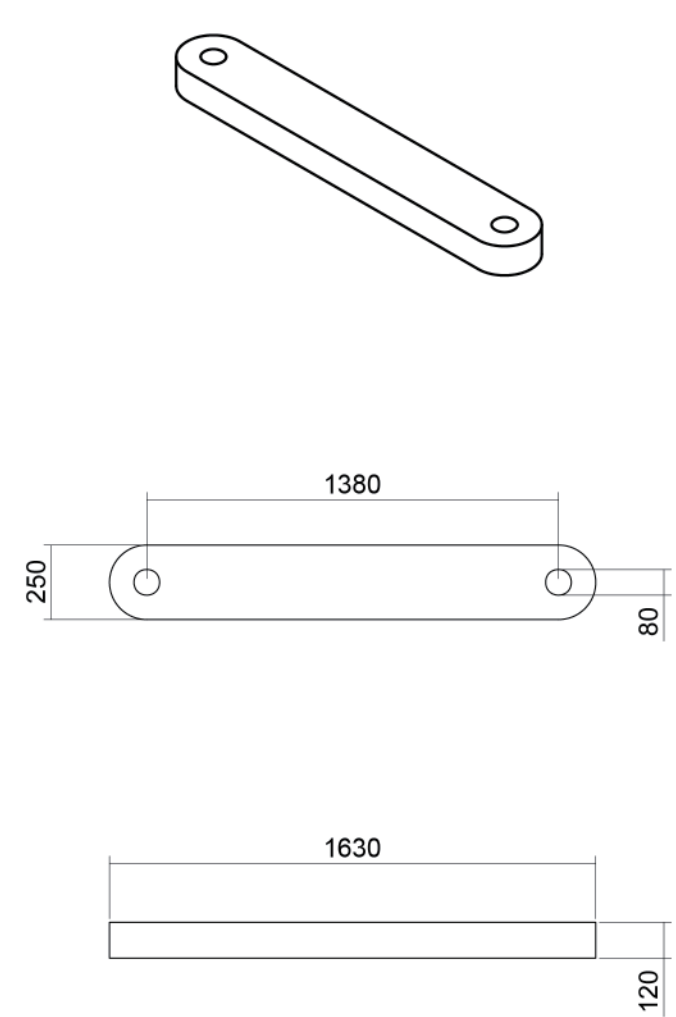
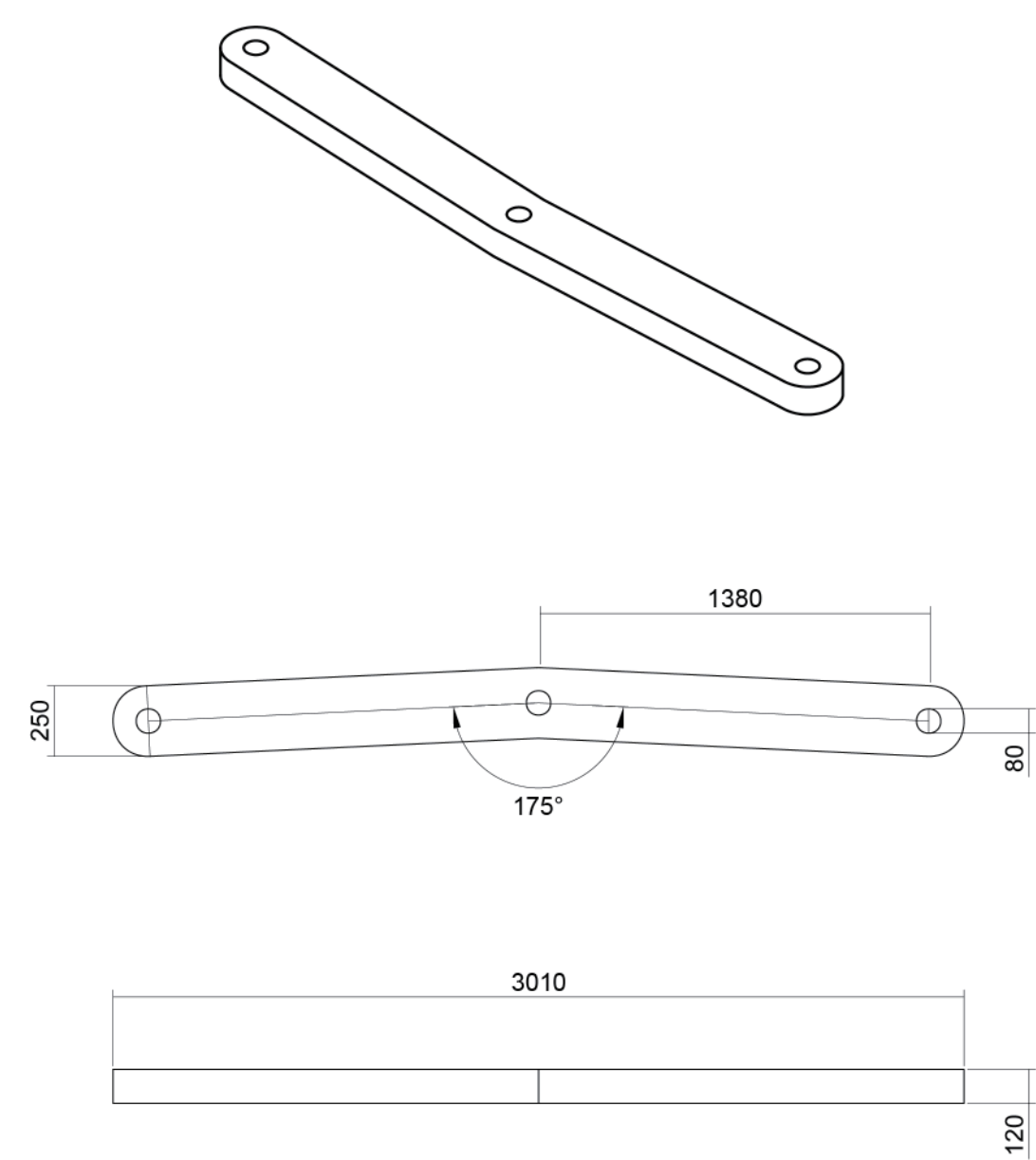
"The illustration depicts how the system works when the units rotate."



Ratchet Gear

Spring

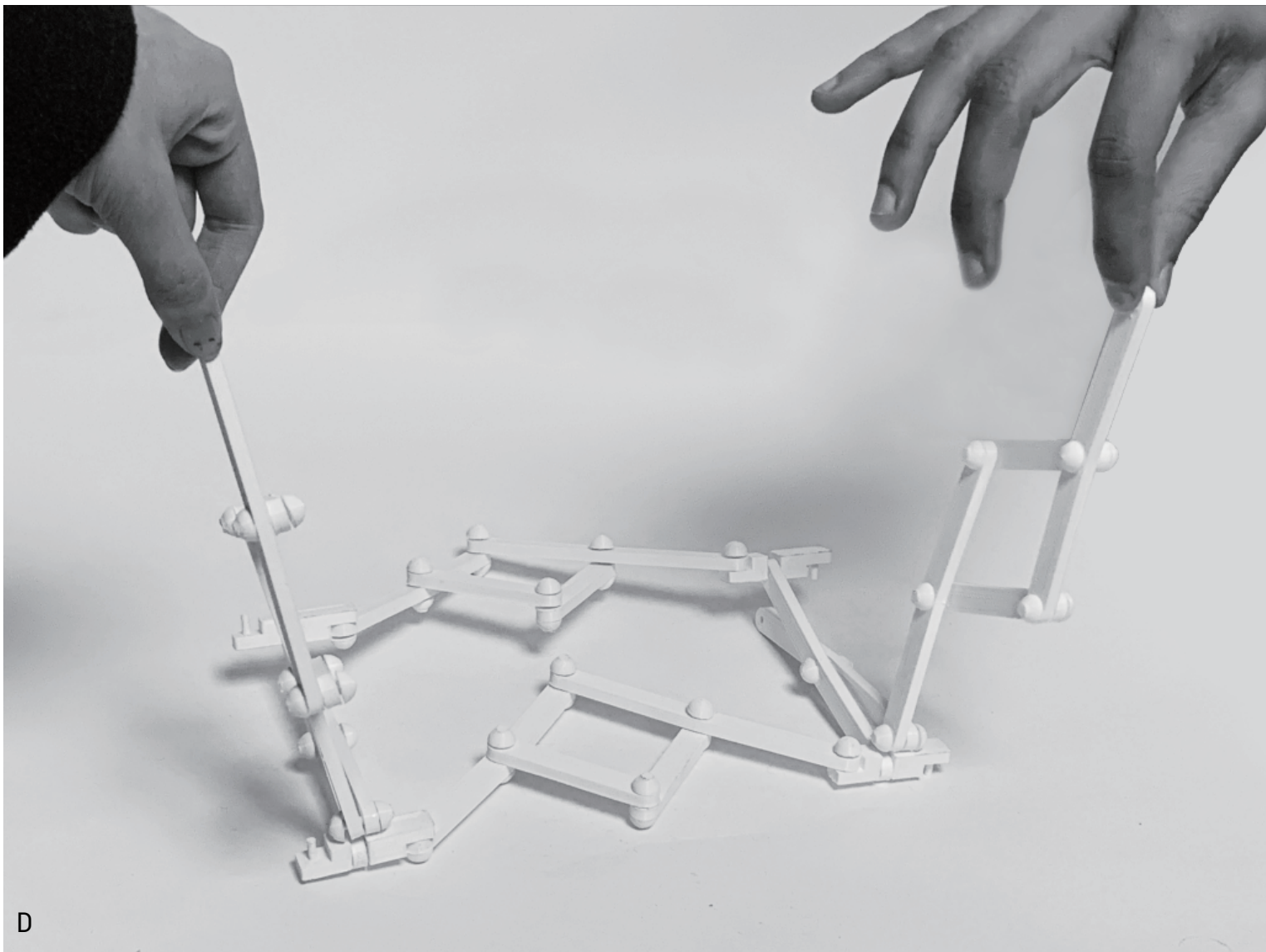
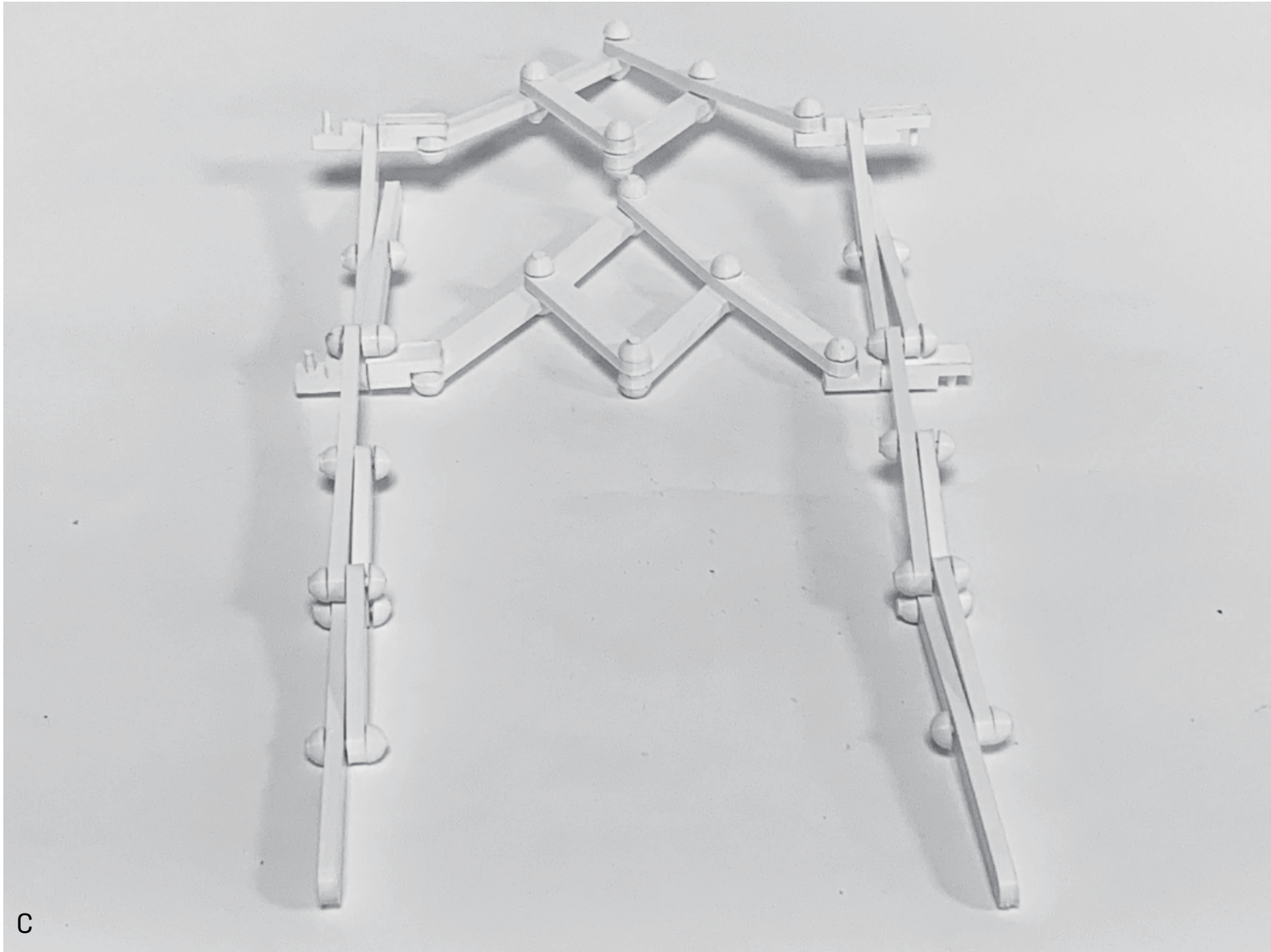
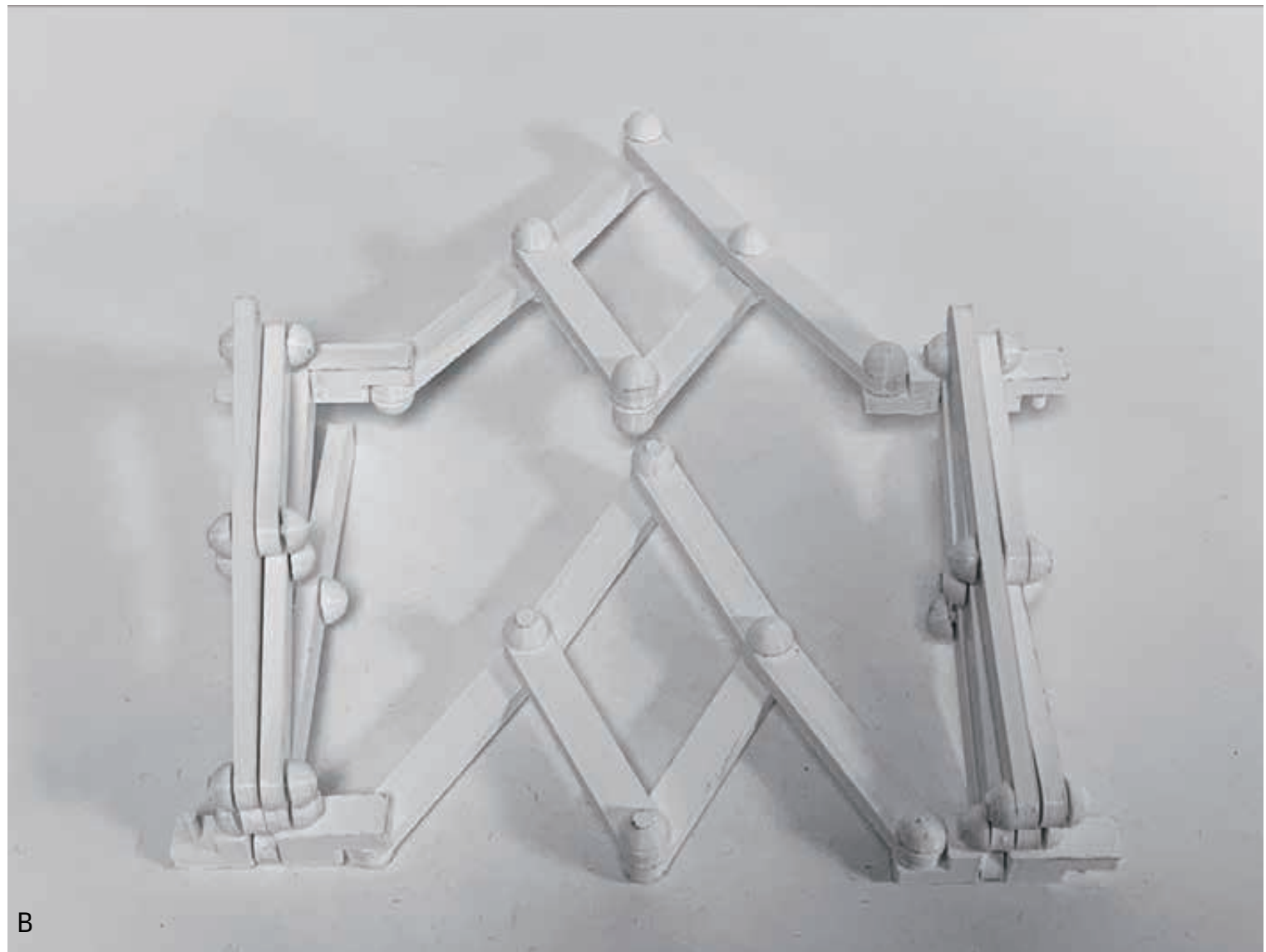
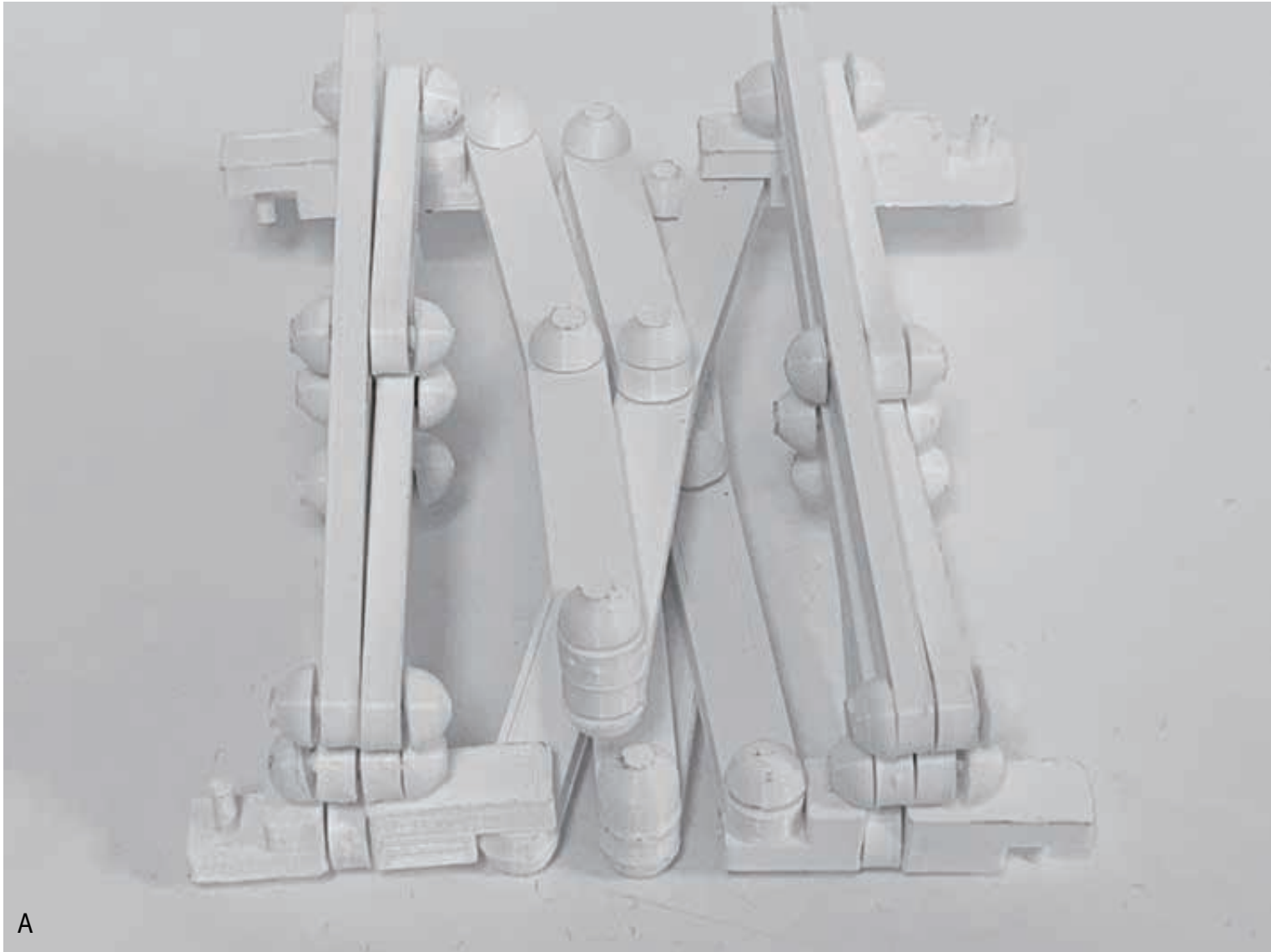




Part V: Main Structure

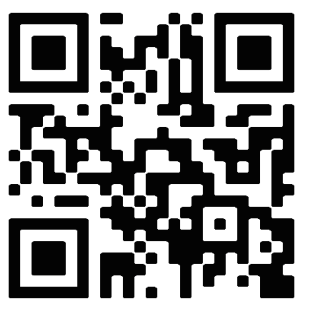
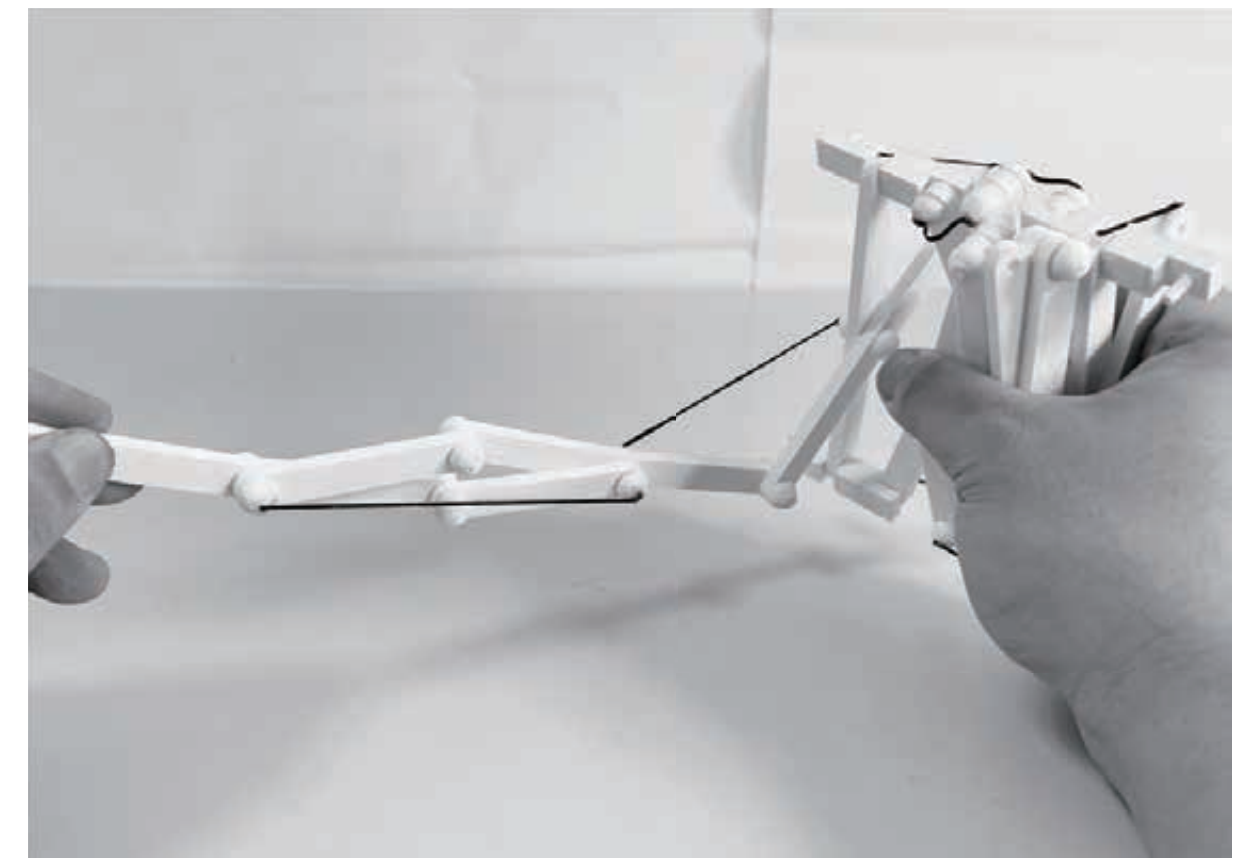
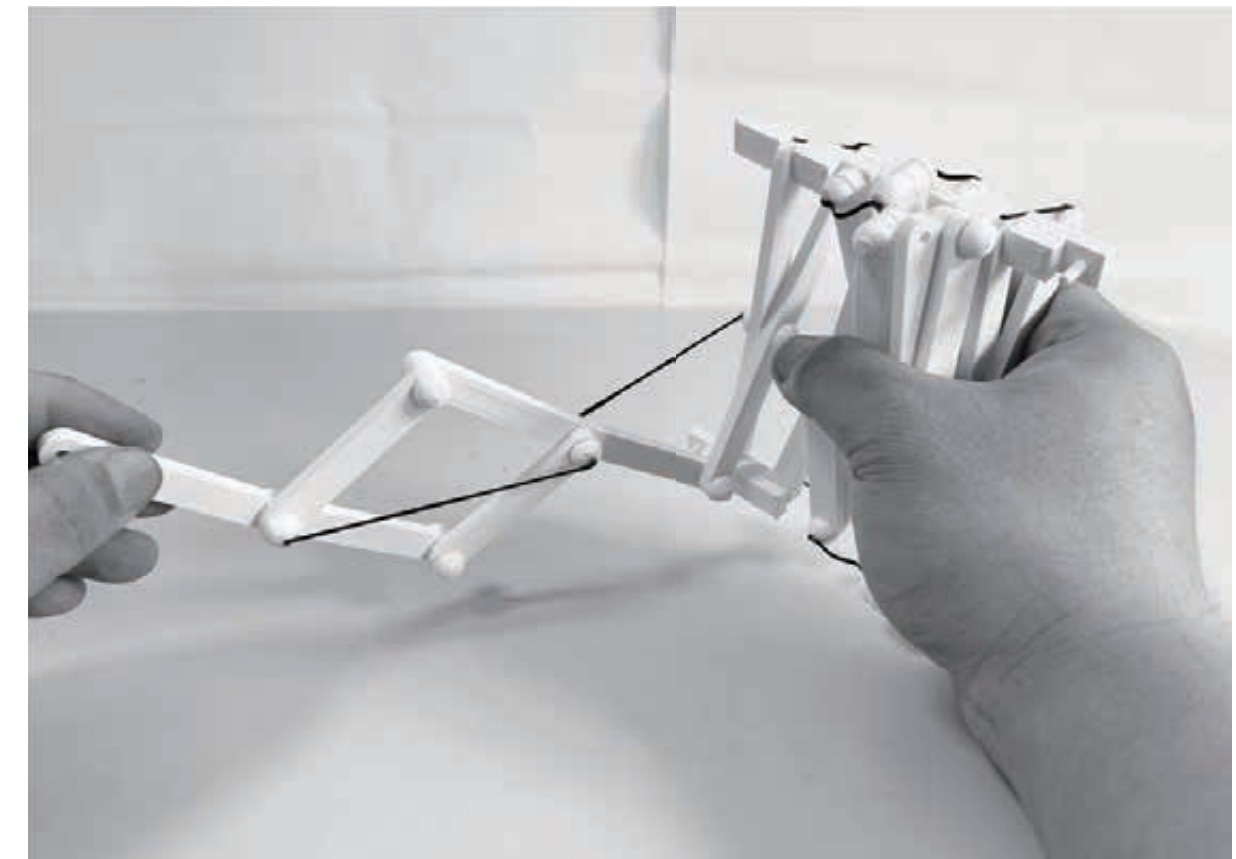
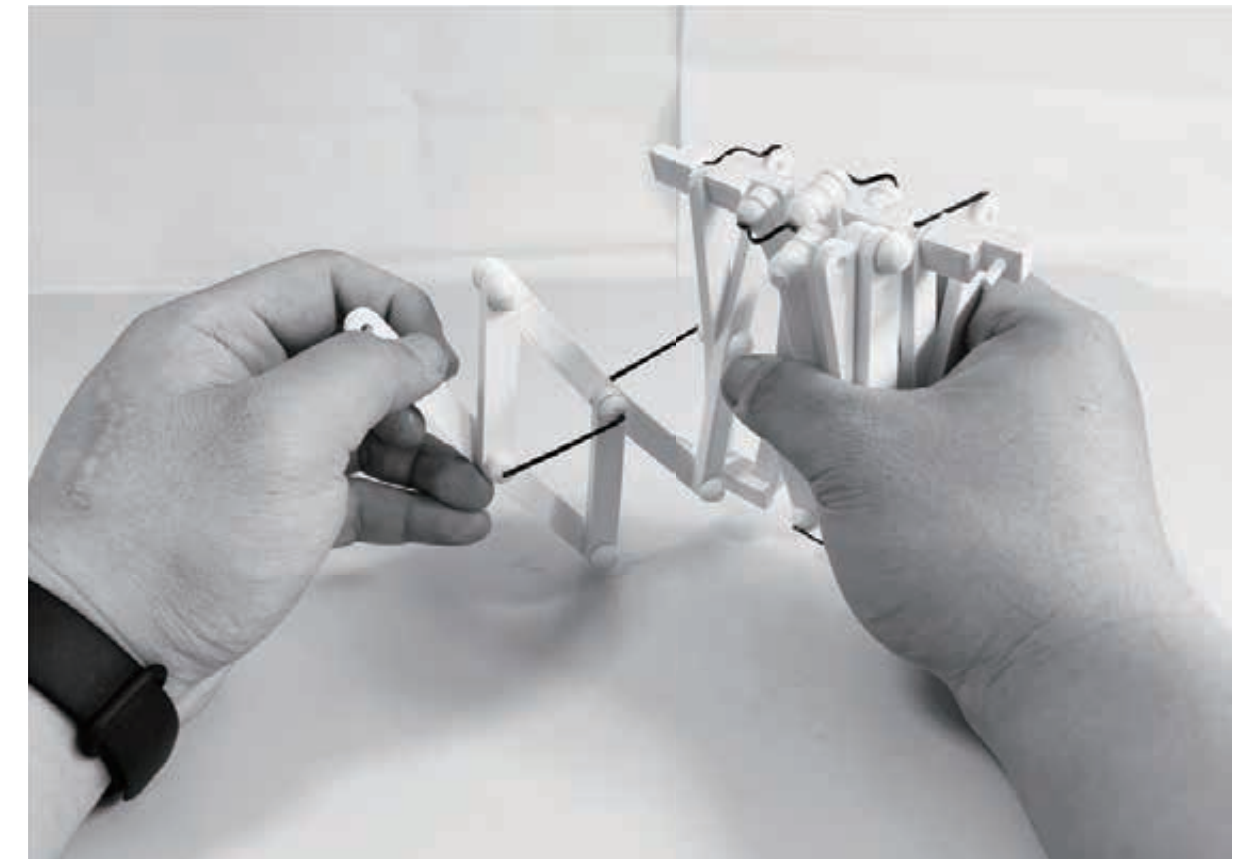
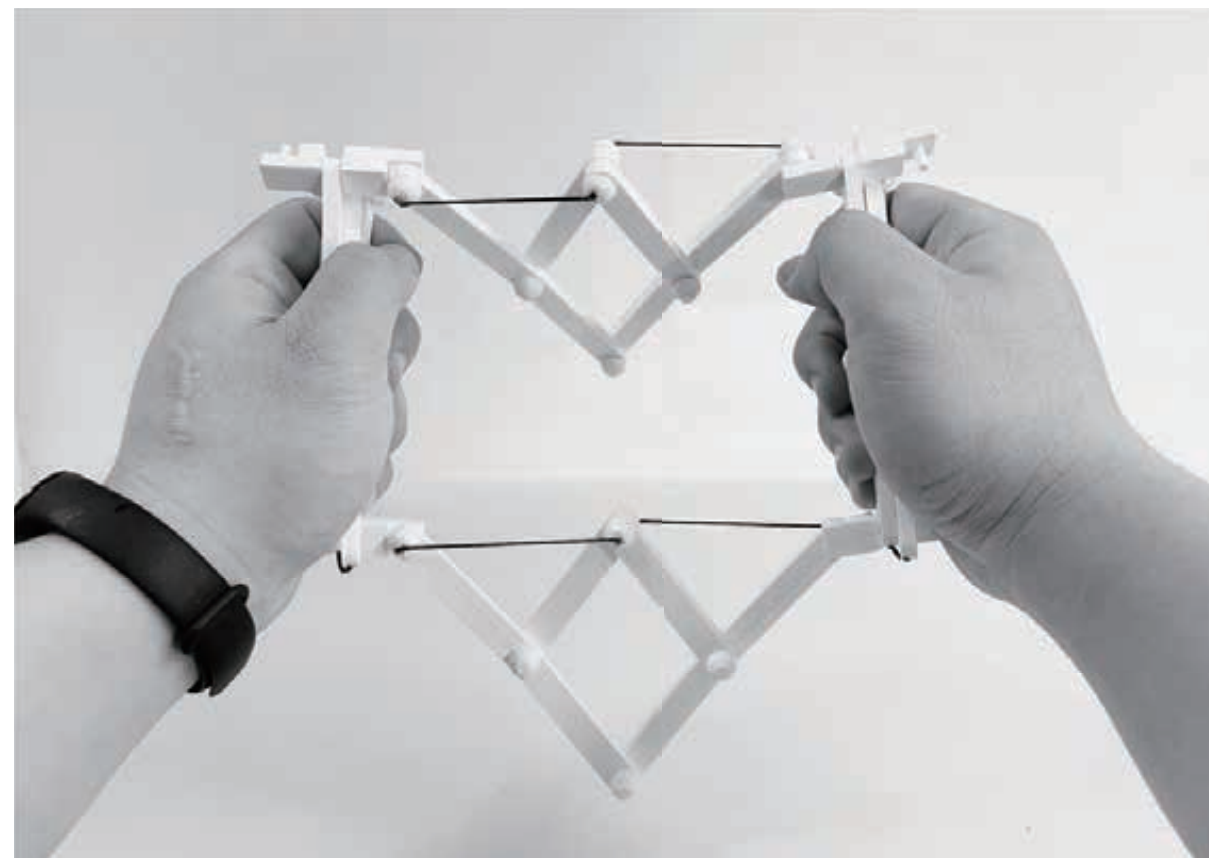
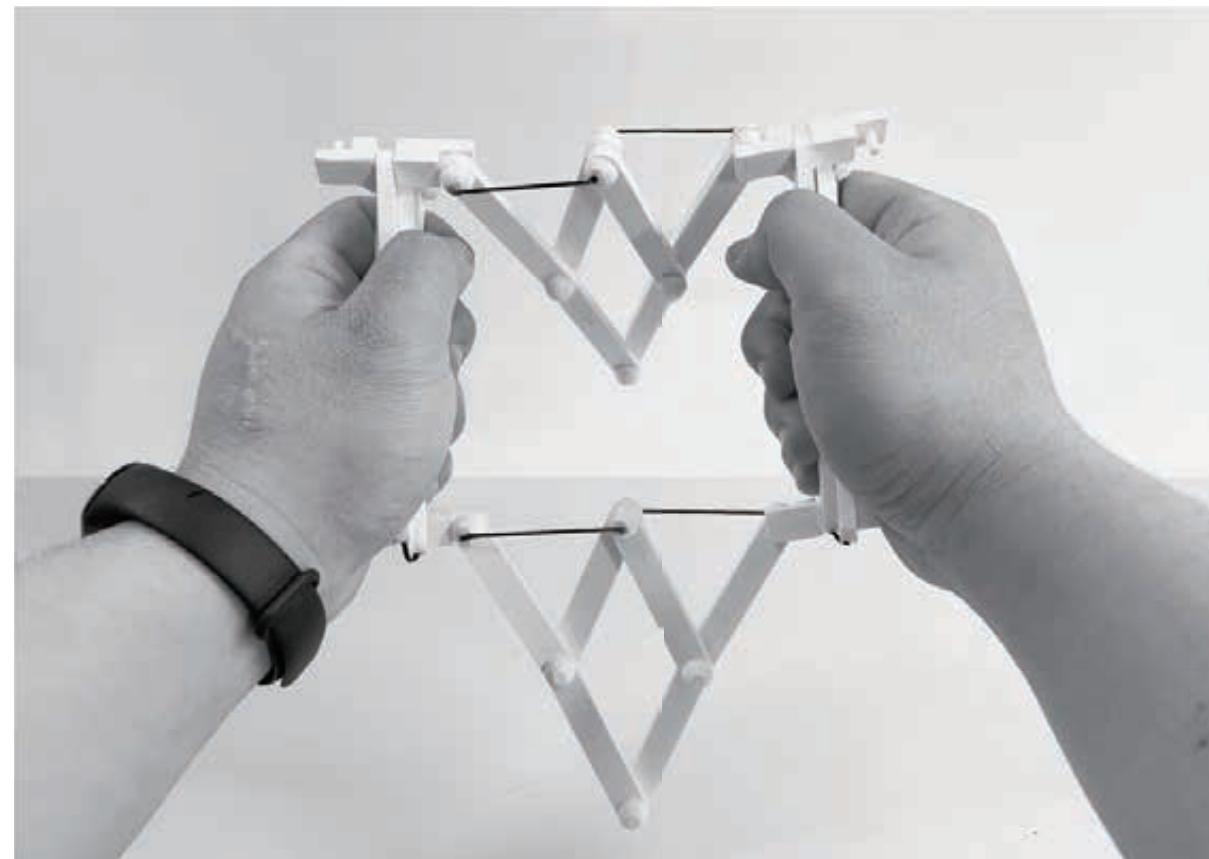
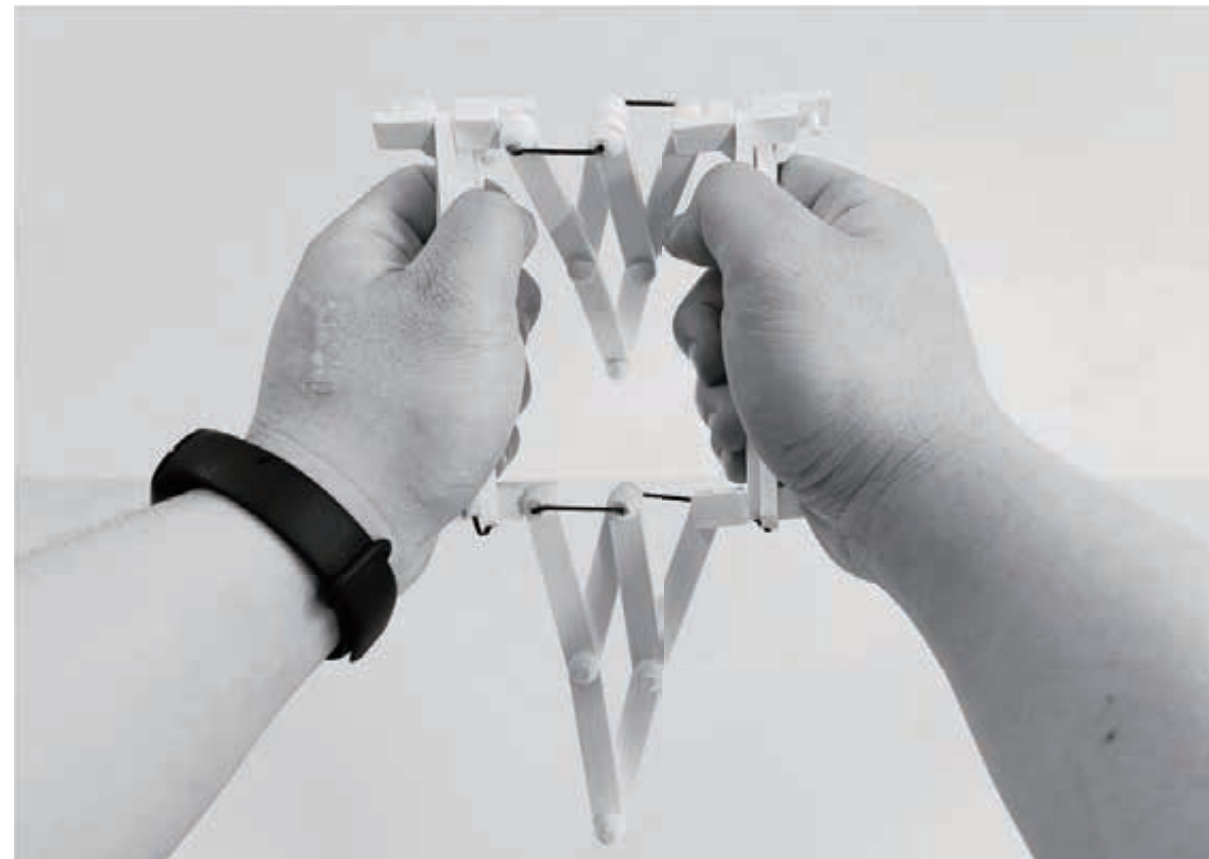
Main Structure Unit Model - Unfolding Sequence

"The Main Structure Unit Model shows, in order A to D, how the smallest module unfolds from the fold."



Part V: Main Structure

Main Structure Unit Model - With Supports



<https://youtu.be/1PcNwaHC9w>

"From left to right, it shows separately how the spring + ratchet system works together in the movement of transverse and the longitudinal structure."

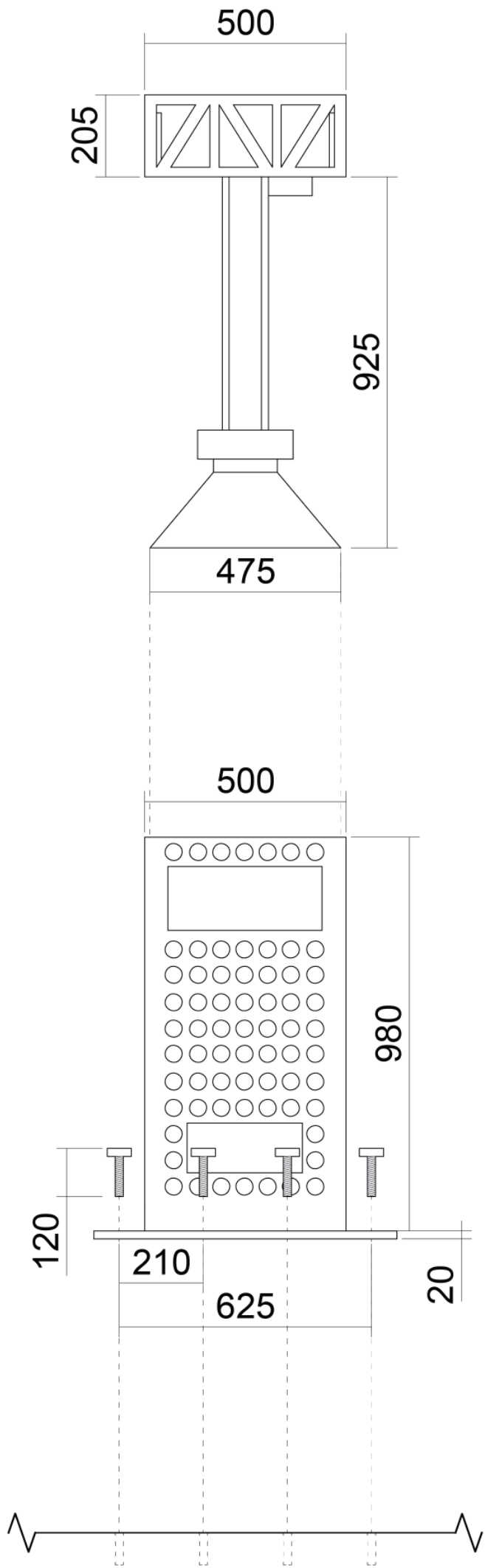
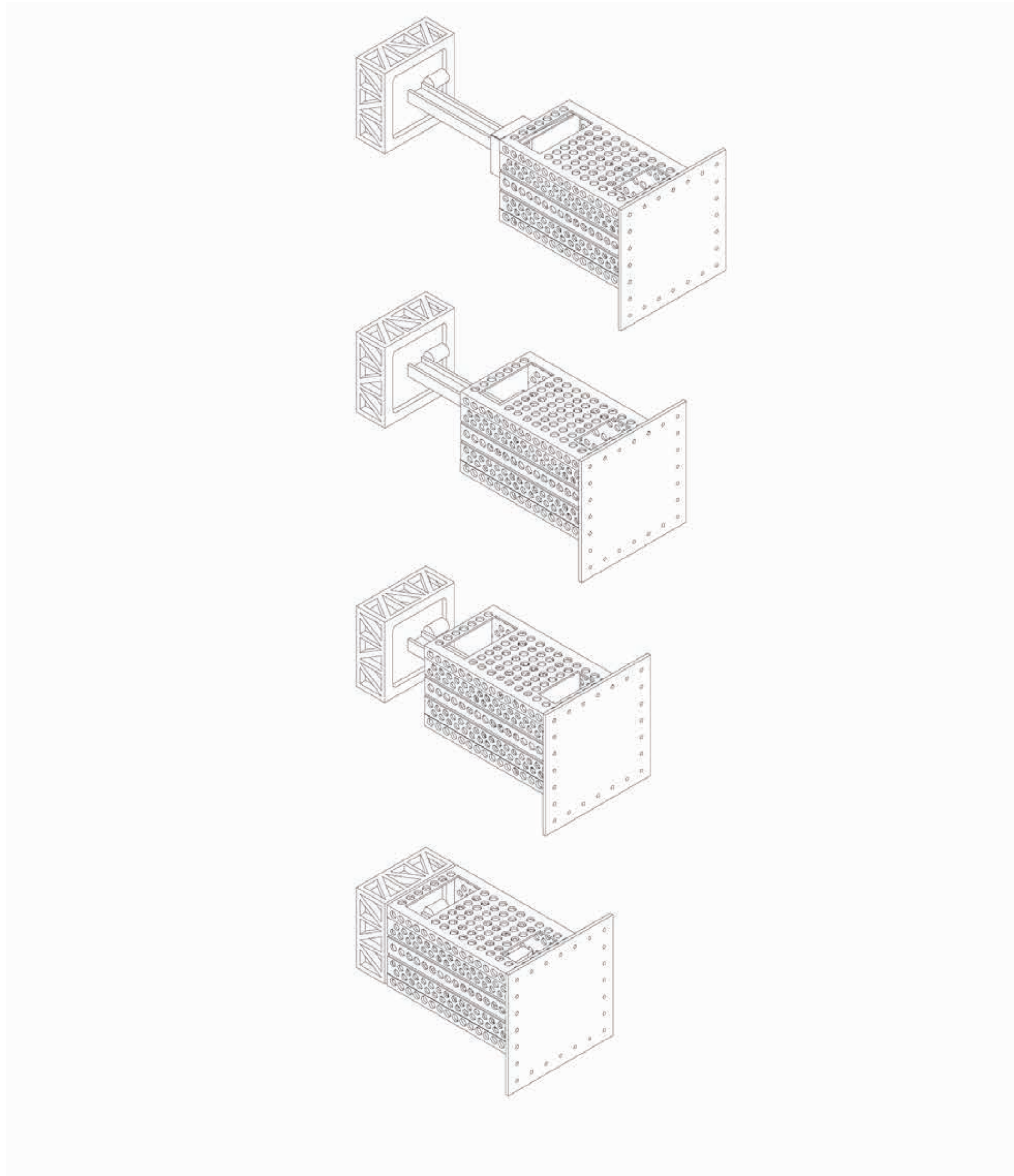
Part VI: Secondary Structure

As the secondary structure that is directly related to the movement of billboards, the Track-moving Self-feedback Billboard Units are extremely important to ensure its stable and safe operation.

Part VI: Secondary Structure

Billboard Arms - Dynamic & Dimensions

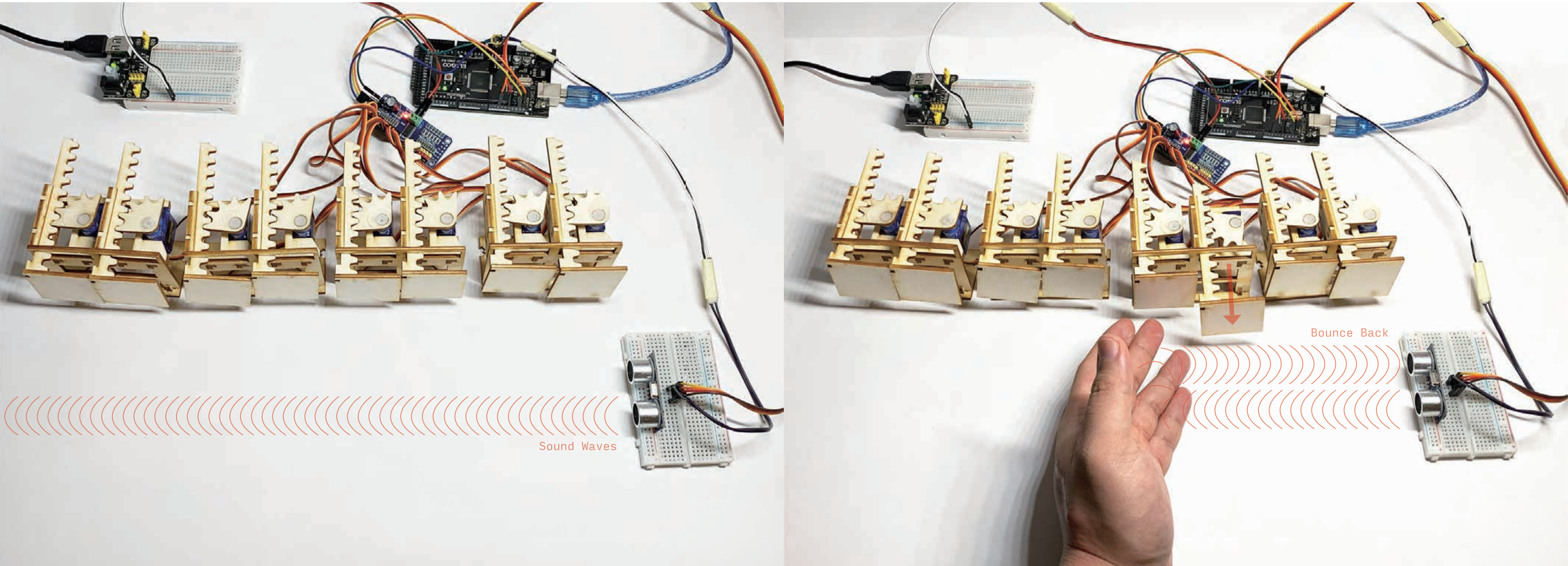
"As a secondary structure that is directly related to the movement of billboards, the billboard arms origin from Coca-Cola 3D billboards."



Part VI: Secondary Structure

Dynamic Feedback System

"The first Arduino model is intended for the study of the dynamic feedback of movable billboards to human actions, which provides greater safety and predictability of the structure."



Part VI: Secondary Structure

Arduino Coding

```
1 #include <Wire.h>
2 #include <Adafruit_PWMServoDriver.h>
3 #include "NewPing.h"
4
5 #define TRIGGER_PIN 10
6 #define ECHO_PIN 13
7 #define MAX_DISTANCE 400
8 #define SERVOMIN 150
9 #define SERVOMAX 375
10
11 Adafruit_PWMServoDriver myServo = Adafruit_PWMServoDriver();
12 NewPing sonar(TRIGGER_PIN, ECHO_PIN, MAX_DISTANCE);
13
14 uint8_t servonum = 0;
15 uint8_t numberOfServos = 8;
16
17 float duration, distance;
18
19 int iterations = 5;
20
21 void setup() {
22   Serial.begin(9600);
23   Serial.begin(9600);
24   myServo.begin();
25   myServo.setPWMFreq(60);
26   delay(10);
27 }
28
29 void loop() {
30   duration = sonar.ping_median(iterations);
31
32   // Determine distance from duration
33   // Use 343 metres per second as speed of sound
34
35   distance = (duration / 2) * 0.0343;
36
37   // Send results to Serial Monitor
38   Serial.print("Distance = ");
39   if (distance >= 400 || distance <= 2) {
40     Serial.println("Out of range");
41   }
42   else if (distance > 2 && distance <= 6) {
43     Serial.print(distance);
44     Serial.println(" cm");
45     for (uint16_t pulselen = SERVOMIN; pulselen < SERVOMAX; pulselen++){
46       myServo.setPWM(0, 0, pulselen);
47     }
48     delay(300);
49     for (uint16_t pulselen = SERVOMAX; pulselen > SERVOMIN; pulselen--){
50       myServo.setPWM(0, 0, pulselen);
51     }
52     delay(200);
53   }
54   else if (distance > 6 && distance <= 10) {
55     Serial.print(distance);
56     Serial.println(" cm");
57     for (uint16_t pulselen = SERVOMIN; pulselen < SERVOMAX; pulselen++){
58       myServo.setPWM(1, 0, pulselen);
59     }
60     delay(300);
61     for (uint16_t pulselen = SERVOMAX; pulselen > SERVOMIN; pulselen--){
62       myServo.setPWM(1, 0, pulselen);
63     }
64     delay(200);
65   }
66   else if (distance > 10 && distance <= 14) {
67     Serial.print(distance);
68     Serial.println(" cm");
69     for (uint16_t pulselen = SERVOMIN; pulselen < SERVOMAX; pulselen++){
70       myServo.setPWM(2, 0, pulselen);
71     }
72     delay(300);
73     for (uint16_t pulselen = SERVOMAX; pulselen > SERVOMIN; pulselen--){
74       myServo.setPWM(2, 0, pulselen);
75     }
76     delay(200);
77   }
```

Required language library to support ultrasonic sensors and servo motor servers.

Ultrasonic emission control pin.

Ultrasonic receive control pin.

Maximum effective ranging.

Servo motor minimum angle and maximum angle calibration.

Define important variables.

Initial motor and total number of motors.

Define the sound wave rebound duration and the measured distance as floating point numbers.

Required number of checks.

The program starts and the components start.

Program loop starts.

Calculate actual distance by formula and measurement result (time).

Measurement result feedback. Limit the maximum and minimum distances.

When the measured distance is between 2 and 6 cm, the No.0 servo motor gives feedback.

When the measured distance is between 6 and 10 cm, the No.1 servo motor gives feedback.

When the measured distance is between 10 and 14 cm, the No.2 servo motor gives feedback.

```
78
79 } else if (distance > 14 && distance <= 18) {
80   Serial.print(distance);
81   Serial.println(" cm");
82   for (uint16_t pulselen = SERVOMIN; pulselen < SERVOMAX; pulselen++){
83     myServo.setPWM(3, 0, pulselen);
84   }
85   delay(300);
86   for (uint16_t pulselen = SERVOMAX; pulselen > SERVOMIN; pulselen--){
87     myServo.setPWM(3, 0, pulselen);
88   }
89   delay(200);
90
91 } else if (distance > 18 && distance <= 22) {
92   Serial.print(distance);
93   Serial.println(" cm");
94   for (uint16_t pulselen = SERVOMIN; pulselen < SERVOMAX; pulselen++){
95     myServo.setPWM(4, 0, pulselen);
96   }
97   delay(300);
98   for (uint16_t pulselen = SERVOMAX; pulselen > SERVOMIN; pulselen--){
99     myServo.setPWM(4, 0, pulselen);
100   }
101   delay(200);
102
103 } else if (distance > 22 && distance <= 26) {
104   Serial.print(distance);
105   Serial.println(" cm");
106   for (uint16_t pulselen = SERVOMIN; pulselen < SERVOMAX; pulselen++){
107     myServo.setPWM(5, 0, pulselen);
108   }
109   delay(300);
110   for (uint16_t pulselen = SERVOMAX; pulselen > SERVOMIN; pulselen--){
111     myServo.setPWM(5, 0, pulselen);
112   }
113   delay(200);
114 } else if (distance > 26 && distance <= 30) {
115   Serial.print(distance);
116   Serial.println(" cm");
117   for (uint16_t pulselen = SERVOMIN; pulselen < SERVOMAX; pulselen++){
118     myServo.setPWM(6, 0, pulselen);
119   }
120   delay(300);
121   for (uint16_t pulselen = SERVOMAX; pulselen > SERVOMIN; pulselen--){
122     myServo.setPWM(6, 0, pulselen);
123   }
124   delay(200);
125
126 } else if (distance > 30 && distance <= 34) {
127   Serial.print(distance);
128   Serial.println(" cm");
129   for (uint16_t pulselen = SERVOMIN; pulselen < SERVOMAX; pulselen++){
130     myServo.setPWM(7, 0, pulselen);
131   }
132   delay(300);
133   for (uint16_t pulselen = SERVOMAX; pulselen > SERVOMIN; pulselen--){
134     myServo.setPWM(7, 0, pulselen);
135   }
136   delay(200);
137
138 } else {
139   Serial.print(distance);
140   Serial.println(" cm");
141 }
142
143 delay(500);
144
145 servonum ++;
146 if (servonum > numberOfServos-1)
147   servonum = 0;
148
149
150 }
```

When the measured distance is between 14 and 18 cm, the No.3 servo motor gives feedback.

When the measured distance is between 18 and 22 cm, the No.4 servo motor gives feedback.

When the measured distance is between 22 and 26 cm, the No.5 servo motor gives feedback.

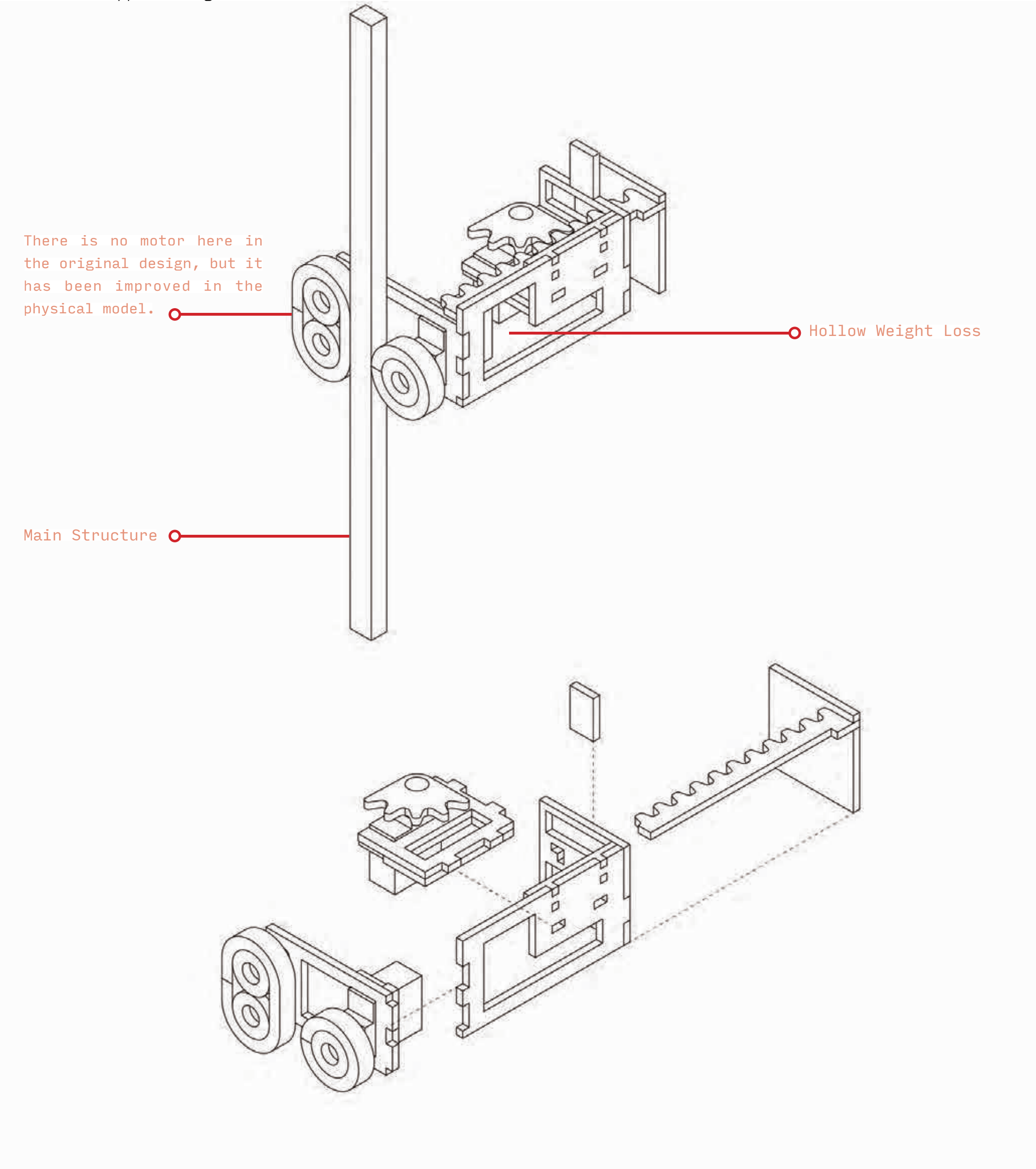
When the measured distance is between 26 and 30 cm, the No.6 servo motor gives feedback.

When the measured distance is between 30 and 34 cm, the No.7 servo motor gives feedback.

Part VI: Secondary Structure

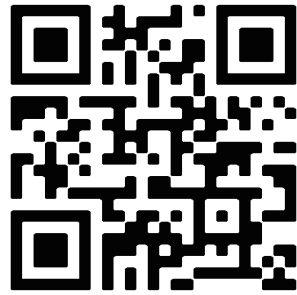
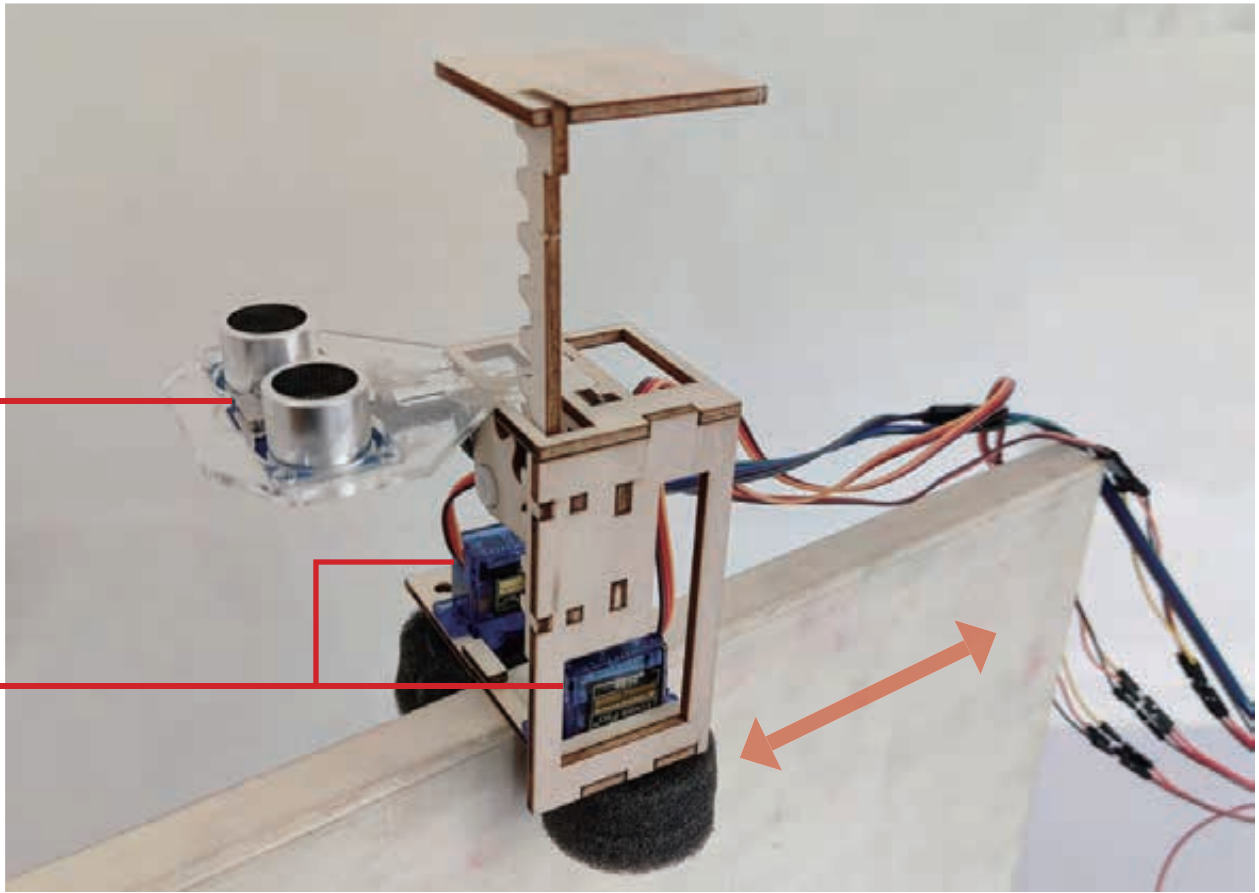
Track-moving Self-feedback Billboard Unit

"This model is a more mature model exploration for setting the moving billboards on floats. It supports movement along the outer edge of floats and is equipped with a sensor to avoid human approaching."

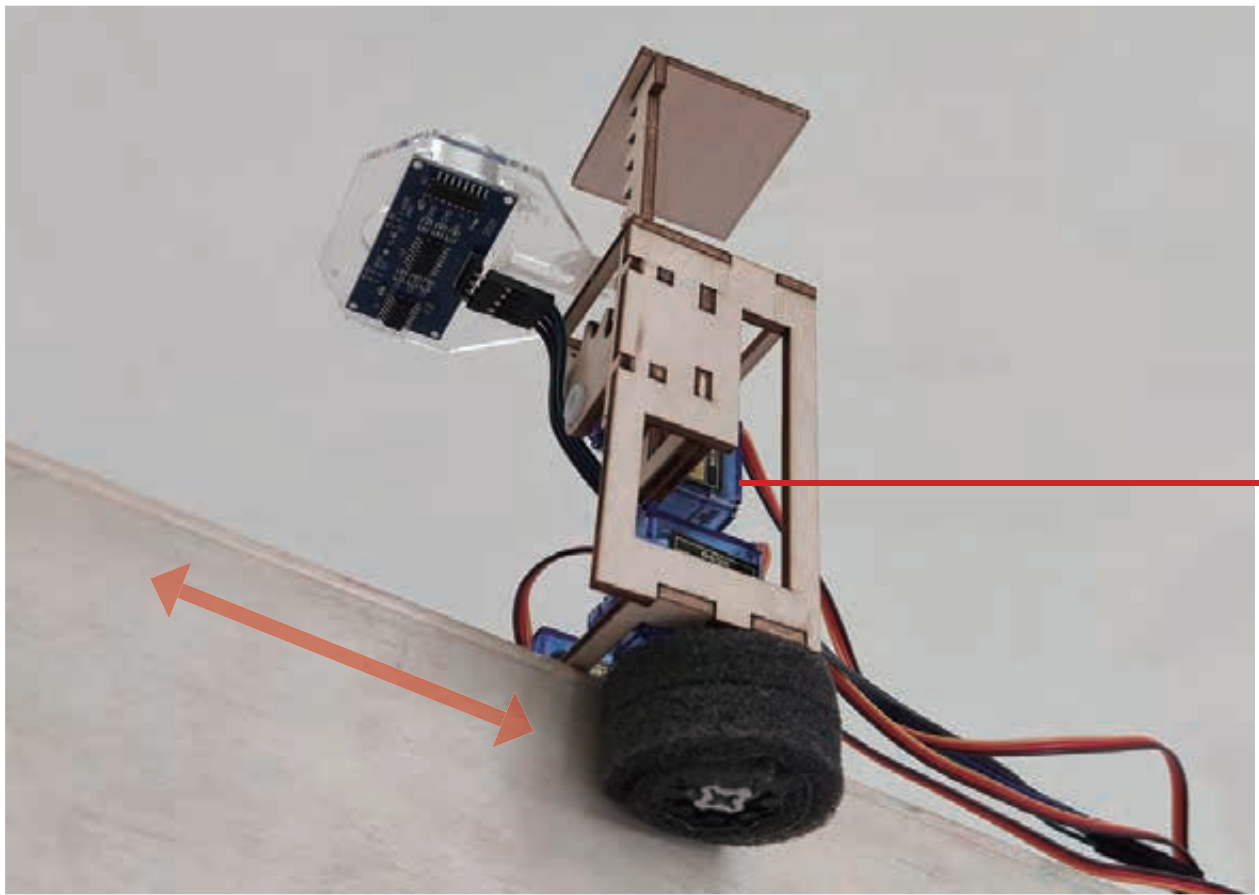


Sonic Sensor

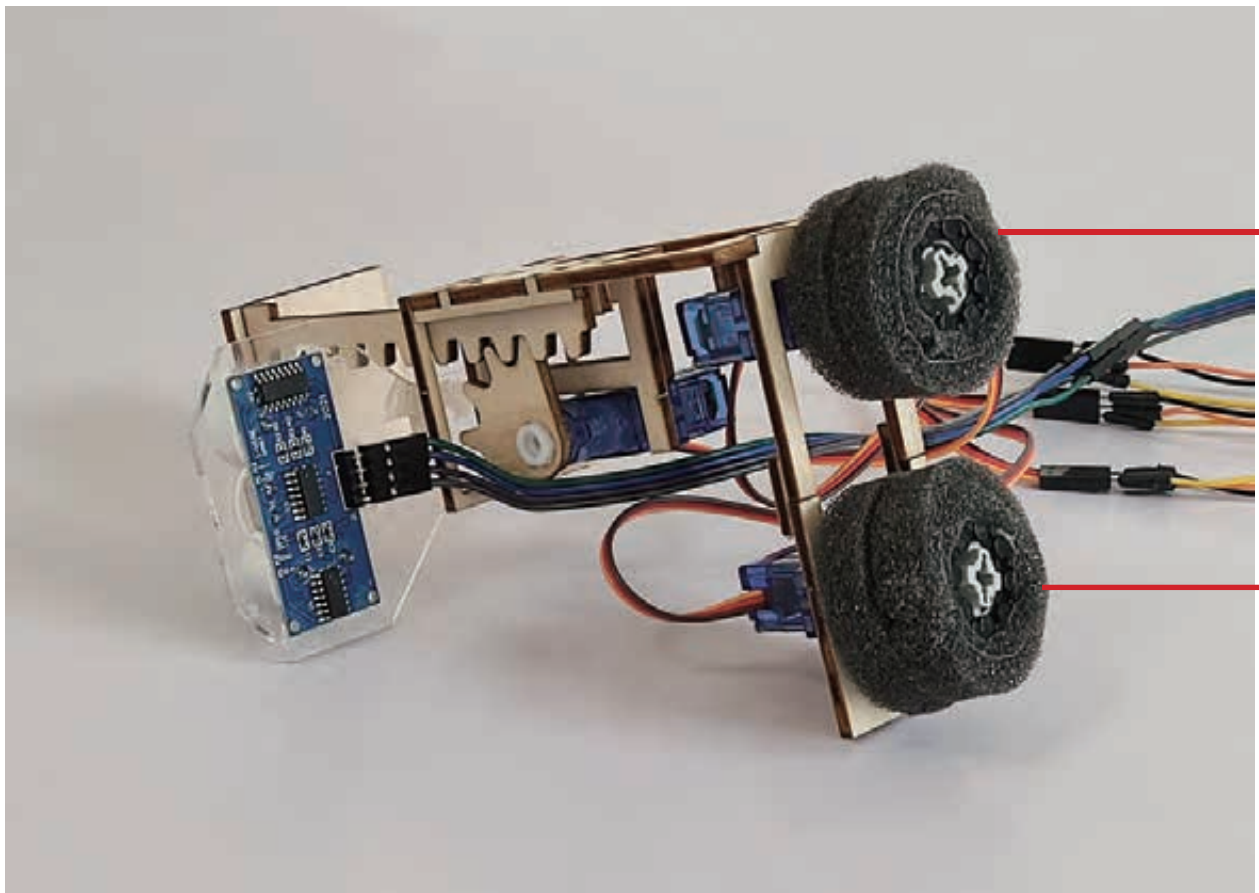
2 Motors



<https://youtu.be/fZYf9TbIOw4>



1 Motor for Feedback



Tight Wheel Set

Part VII: Power System

Although the building is made by making full use of recycled materials existing on the site, the manufacturing and running costs of a brand new giant float are indeed worth considering. But it is an annual, reusable, mobile building with infinite transformation possibilities. This chapter attempts to explain the power sources for the general energy consumption.



(1KW/M²*Area*18h*365D/1000)

"At present, the billboards on Times Square cover building surface area of 18180 square meters, which will cause an annual energy consumption of at least 119422 Mega Watt/Year."



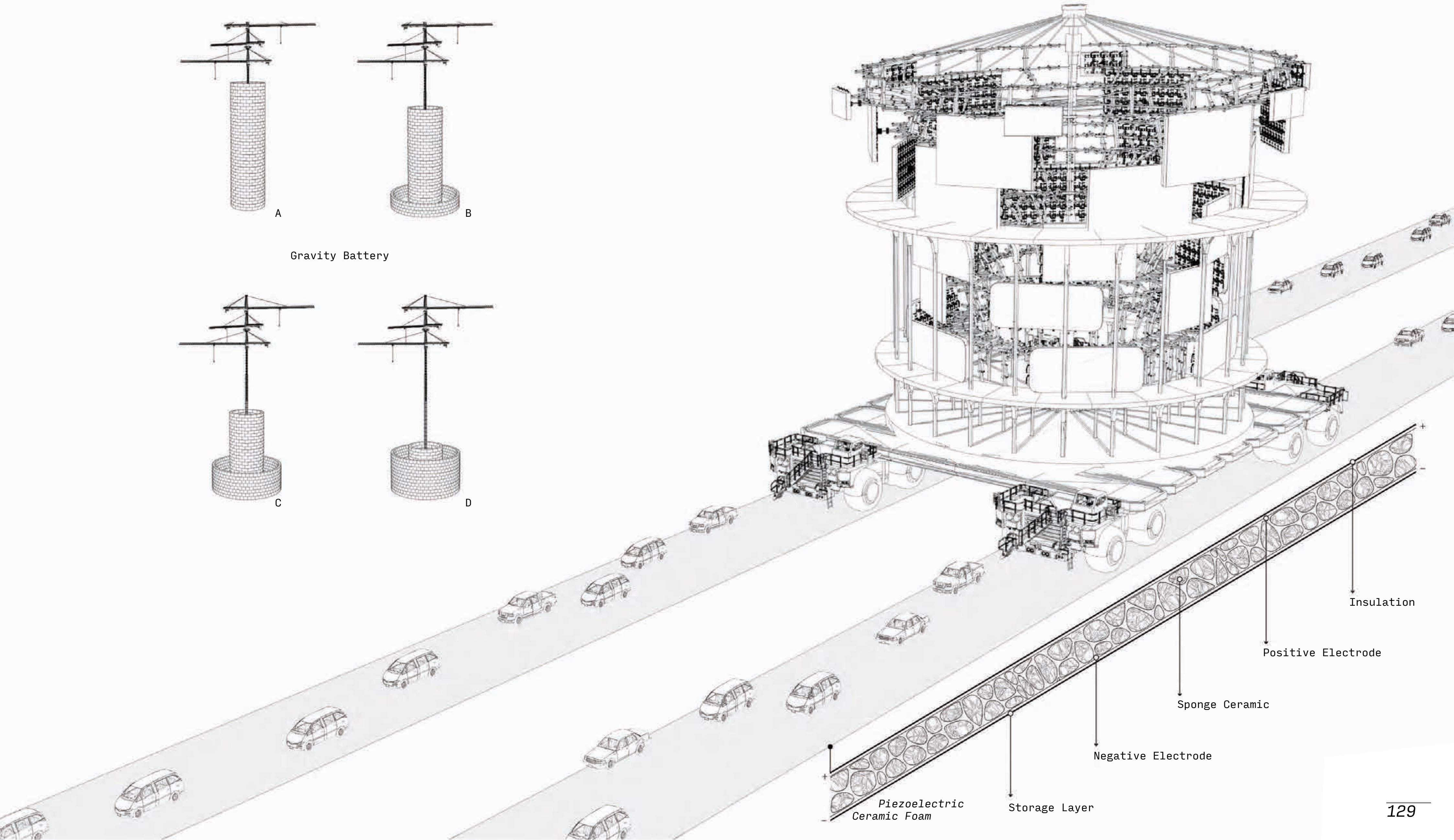
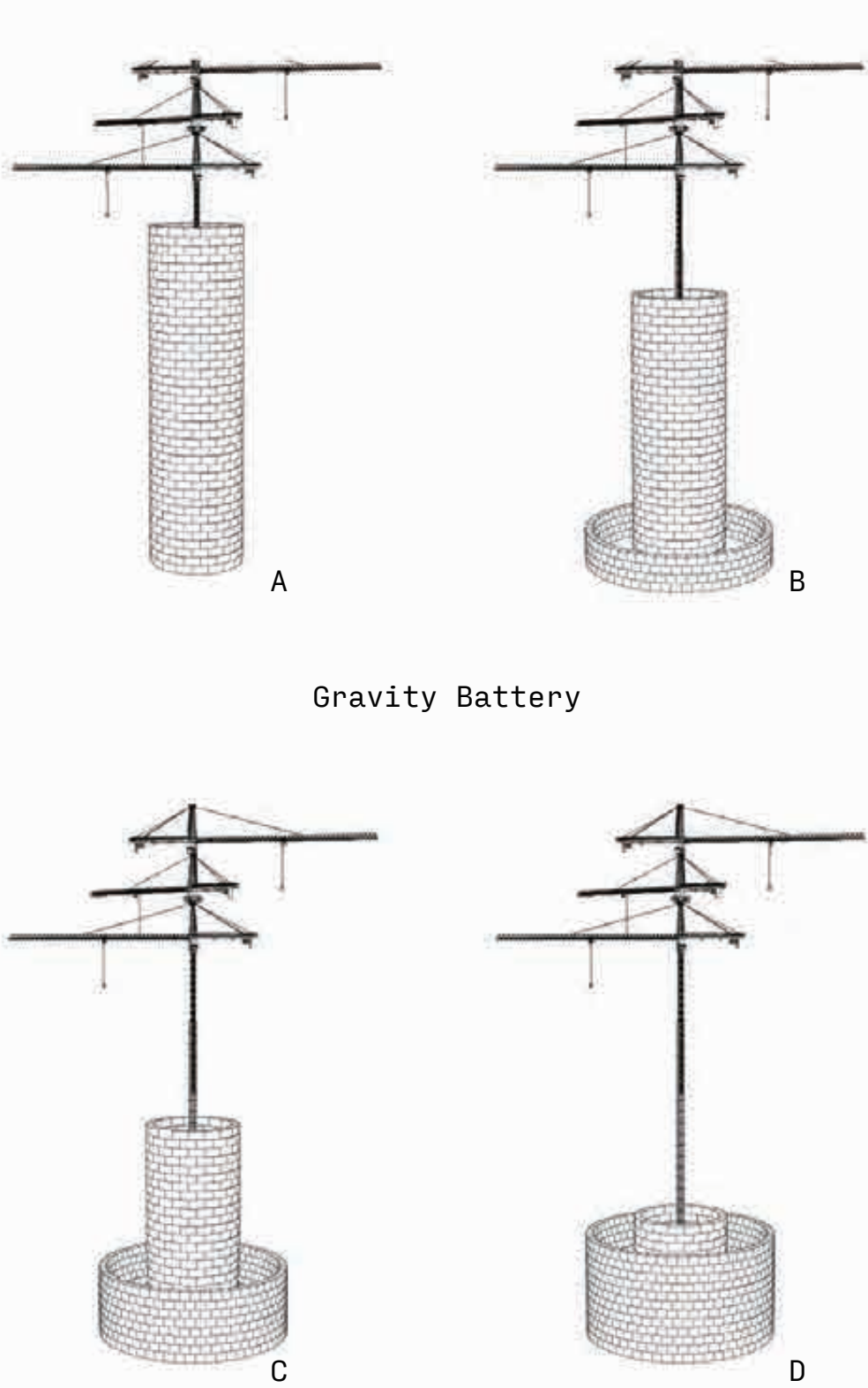
(0.125KW/M²*Area*8h*365D/1000)

"After the demolition of the billboards, the scaffolds that was originally used as support will not be removed, and 1/3 of them will be re-laid with solar panels, which can generate electricity of 6635 Mega Watt/Year, part of which can Electricity savings as a float parade."

Part VII: Power System

Gravity Battery & Piezoelectric Ceramic Foam

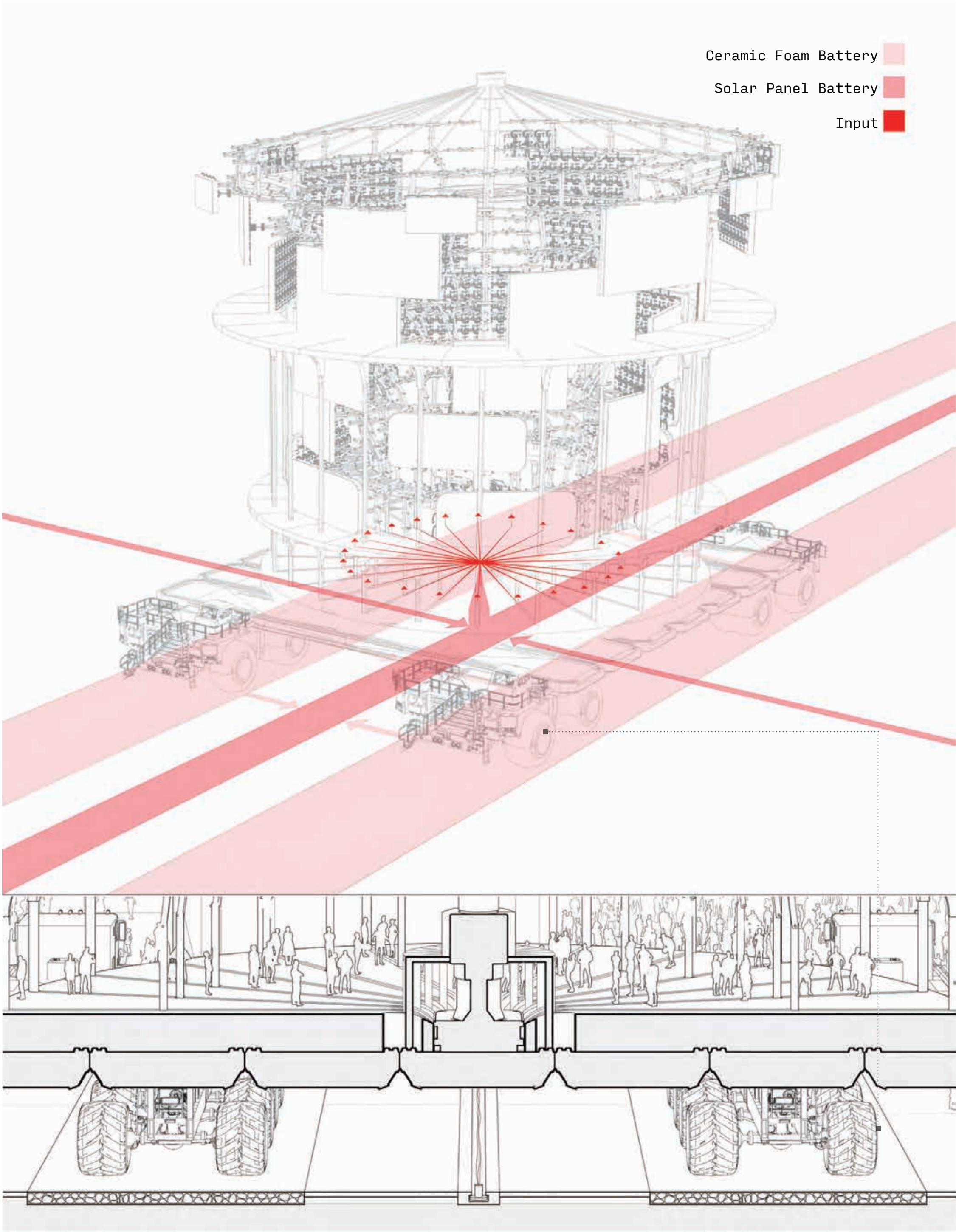
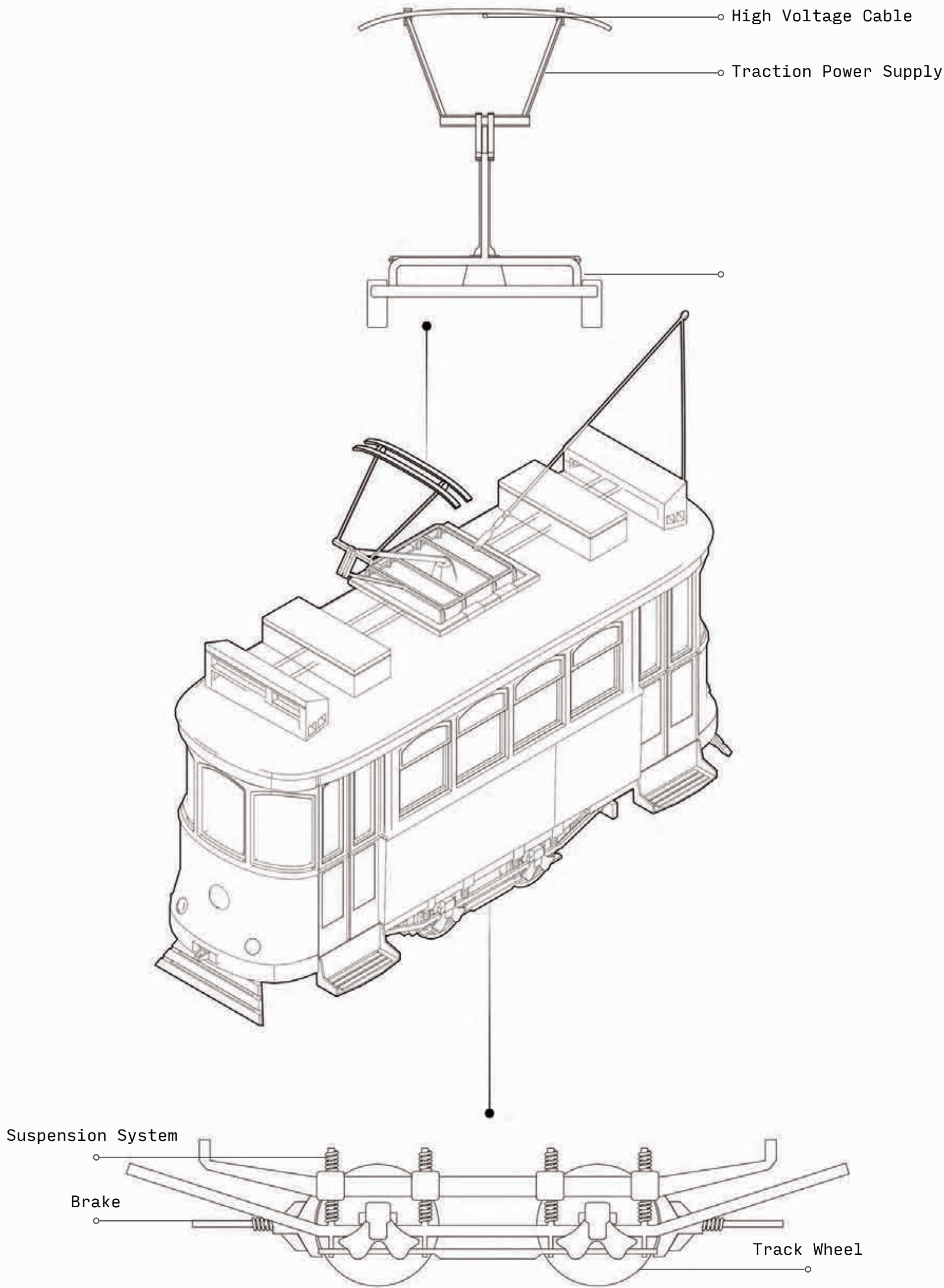
"By learning about Gravity Battery (a technology that converts renewable energy such as solar energy into potential energy and stores it), I discovered the idea that gravity provides an auxiliary power reserve. With the help of a new material, Piezoelectric Ceramic Foam, to pave the road, the all year round continuous driving in Times Square will slowly accumulate electricity."



Part VII: Power System

Power Transmission Design

"By studying the power transmission structure of trolleybuses and trams, I designed the track ground power transmission line at the bottom of the float."



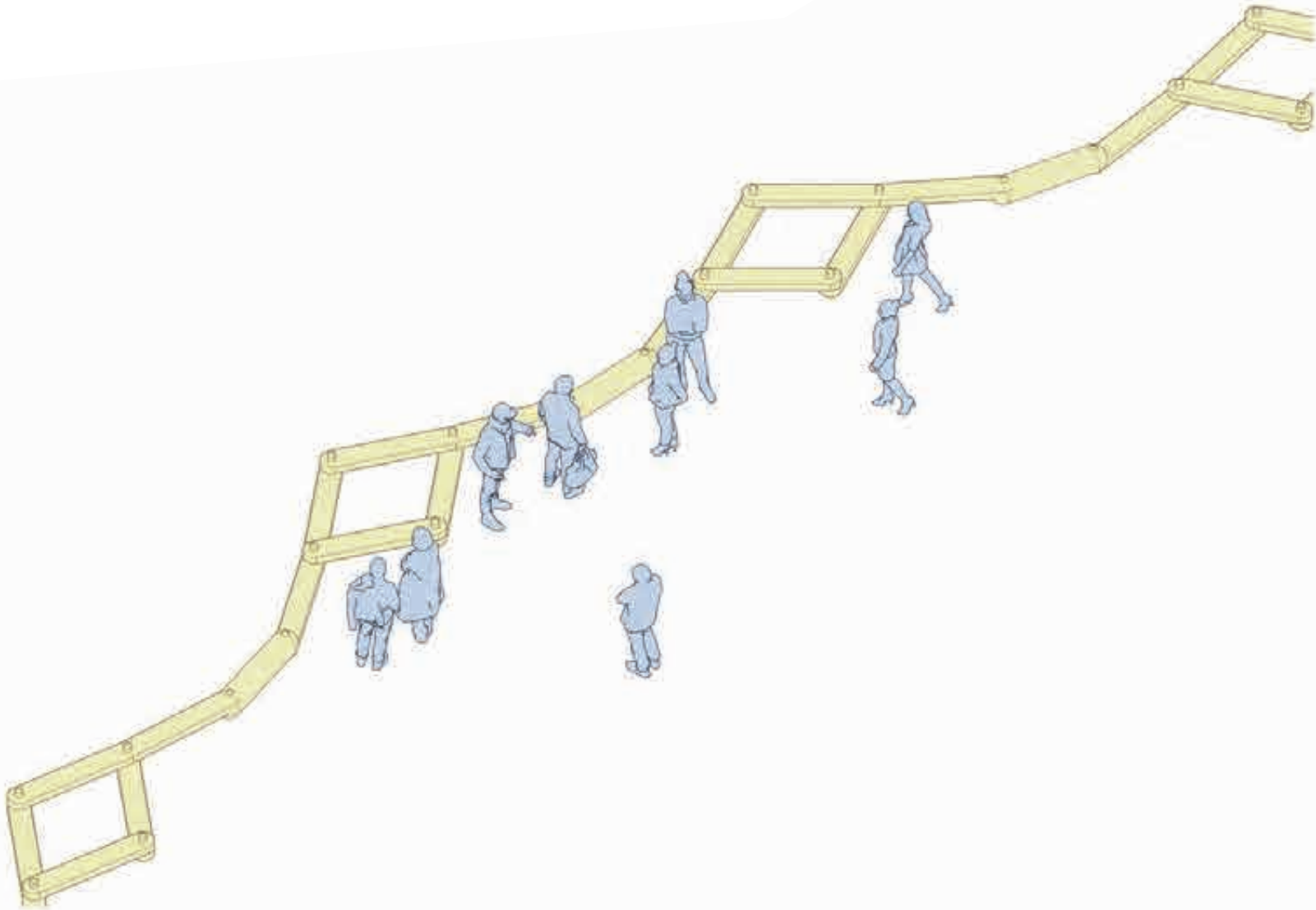
Part VIII: Avoidance of Hazards

A large dynamic building is risky. In this chapter, through conception and investigation, a series of risk early warning systems and protection measures are established.

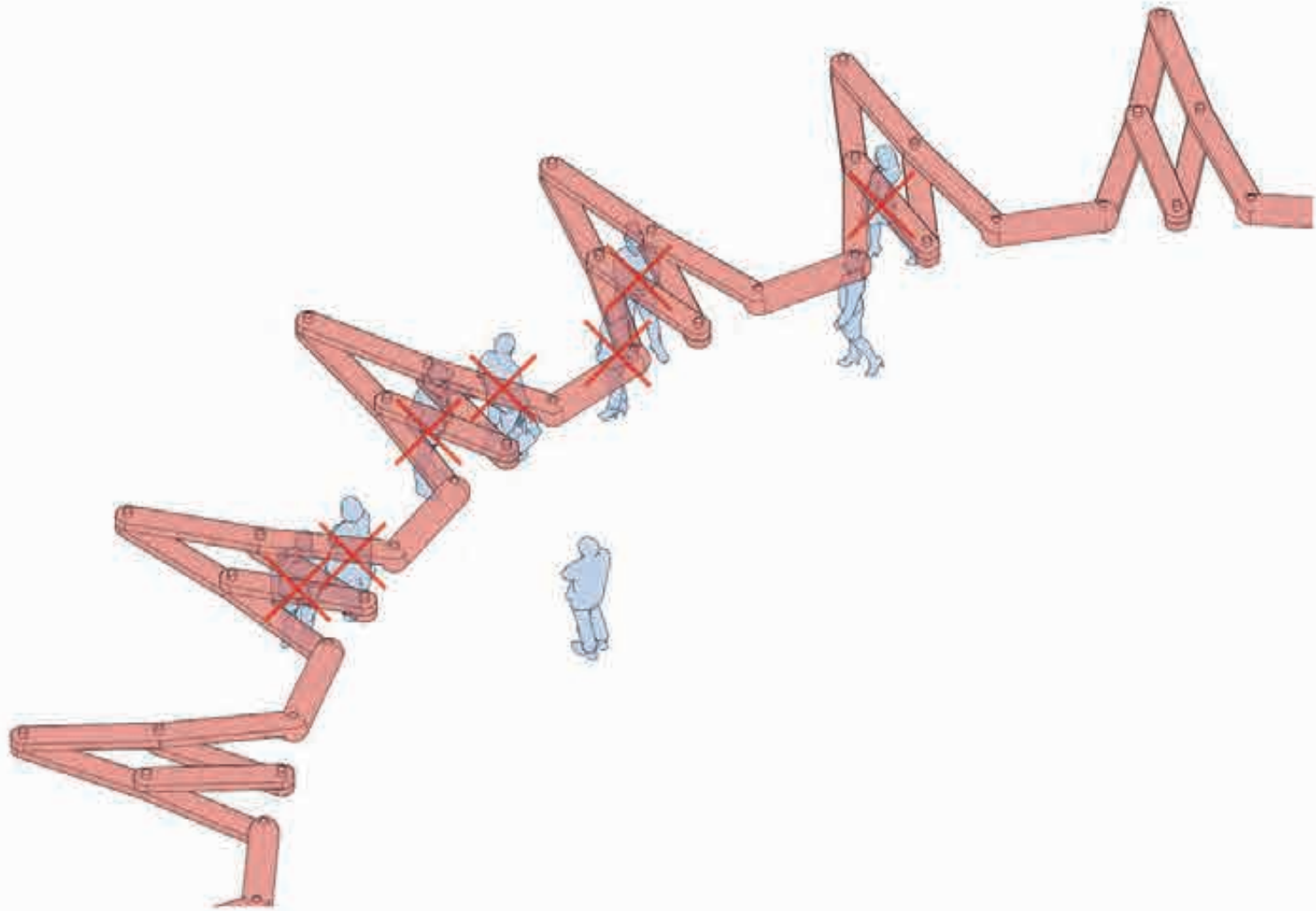
Part VIII: Avoidance of Hazards

Compression Hazard

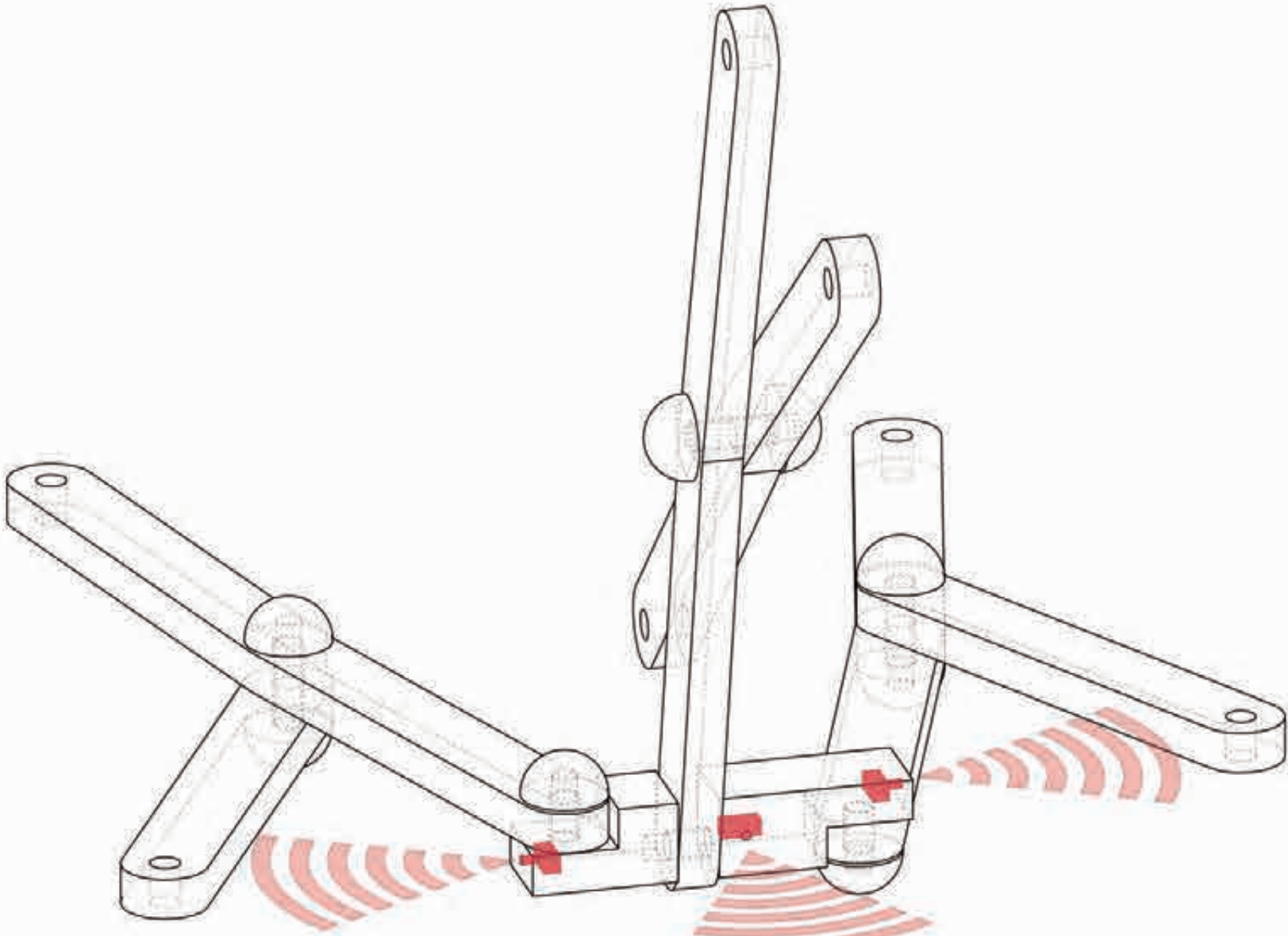
"In order to prevent the dynamics of the floats from accidentally injuring the audience during the performance, I designed a multi-directional sensing system that can detect and stop potential crises before they occur."



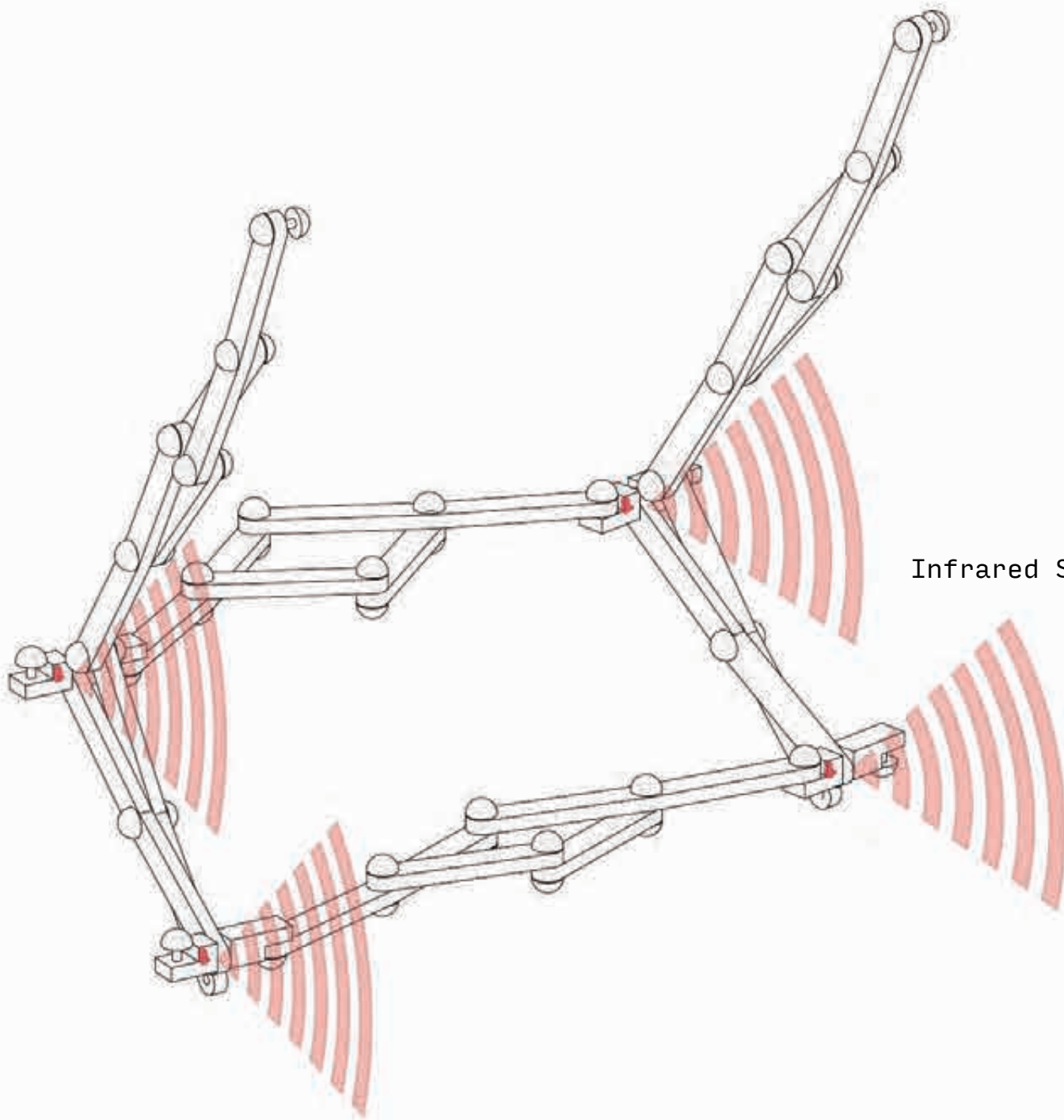
Warning



Hazard



Infrared Signal

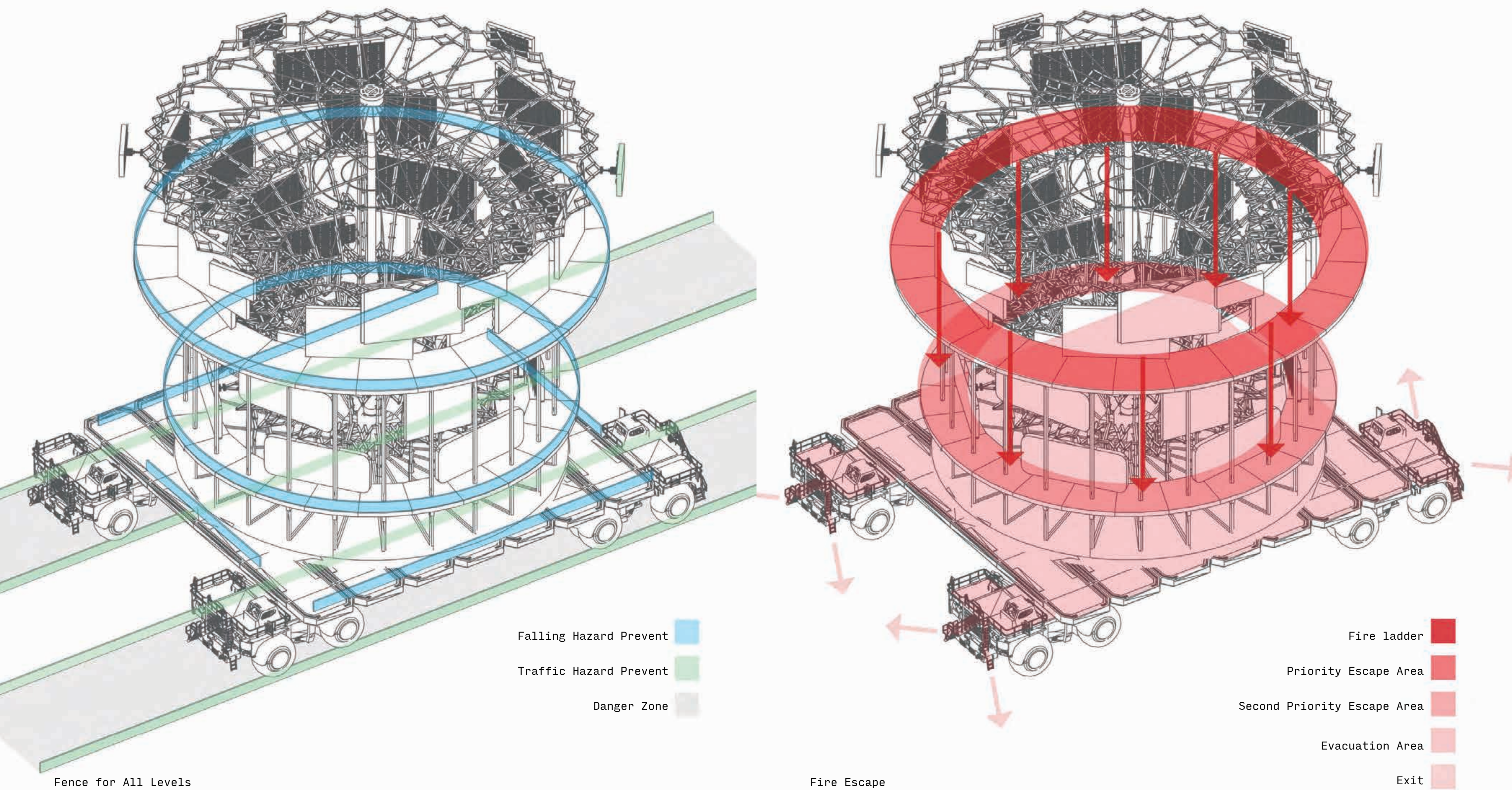


Infrared Signal

Part VIII: Avoidance of Hazards

Fence & Fire Hazard

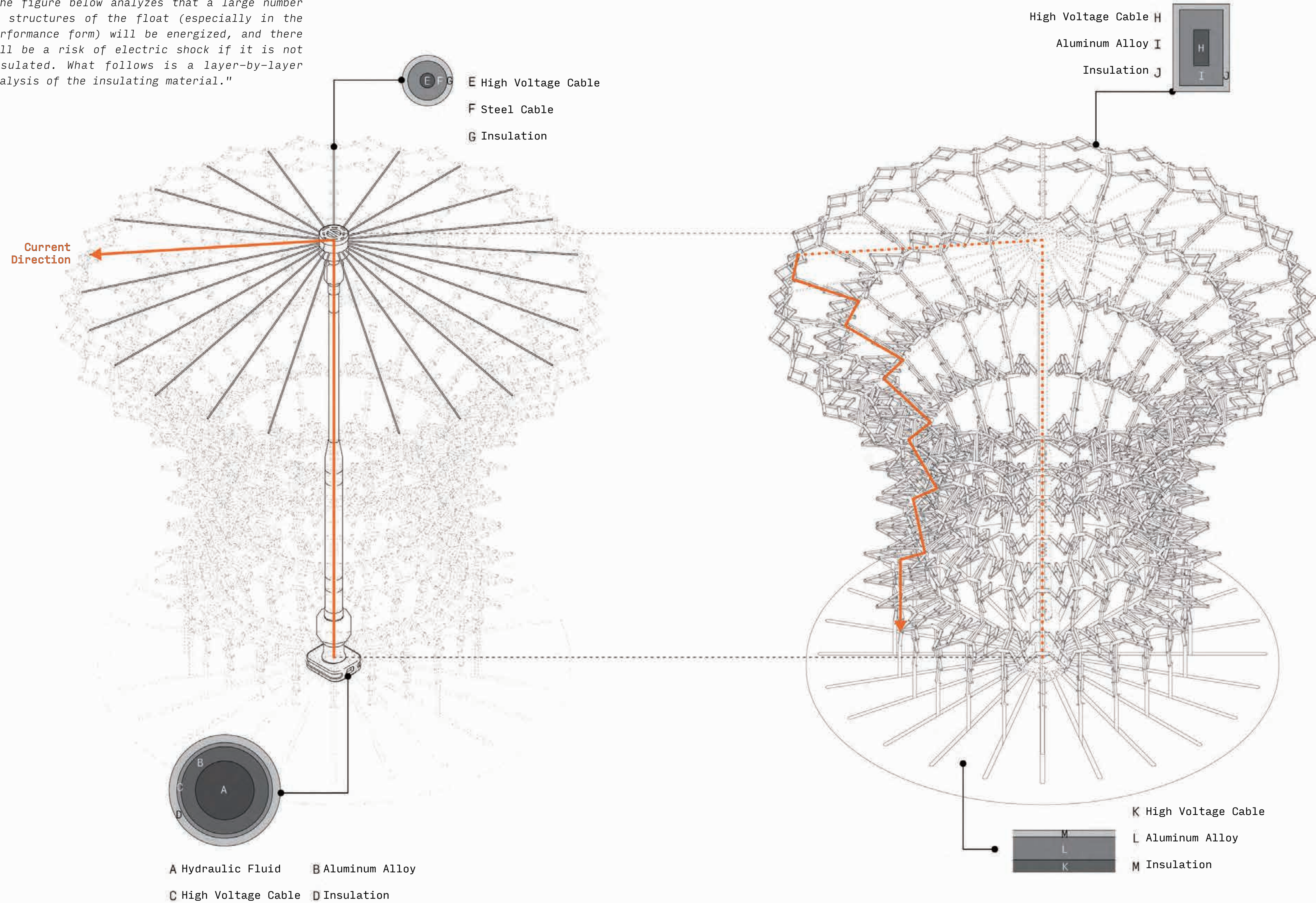
"As the basic safety facilities of the building, the designed anti-fall fences, traffic fences and fire escape channels are indispensable."



Part VIII: Avoidance of Hazards

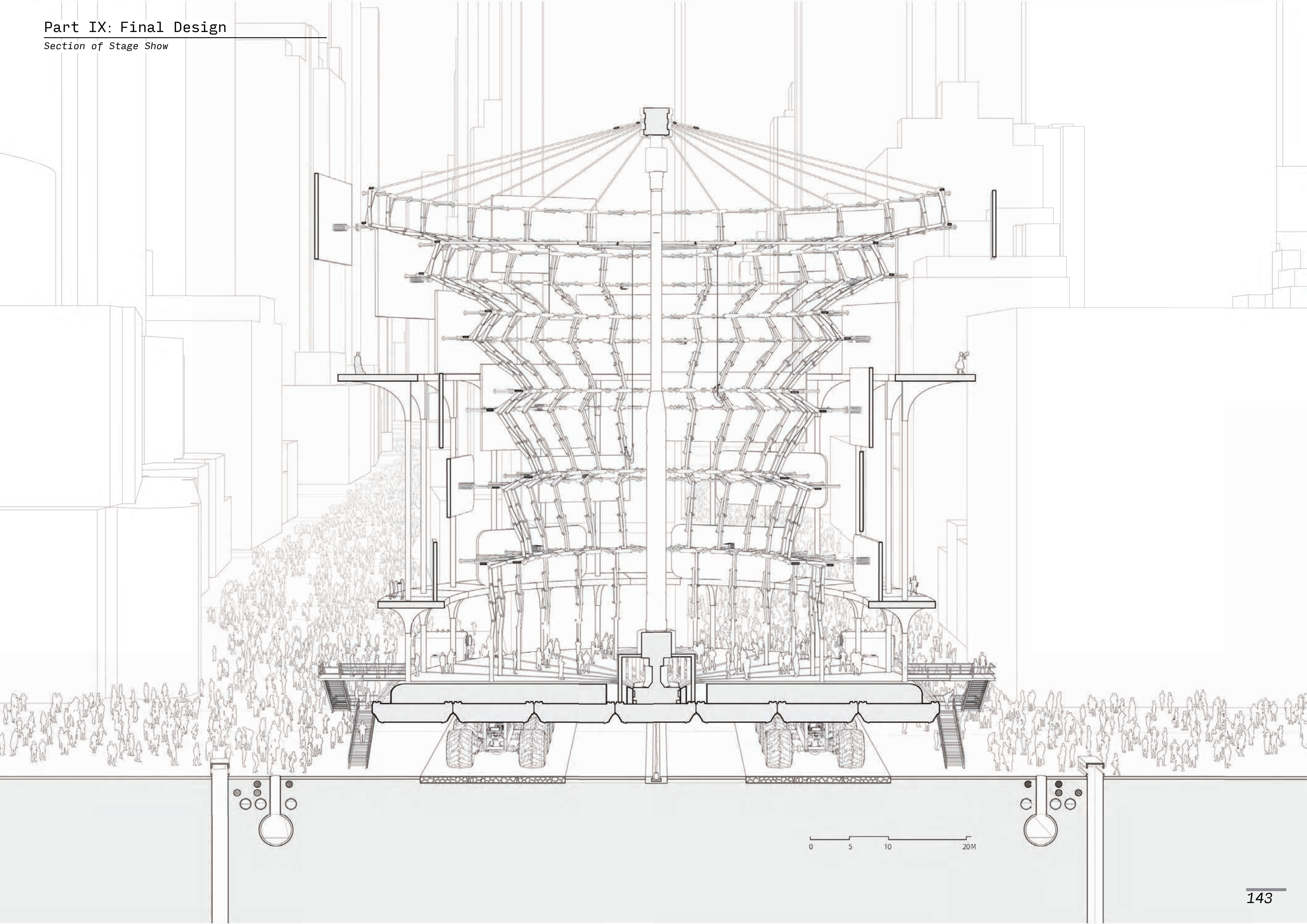
Electric Path & Insulation

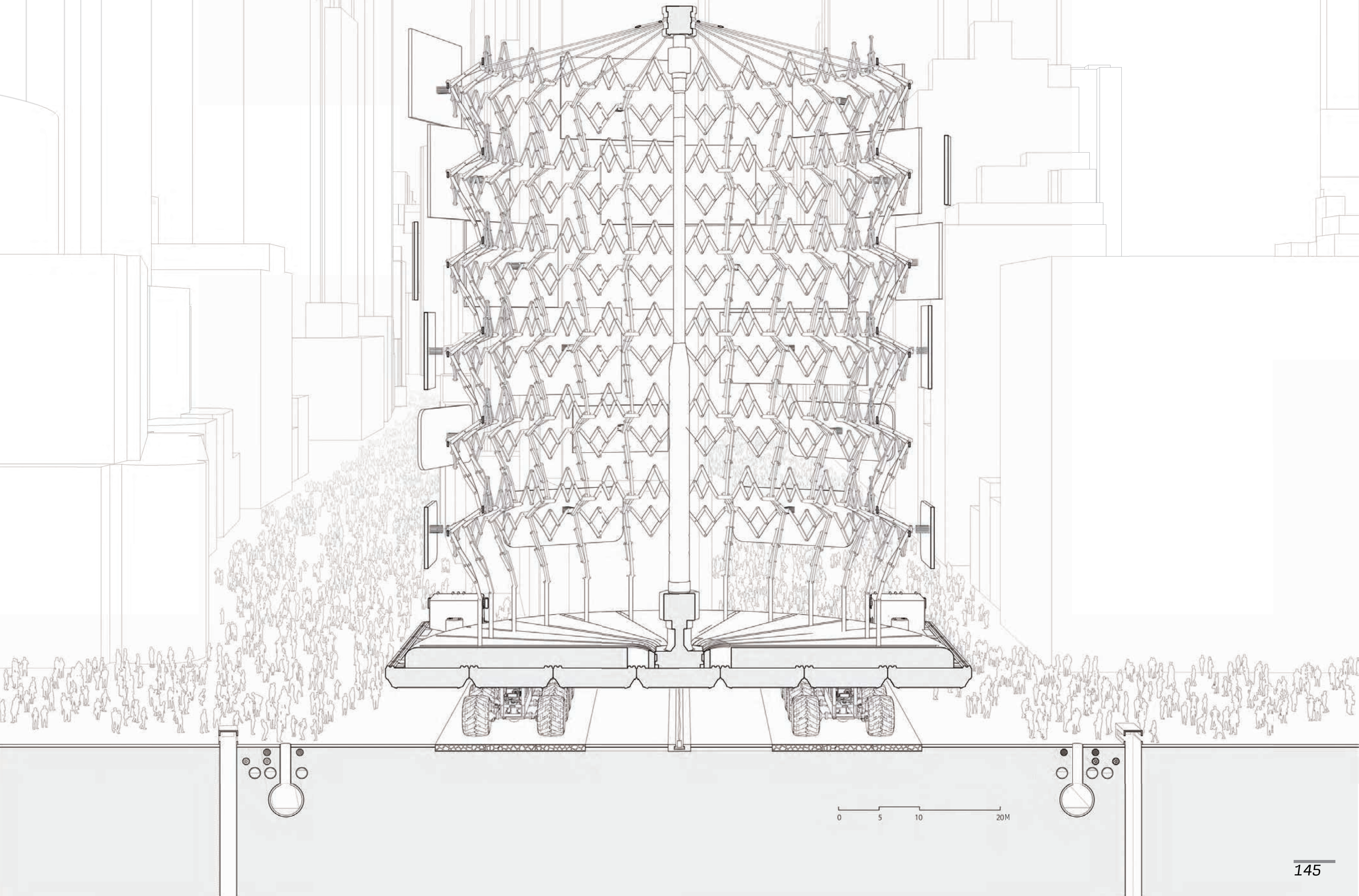
"The figure below analyzes that a large number of structures of the float (especially in the performance form) will be energized, and there will be a risk of electric shock if it is not insulated. What follows is a layer-by-layer analysis of the insulating material."

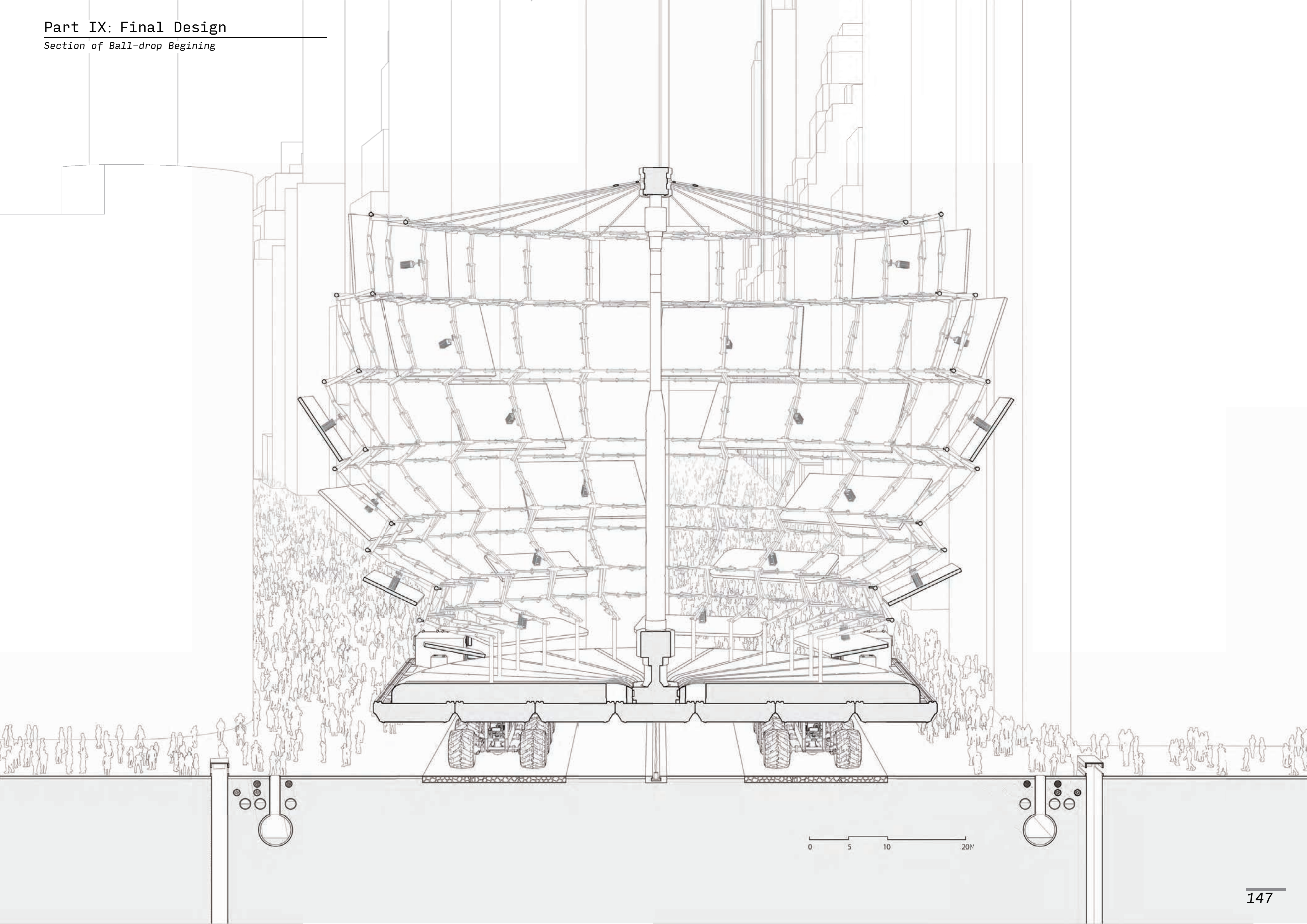


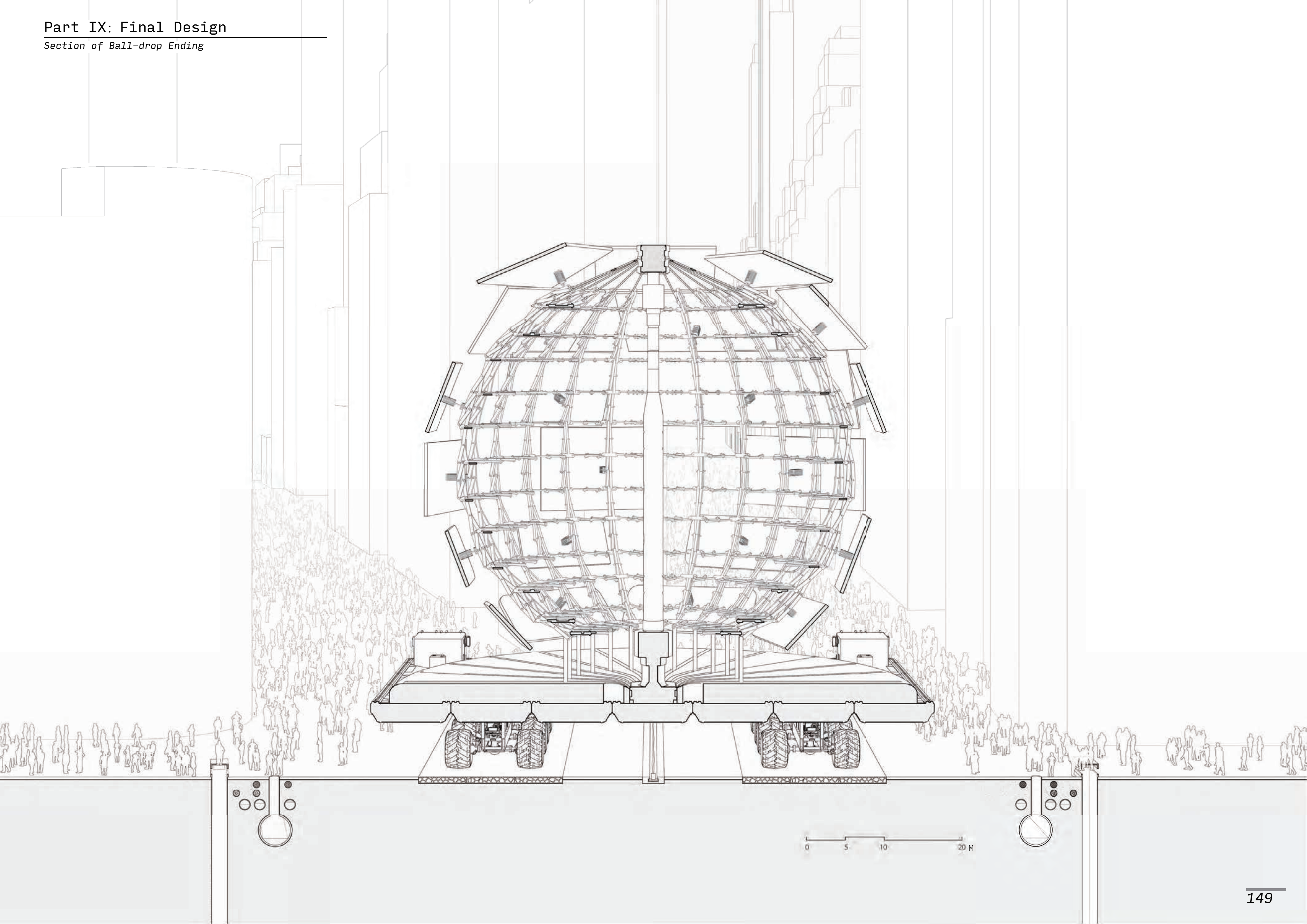
Part IX: Final Design

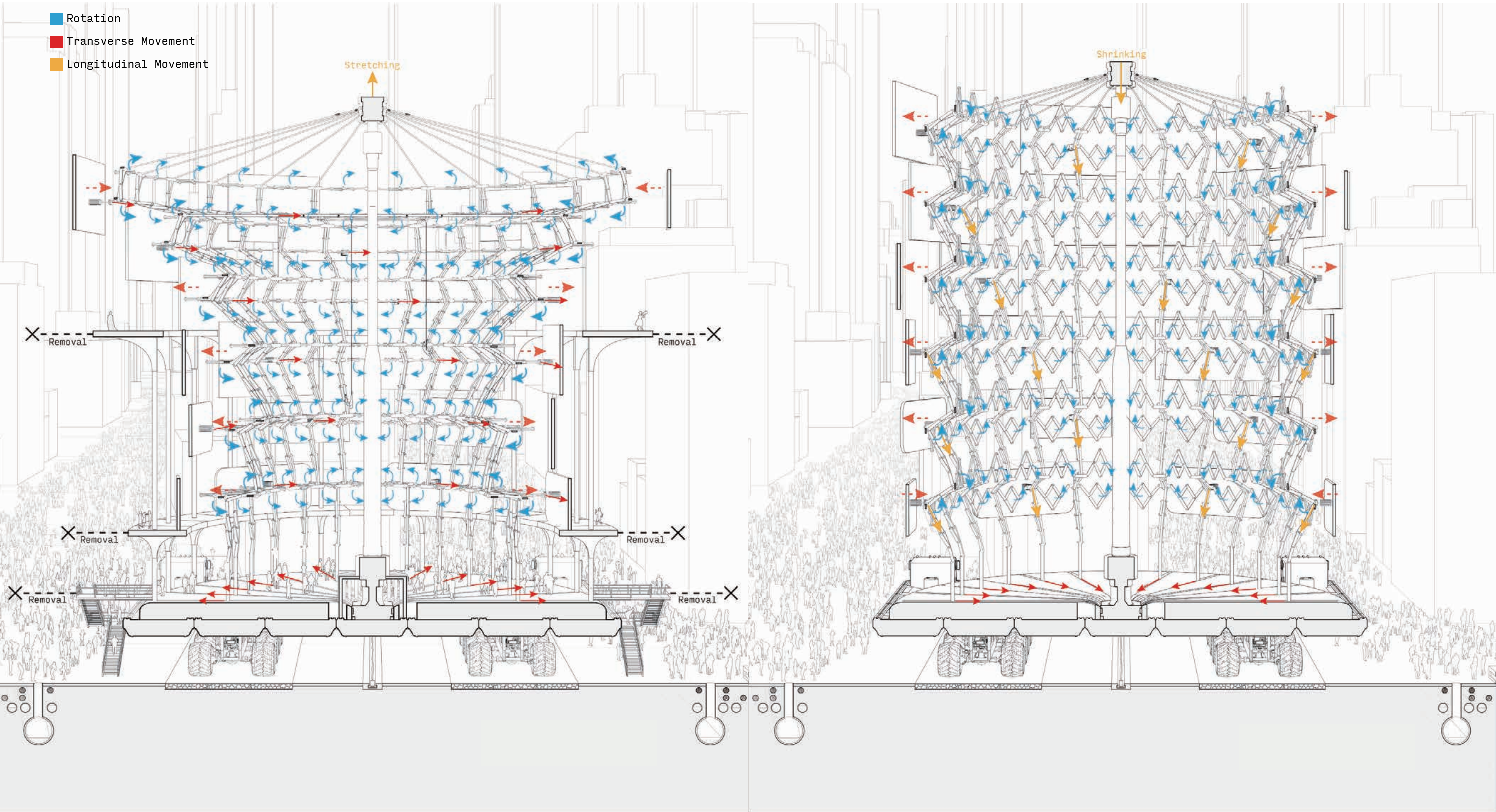
The final chapter will consist of perspective sections of the four spatial forms of the building, a long sectional diagram, as well as a summary.

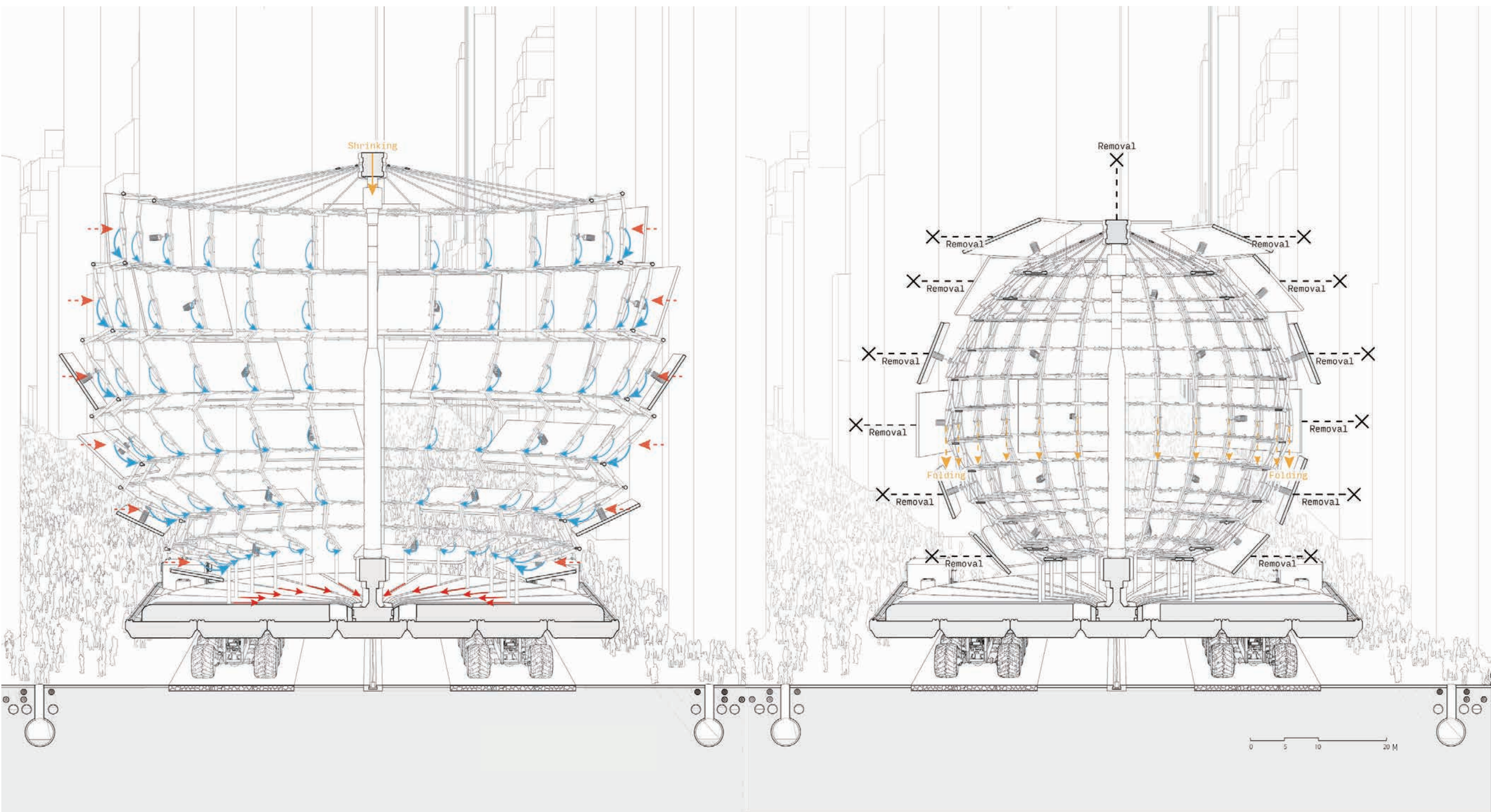








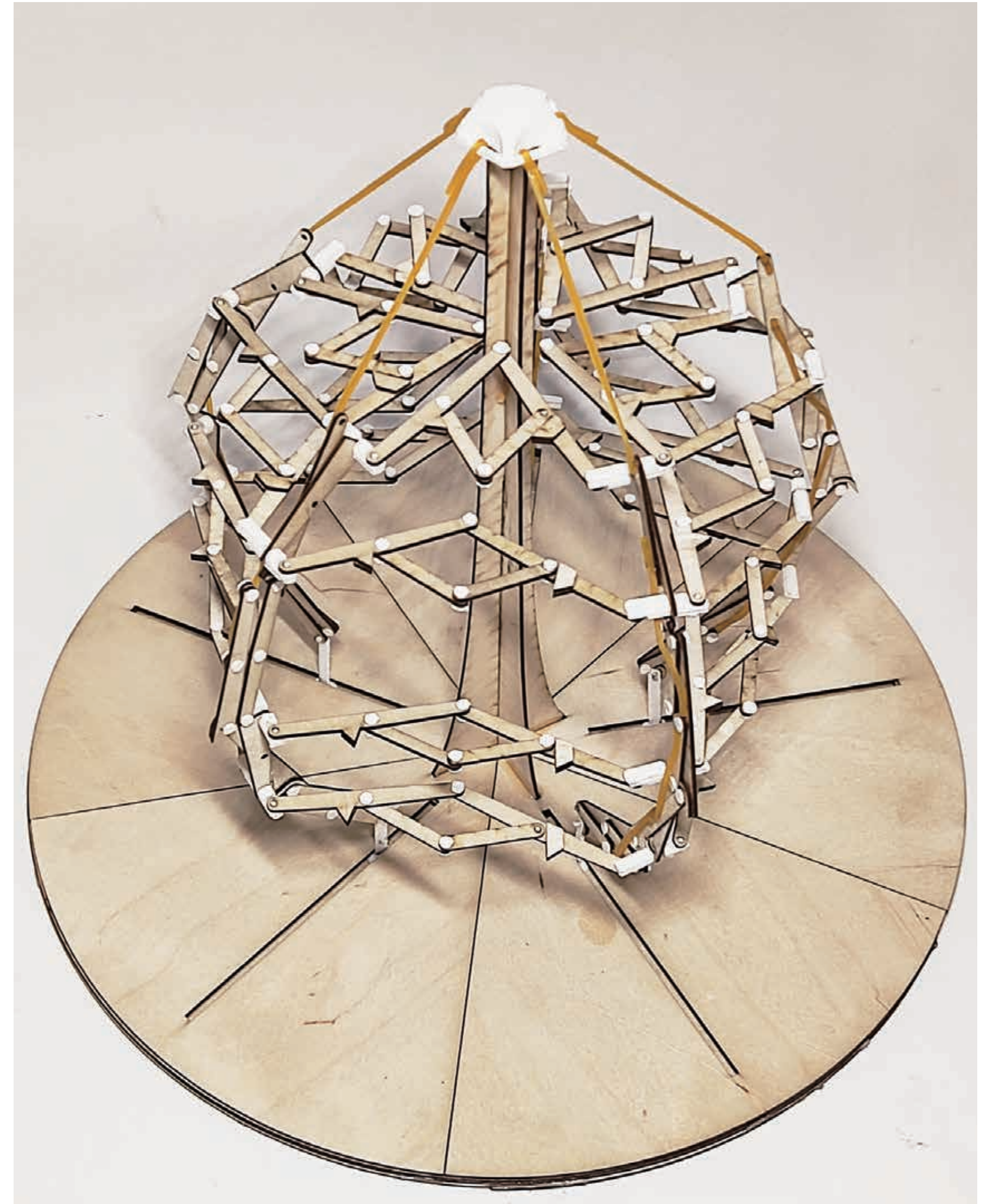




Part IX: Final Design

4-level Physical Model

"The final model simplifies the large number of repeated modules of the design, retains key design elements, and proves the dynamic and load-bearing feasibility of the design scheme."



This technical study started with a detailed investigation of the design site, and then passed through a large number of case studies as the feasibility evidence and design reference for the dynamic design. The design focused and began with dynamic architecture, in specific, the study of a dynamic structure that allows transforming from a gig stage to a telescoping ball. In the process of further deepening the design, many other design factors that have not been considered have gradually come into view. Each newly discovered design problem triggers deeper practical thinking, such as: Can the entire dynamic building not rely on the main support? How to balance the flexibility and stability of local units? Where does the energy consumption of the parade come from? ...These profound questions also imperceptibly urged me to further improve the project.

It is true that the design still has many deficiencies, such as yet to fully verify the feasibility of the whole dynamic system, the imperfections of the physical models, the relatively low participation of environmentally friendly design ideas, and the not fully expressed internal mechanical structure of the float, etc. But I believe in the correctness and forward-looking of the overall design ideas so far, and I also believe in the rationality of the general direction of research at various technical levels in each chapter.

As a cultural symbol of Times Square, electronic billboards have occupied more and more exterior surfaces of buildings over time. For a long time, the buildings on the side of the square have been more or less abandoned and abused. A brave and daring new design proposal, plans to remove these long-term sun-blocking, noisy, light-polluting and power-hungry technological products. Drawing on the historic, annual ball-drop event in Times Square, the billboards have been reimagined as an annual, extravagant event float. The event was so big that Times Square no longer needed billboards to provide the fame. Learning from Robert Venturi, floats replaced these billboards and became the new duck of Times Square. Or just to provide a critical thinking.